Online Appendix

Water Troubles Downstream: How Rivers Carry Rainfall Shocks to Conflict in India

Mahiro Kobayashi*†

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*Graduate School of Public Policy, University of Tokyo.
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Appendix A. Data Construction

Appendix A documents how the data and the treatment are constructed: the unit of analysis, the upstream catchment of each district, the weights on its sub-basins, and the rainfall measure.

A.1. Unit of Analysis

- **Why districts?** They are the finest unit at which UCDP events can be placed without dropping or arbitrarily assigning a third of them.
- **Does leaving out Pakistan- and China-administered Kashmir matter?** No (Table E.4).

All series are harmonized to 2001-census district boundaries. The census map has 594 units: 593 districts, and the part of the former state of Jammu and Kashmir administered by Pakistan and China (Gilgit-Baltistan, Azad Jammu and Kashmir and the Shaksgam tract), which the map draws but the census did not enumerate. That territory holds 4.0 percent of UCDP events, mostly India–Pakistan fighting along the Line of Control, and is left out of the panel. Table E.4 shows that the headline estimates do not depend on leaving it out: they keep their sign and significance whether its events are reassigned to the nearest Indian district or it enters the panel as a district of its own. The district is the finest unit at which UCDP events can be used without loss: of the 89.7 percent of events UCDP locates to a district or better, 34.8 percent carry exact coordinates, 33.1 percent are placed within 25 km of a known point, and 32.1 percent are known only to lie in a district and are coded at its centroid. A finer unit — a grid cell or a river sub-basin — would have to drop that third of the events or assign it arbitrarily, and would misassign more of the 25 km events: placing each at random within its stated radius, the true location falls outside the assigned unit for 33 percent of them under districts and 67 percent under 0.25-degree cells.

Every located event belongs to one of 64 UCDP conflicts, and Table A.1 lists them with the two classifications the paper uses. The conflict-system tables (Tables E.6 and E.5) assign each conflict by name to the Maoist (4,246 events), Kashmir (8,205), Northeast (2,524), communal (360) or India–Pakistan interstate (751) system, or to a residual (474), so an event follows its conflict wherever it occurs. Three assignments are judgements rather than UCDP labels: the Punjab insurgency and Sikh attacks on civilians (252 events) are a separate insurgency and sit in the residual rather than with Kashmir; Ranvir Sena, an anti-Naxal militia, sits with the Maoist system; and fights between Naga factions (225 non-state events) with the Northeast. The shares in Section II (91 and 4 percent) come instead from an event-level rule on the actors, so that a one-sided attack by a Kashmir group and a battle between the government and the Maoists both count as insurgency whatever their conflict name. The two schemes agree on all but 609 events (3.7 percent): the Punjab insurgency is insurgency under the actor rule but belongs to no system in the tables; the Naga faction fights, Ranvir Sena and a few clashes within an insurgency are system events the actor rule files elsewhere; and the Islamic State, Indian Mujahideen and VHP attacks go the other way. A further 654 located events fall in the unenumerated part of Jammu and Kashmir and enter no regression.

Table A.1: UCDP conflicts in the located record, by conflict system and type of violence

UCDP conflict	Type of violence	Actor-rule class	Located events
Maoist (8 conflicts, 4,246 events)			
India: Government	State-based	Insurgency	2,392
CPI-Maoist – Civilians	One-sided	Insurgency	1,662
PLFI – Civilians	One-sided	Insurgency	79
PWG – Civilians	One-sided	Insurgency	50
MCC – Civilians	One-sided	Insurgency	40
3 other conflicts	Non-state, One-sided	Communal, Insurgency, Other	23
Kashmir (4 conflicts, 8,205 events)			
India: Kashmir	State-based	Insurgency	6,661
Kashmir insurgents – Civilians	One-sided	Insurgency	1,529
2 other conflicts	Non-state, One-sided	Insurgency, Other	15
Northeast (25 conflicts, 2,524 events)			
India: Assam	State-based	Insurgency	474
India: Manipur	State-based	Insurgency	370
India: Bodoland	State-based	Insurgency	329
ULFA – Civilians	One-sided	Insurgency	182
India: Tripura	State-based	Insurgency	173
NSCN-IM – NSCN-K	Non-state	Other	167
India: Western South East Asia	State-based	Insurgency	138
NLFT – Civilians	One-sided	Insurgency	114
NDFB – Civilians	One-sided	Insurgency	110
India: Nagaland	State-based	Insurgency	92
India: Garoland	State-based	Insurgency	74
ATTF – Civilians	One-sided	Insurgency	51
NSCN-K – NSCN-K-K	Non-state	Other	43
NSCN-IM – Civilians	One-sided	Insurgency	41
NDFB-S – Civilians	One-sided	Insurgency	36
DHD – BW – Civilians	One-sided	Insurgency	21
KRA – Civilians	One-sided	Insurgency	21
UPDS – Civilians	One-sided	Insurgency	21
7 other conflicts	Non-state, One-sided, State-based	Communal, Insurgency, Other	67
Communal (10 conflicts, 360 events)			
Hindus (India) – Muslims (India)	Non-state	Communal	191
Kuki – Meitei	Non-state	Communal	41
Bangladeshi Immigrants – Bodo	Non-state	Communal	26
Bodo – Santhal	Non-state	Communal	23
Dimasa – Zeme Naga	Non-state	Communal	23
Tripura Non-Tribal Communities – Tripura Tribal Communities	Non-state	Other	20
4 other conflicts	Non-state	Communal	36
Interstate (2 conflicts, 751 events)			
India – Pakistan	State-based	Other	750
1 other conflicts	State-based	Other	1
Unaffiliated one-sided (7 conflicts, 165 events)			
Government of India – Civilians	One-sided	Other	121
VHP – Civilians	One-sided	Communal	26
5 other conflicts	One-sided	Insurgency, Other	18
Other (8 conflicts, 309 events)			
Sikh insurgents – Civilians	One-sided	Insurgency	148
India: Punjab/Khalistan	State-based	Insurgency	104
India: Islamic State	State-based	Insurgency	41
5 other conflicts	Non-state, One-sided, State-based	Insurgency, Other	16
Total			16,560

Notes: Every UCDP event located to a district or better, 1990–2025 (16,560 events). A conflict system is assigned to each UCDP conflict by name, so an event belongs to its conflict's system wherever it occurs; conflicts with fewer than 20 located events are pooled within their system. Type of violence is UCDP's: state-based (a government against an organized group), non-state (two organized groups), one-sided (an organized actor against civilians). The actor-rule class is the event-level rule behind the shares in Section II: insurgency is state-based violence between the Government of India and a non-government actor, or one-sided violence by a non-government actor; communal is non-state violence between named communal groups or one-sided violence by VHP, Ranvir Sena or ABSU; other is the rest (interstate violence, one-sided violence by governments, fights between insurgent factions). The part of Jammu & Kashmir the census does not enumerate holds 654 of the located events, which enter no regression.

Table A.2: The HydroBASINS hierarchy over India

Level	Number	Median km^2	Unit	Use in this paper
HydroBASINS sub-basins				
1	1	20,827,400	Continent (Asia)	
2	3	2,255,600	Continental regions	
3	9	748,800	Largest river basins	
4	34	75,500	Whole river systems	Inference clusters (Sec. B)
5	120	23,100	Main river branches	Inference clusters (Sec. B)
6	431	6,200		Rainfall coarsened (Fig. A.2)
7	1,486	1,600		Rainfall coarsened (Fig. A.2)
8	4,662	502		Rainfall coarsened (Fig. A.2)
9	12,393	204		Rainfall coarsened (Fig. A.2)
10	22,945	139		Treatment catchments
11	24,976	136		
12	25,029	136		
For reference				
	1	3,186,000	India	
	593	4,060	Districts (2001 census)	

Notes: HydroBASINS v1c (Lehner and Grill, 2013), Asia layer. Sub-basins are counted if they intersect India (2001 census districts); median areas are of those sub-basins, including any part outside India. Levels 1–3 were drawn by hand; from level 4 on, each sub-basin is divided into nine: its four largest tributaries and the five stretches of main river between them. Reference rows: the area of India and of the median district on the 2001 census map, without the unenumerated Pakistan- and China-administered unit, on an equal-area projection; for India the area column gives its total area.

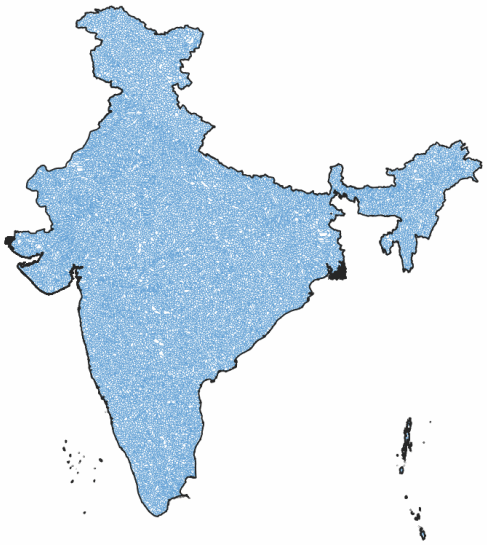
A.2. Defining Upstream Catchment

- **How is a district's upstream catchment drawn?** Every sub-basin whose water reaches the district, minus those overlapping it or a neighbour.
- **Does the result depend on how finely rainfall is measured?** No (Figure A.2).
- **Does it depend on where the catchment boundary is drawn?** No: across buffer radii of 0 to 100 km, all 44 estimates are positive (Figure A.3).

The catchments are built from HydroBASINS (Lehner and Grill, 2013), a global map that divides land into sub-basins and records, for each, the one sub-basin its water flows into next, so that water can be followed from any sub-basin down to the sea (Figure A.1). The map is nested in twelve levels: each level splits every sub-basin of the level above into smaller ones, until they reach roughly 100–250 km^2, below which the map stops splitting (Table A.2). The paper builds catchments from level 10, and uses levels 4 and 5, whole river systems and their main branches, as clusters for inference (Section B).

Level 10 is fine enough to draw a catchment around each district, but no finer than rainfall is measured. At level 10 the median sub-basin covers 139 km^2, against 4,060 km^2 for the median district (about thirty sub-basins per district), and is 4.9 times a cell of the 0.05-degree CHIRPS rainfall grid; only 7.2 percent of sub-basins are smaller than one cell. A finer level would add almost nothing: level 11 has only 9 percent more sub-basins over India than level 10, and the same median size (136 against 139 km^2), since level-10 sub-basins are already close to the size at which HydroBASINS stops splitting.

(a) Level-10 sub-basins



(b) The Kosi–Ganga confluence, district boundaries in black

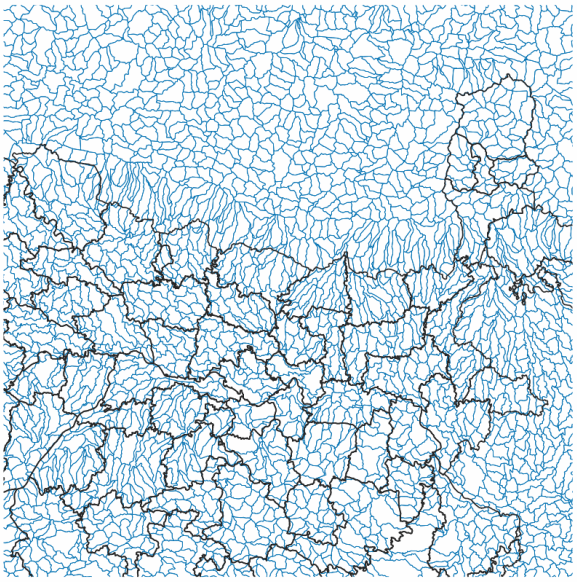


Figure A.1: The HydroBASINS level-10 delineation

Notes: Panel (a): level-10 sub-basin boundaries (blue) within India's bounding box, national outline in black. Panel (b): the Kosi–Ganga confluence at full resolution, with 2001-census district boundaries in black.

A district's upstream catchment is every level-10 sub-basin whose water reaches it, found by following the chain of downstream sub-basins until it enters the district. HydroBASINS records a single downstream sub-basin for each, so it cannot represent a river that divides.[1] The construction therefore adds the points where rivers split, taken from a hydrological routing model (Section A.3), and follows the water down every branch, each carrying its share of the flow.

Two kinds of sub-basin are then removed, so that the catchment holds only rain that reaches the district and that neither the district nor its neighbours received. The first drain to an inland lake rather than the sea, such as Sambhar Lake in Rajasthan or the closed lakes of southern Tibet: HydroBASINS links them to a nearby river only for bookkeeping, and their water reaches no district outside their own closed basin. The second are caught by the neighbour exclusion of Section III, which removes every sub-basin that overlaps the district or a bordering district. Across an international border the neighbours are the adjacent administrative units of Bangladesh, Nepal, Pakistan, China, Myanmar and Bhutan, at the level closest in size to an Indian district.

The result does not depend on how finely rainfall is measured. Figure A.2 keeps the catchment, neighbour exclusion and weights of Table 2 and coarsens only the rainfall: each level-10 sub-basin is given the average rainfall of the catchment's sub-basins that lie in the same sub-basin at level 9, 8, 7 or 6 (Table A.2 gives their sizes). Even at level 6, where a sub-basin is larger than the median district, no coefficient moves by more than 0.16; columns 2 to 5 stay significant at the one-percent level throughout, and column 1, without the spatial controls, at the ten-percent level; column 4 is 1.344 against 1.460.

[1] HydroBASINS is built from a flow-direction grid in which every cell drains to exactly one neighbour. At a split, all of the river's water therefore continues down one branch, whichever the elevation model favours, and the other branch appears as a new river that starts at or near the split and collects only the land draining directly into it. Below Farakka, for instance, the whole Ganga continues into Bangladesh as the Padma, while the Bhagirathi–Hooghly is recorded as a separate river whose catchment at Berhampore is a few thousand square kilometres, against nearly a million for the Ganga; in the Godavari delta the river continues down the Gautami branch, and the Vasishta starts afresh.

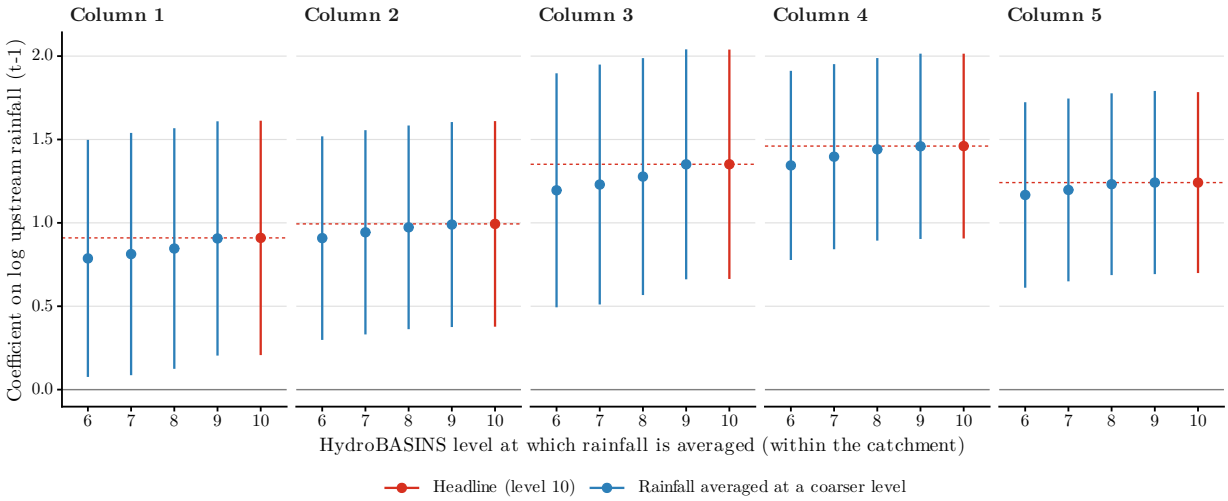


Figure A.2: The lagged upstream coefficient with rainfall averaged over coarser sub-basins

Notes: Each panel is one column of Table 2. The catchment, neighbour exclusion and river-distance weights are Table 2's, built from level-10 sub-basins. Each level-10 sub-basin's monsoon rainfall and temperature are replaced by the area-weighted average over the catchment's level-10 sub-basins that share its larger sub-basin at the level on the horizontal axis, so sub-basins removed by the neighbour exclusion never enter. Level 10 is the paper's estimate, marked by the dashed line. Bars are 95 percent intervals on Conley spatial-HAC standard errors (150 km).

Figure A.3 replaces the neighbour exclusion with a buffer radius, keeping only sub-basins farther than a given distance from the district, and re-estimates the elasticity at every radius from 0 to 100 kilometres. It does so four ways: with distance measured to the sub-basin's polygon or to its representative point, and with the district set either changing with the radius or held fixed. Under the specification of Table 2, column 4, all 44 estimates are positive, ranging from 0.52 to 2.28, and 37 of them are significant at five percent (panels a–d); every estimate under column 1's specification is positive as well.

The district's own catchment also matters more than other catchments from the same state. Panels (f) and (g) give each district another district's catchment from its state, 5,000 times at each radius, on the specification and districts of panels (c) and (d).[2] The true map beats these counterfeits at every radius under both distance measures: the exact p-value is at most .005 with polygon distance and .0002, the smallest 5,000 draws allow, with representative-point distance.

Two further cautions apply. First, the eleven radii are not separate tests: each catchment is the one at the previous radius minus its nearest sub-basins, so neighbouring estimates share most of their data. Second, the districts that keep a catchment at 100 km are not a random subset. Their catchments are larger (62 against 3.2 thousand km^2), their sub-basins farther away (158 against 13 km) and their rivers flatter (0.91 against 0.62 flat share): they are large floodplain districts whose water comes from far away, and they are not disproportionately in the Northeast.

[2]Section B.5 describes the test.

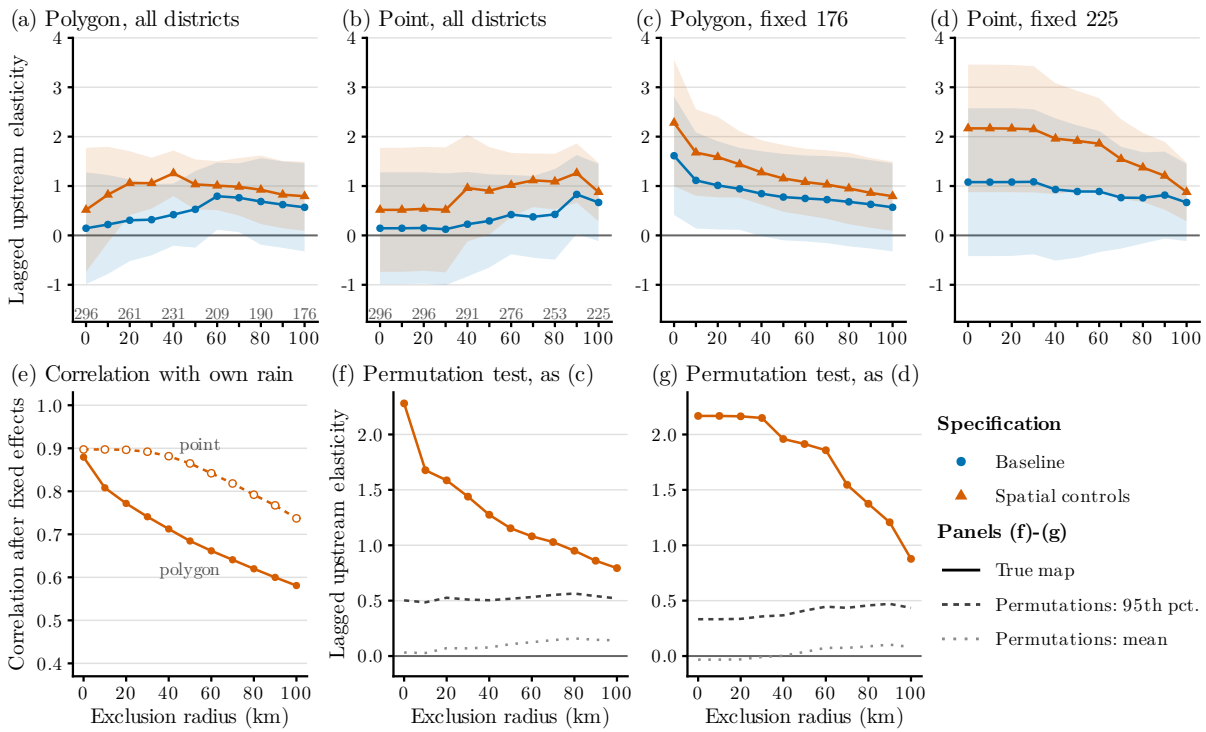


Figure A.3: The lagged upstream elasticity across buffer radii

Notes: Each radius keeps only upstream basins farther than that distance from the district. Panels (a)–(d) re-estimate columns 1 (Baseline) and 4 (Spatial controls) of Table 2, with Conley (150 km) 95 percent intervals; distance is measured to the basin polygon (a, c) or its representative point (b, d). In (a)–(b) the district set shrinks as the radius grows (districts retained along the bottom); (c)–(d) hold it at the districts that keep a catchment at 100 km. Panels (e)–(g) use the fixed district sets of (c)–(d): (e) is the correlation, after district and year effects, between the lagged treatment and the district's own lagged monsoon rainfall; (f) and (g) compare the column-4 estimate under the true district–catchment map with 5,000 permutations of that map within states, on the cells of (c) and (d) (filled points: exact two-sided $p < 0.05$; every p is at most .005). The neighbour exclusion is not a distance rule, so no cell reproduces Table 2.

A.3. The Treatment Weights

- **How are upstream sub-basins weighted?** By how far their water travels along the river to the district, with shares where rivers split.
- **Does the result depend on the weighting?** No: under five alternatives, column 4 stays between 1.41 and 1.54 (Figure A.6).

The treatment weights each upstream sub-basin by how far its water travels along the river to reach the district: $\omega_{ib} = \sum_p s_p/(1 + r_p)$, summed over the paths p from sub-basin b to district i, where r_p is the path's length and s_p the share of b's flow that follows it (Section III). Figure A.4 draws each piece of this notation. On a single channel there is one path with $s_p = 1$.

The length r_p is how far the water travels along the river, from where it leaves sub-basin b to where it enters the district. HydroBASINS records each sub-basin's distance to the river mouth, so on a single channel r_p is the difference between the two. A path through a river split is measured in pieces and added up.

The shares s_p come from the CaMa-Flood routing model, which gives, at each of the 85 points where a river divides, the share of the river's June–November flow that leaves through each branch, averaged over the estimation sample, 1989–2025. A path's share is the product of the shares of

the branches it takes,

$$s_p \;=\; \prod_{k \in p} \phi_k,$$

where ϕ_k is the share of the flow arriving at a split that leaves through branch k. In Figure A.4 these are the labels beside the branches: the river leaving b first splits three ways, into ϕ_1, ϕ_2 and $1-\phi_1-\phi_2$, and the branch carrying ϕ_2 splits again, into ϕ_3 and $1-\phi_3$. Path p_1 takes one branch, so $s_{p_1} = \phi_1$; path p_2 takes two, so $s_{p_2} = \phi_2 \times \phi_3$, less than either of its branch shares. A path through no split has $s_p = 1$. A sub-basin that reaches the district by several paths adds them up, here $\omega_{ib} = \phi_1/(1+r_{p_1}) + \phi_2\phi_3/(1+r_{p_2})$, as in the figure's key. The branches marked $1-\phi_1-\phi_2$ and $1-\phi_3$ never reach i, so the path shares sum to less than one, $\phi_1 + \phi_2\phi_3 < 1$.

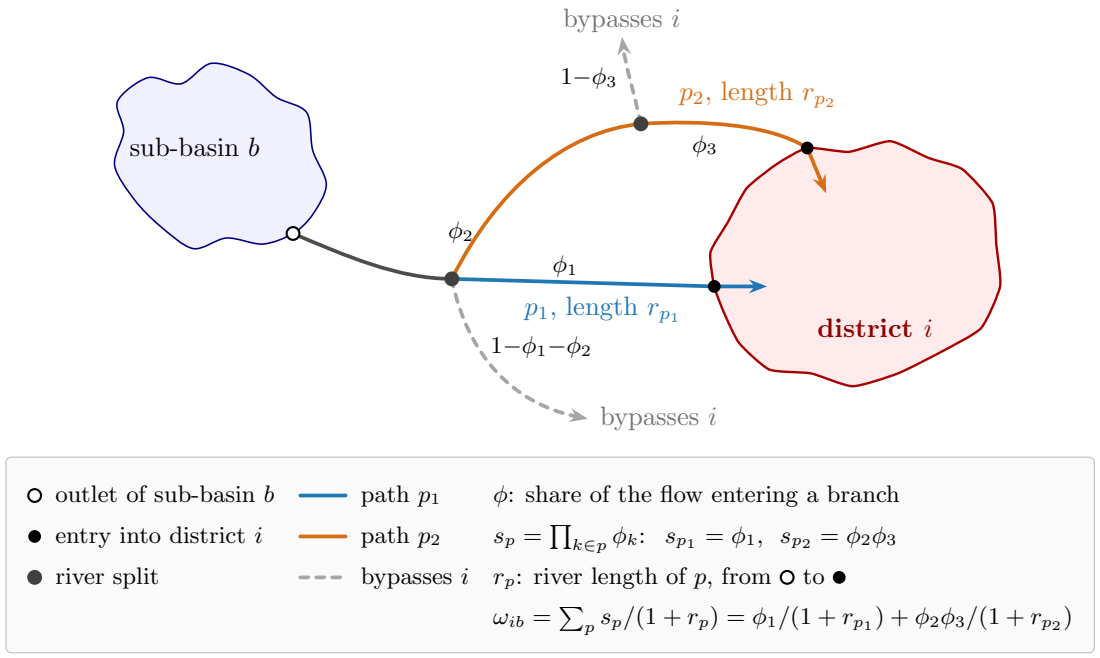


Figure A.4: The notation of the treatment weights

Notes: Schematic, not to scale. Below the outlet of sub-basin b (open circle) the river splits into three branches, with flow shares ϕ_1, ϕ_2 and $1-\phi_1-\phi_2$; the branch carrying ϕ_2 splits again, into ϕ_3 and $1-\phi_3$. Two paths enter district i (filled circles; strictly, the first sub-basin overlapping i by more than 1 km^2). A path's share is the product of the branch shares it takes, $s_{p_1} = \phi_1$ and $s_{p_2} = \phi_2\phi_3$, and its length r_p runs from the outlet to the entry point.

Figure A.5 shows the districts at the 90th percentile, the median and the 10th percentile of catchment size in the estimation sample: Dhubri (3,299 sub-basins; the district of Figure 2), Kokrajhar (217) and Malappuram (7).

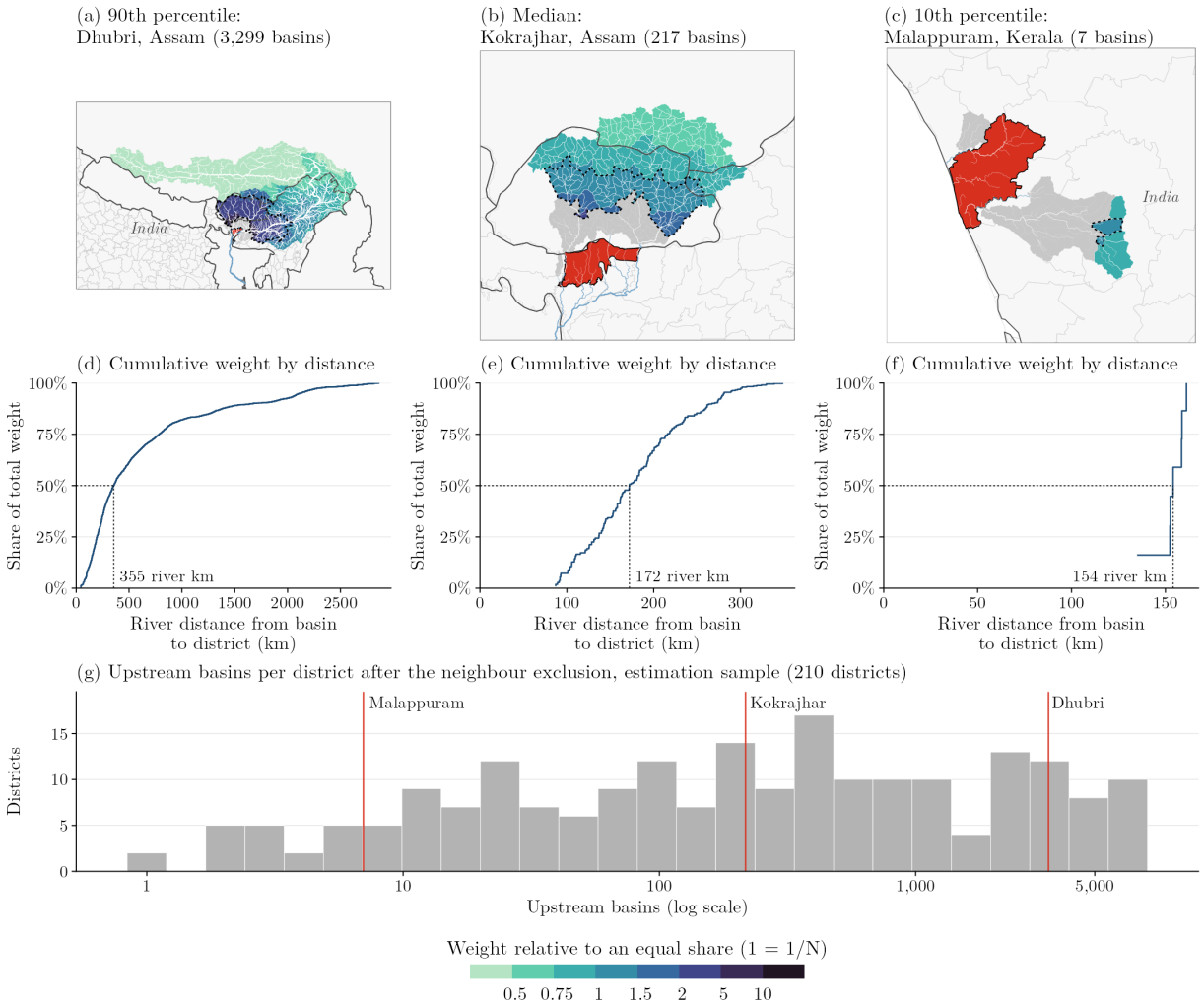


Figure A.5: Upstream weights for three districts

Notes: Districts at the 90th percentile, median and 10th percentile of upstream sub-basins in the estimation sample. Maps shade each basin by Nw_{ib}, its weight relative to an equal share (1 = equal); grey basins are excluded (own, neighbouring or adjacent foreign units); the dashed outline holds the nearest basins carrying half the weight. White lines are rivers flowing into the district (HydroRIVERS, Strahler order ≥ 3), blue lines their course downstream. Panel (g): upstream basins per district.

The estimate does not depend on how the weights are built. Figure A.6 re-estimates columns 1 and 4 of Table 2 with the weights built in each of the six ways below, one per row of the figure. In every case $w_{ib} = \omega_{ib} / \sum_{b' \in U(i)} \omega_{ib'}$; $f_{ib} = \sum_p s_p$ is the share of b's water that reaches i, g_{ib} is the great-circle distance in kilometres from the district's population-weighted centre to the sub-basin's centroid, A_b is the sub-basin's area, and c_b is its marginal runoff per unit of monsoon rain, net of snowmelt.

1. *River distance (the paper's treatment)*: $\omega_{ib} = \sum_p s_p/(1 + r_p)$.
2. *Great-circle distance*: $\omega_{ib} = f_{ib}/(1 + g_{ib})$.
3. *Great-circle distance × basin area*: $\omega_{ib} = f_{ib}A_b/(1 + g_{ib})$.
4. *River distance × basin area*: $\omega_{ib} = \sum_p s_p A_b/(1 + r_p)$.
5. *Runoff volume: area × runoff coefficient*: $\omega_{ib} = f_{ib}A_b c_b$, with c_b measured over June–November.
6. *Runoff volume, decaying with river distance*: $\omega_{ib} = \sum_p s_p A_b c_b/(1 + r_p)$.

Across the five alternatives, column 1 stays between 0.83 and 0.95 and column 4 between 1.41 and 1.54. Figure A.7 varies how fast weight falls with distance, from equal weights to steep decay; column 4 stays between 1.31 and 1.50 throughout.

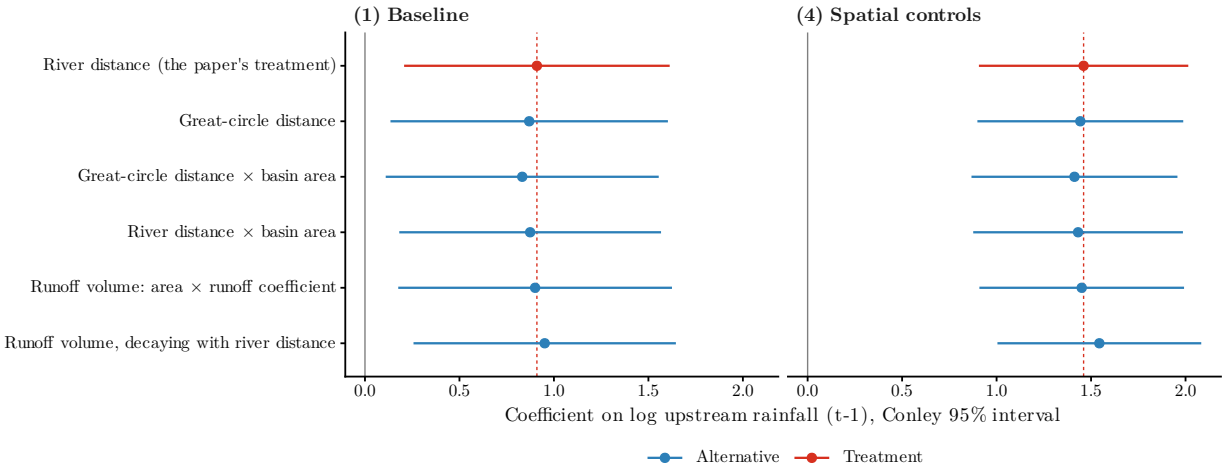


Figure A.6: The lagged upstream coefficient under alternative weightings

Notes: Each row re-estimates columns 1 and 4 of Table 2 with the treatment weights of the corresponding item in Section A.3. Bars are 95 percent intervals on Conley spatial-HAC standard errors (150 km). The dashed line marks the paper's estimate.

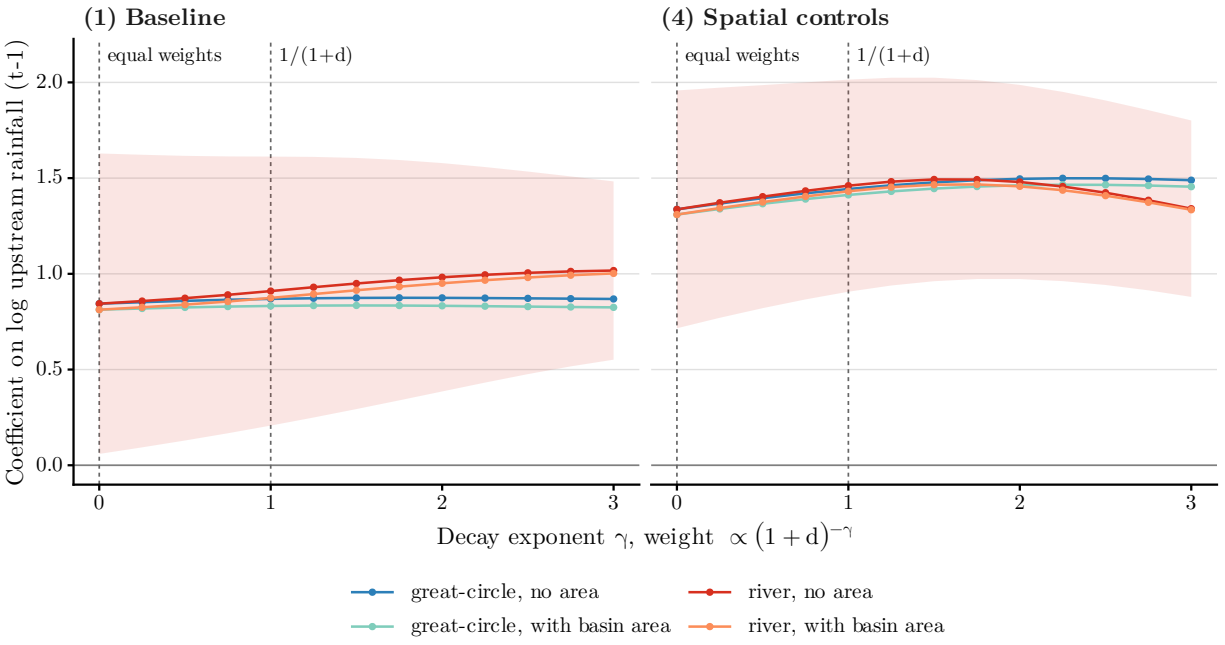


Figure A.7: How fast should weight decay with distance?

Notes: Each point re-estimates column 1 or column 4 of Table 2 with basin weights proportional to $(1+d)^{-\gamma}$, times the basin's area where indicated, normalised within district; d is river or great-circle distance as in Figure A.6. $\gamma = 0$ is equal weighting and $\gamma = 1$ the paper's functional form. The band is the 95 percent Conley interval for river distance without area.

A.4. Measuring the Upstream Shock

- **Does the result depend on the rainfall product?** Its sign does not; its size does (Figure A.8).
- **Why CHIRPS v3?** It reaches the foreign headwaters and resolves rainfall at the scale of the sub-basins.
- **Is the result a measurement artifact?** No: it is as large where gauges are scarcest (Table A.4).

The sign of the result does not depend on the rainfall product; its size does. The treatment is built from CHIRPS v3 (Funk et al., 2026), and Table A.3 sets it beside five alternatives built from gauges alone, a weather model, or an interpolated climatology. Each rebuilds the treatment from the same sub-basins, weights and monsoon months, so only the upstream rainfall changes. Figure A.8 re-estimates Table 2, column 4, under each product. The elasticity is positive under all six, on identical rows (panel (a)) and on each product's full coverage (panel (b)). On the common sample it is 1.50 on CHIRPS v3 and 1.02 on CHIRPS v2, against 0.26, 0.33, 0.24 and 0.26 on GPCC, CRU TS, ERA5-Land and TerraClimate.

Table A.3: The rainfall products

Product	Built from	Resolution	Record	Source
CHIRPS v3 (paper)	Satellite and gauges	0.05°	1981–present	Funk et al. (2026)
CHIRPS v2	Satellite and gauges	0.05°	1981–present	Funk et al. (2015)
GPCC v2022	Gauges only	0.25°	1891–2020	Schneider et al. (2014)
CRU TS 4.08	Gauges only	0.5°	1901–2023	Harris et al. (2020)
ERA5-Land	Reanalysis	0.1°	1950–present	Muñoz-Sabater et al. (2021)
TerraClimate	Interpolated climatology	1/24°	1958–present	Abatzoglou et al. (2018)

Notes: Record is the product's own coverage. The estimates use each product within 1990–2025 as far as the files used here reach (Figure A.8, panel (b)).

The sizes differ because the products disagree about the year-to-year variation the estimate uses. Panel (c) of Figure A.8 regresses each product's upstream rainfall on CHIRPS v3's within district: the products move 0.51 to 0.77 log points per log point of CHIRPS v3, and CHIRPS v3 accounts for only 0.35 to 0.43 of their within-district variation.

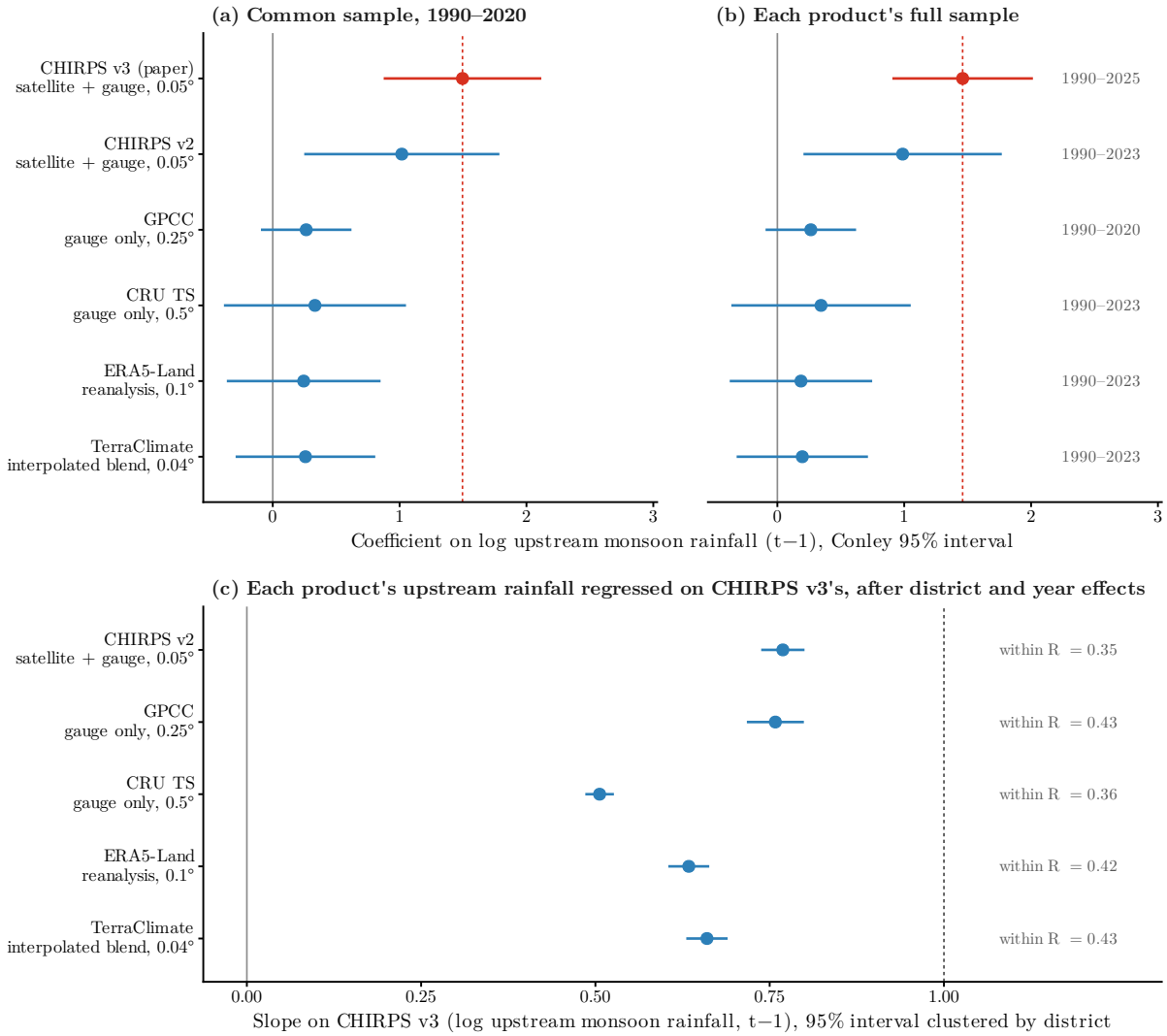


Figure A.8: The lagged upstream elasticity across precipitation products

Notes: Each point re-estimates Table 2, column 4, replacing only the upstream rainfall with the product shown; local rainfall and the spatial block remain CHIRPS v3. Panel (a) uses the 6,324 district-years (204 districts, 1990–2020) that every product covers; panel (b) uses every district-year each product covers within 1990–2025 (years at right), so its CHIRPS v3 point is Table 2, column 4. In panels (a) and (b), bars are 95 percent intervals on Conley spatial-HAC standard errors (150 km) and the dashed line marks the CHIRPS v3 estimate. Panel (c) regresses each product's lagged log upstream rainfall on CHIRPS v3's with district and year fixed effects, on the common sample; bars are 95 percent intervals clustered by district, the grey dashed line marks a one-for-one relationship, and within R^2 is the share of the product's within-district variation that CHIRPS v3 accounts for.

CHIRPS v3 is the product best suited to this design, for three reasons. It combines satellite observation, which reaches the foreign headwaters, with gauges where they exist, whereas the gauge-only products interpolate across sparse Himalayan networks and India's gridded product (IMD) stops at the border, beyond which 24 percent of upstream sub-basins lie. Its 0.05-degree grid matches catchments built from sub-basins of about 139 km^2. It also covers the paper's whole window.

The magnitude should be read as a CHIRPS v3 magnitude, since version 3 differs substantially from version 2. Version 3 blends gauge data from nearly four times as many sources, corrects for gauge undercatch, uses a new 1991–2020 climatology, and replaces the satellite algorithm, which in version 2 understated the year-to-year variance of rainfall and the size of heavy storms (Funk et al., 2026). The two versions correlate only 0.59 after district and year effects, and version 2 gives

the smaller coefficient ($p = 0.010$). Either version 3 resolves rainfall the other products miss, or the difference is noise that happens to correlate with conflict; this paper cannot separate the two.

Over these catchments CHIRPS rests on few gauges, and most of them are foreign. The product reports the gauges it blended in each pixel and month (Figures A.9 and A.10). Few are Indian, since the IMD network does not flow into the product: 96 percent of Indian upstream sub-basins are effectively ungauged, against 83 percent of foreign ones, and the median estimation district draws 96 percent of its upstream weight from pixels with no station. The product blends about 82 gauges a monsoon month across India, and in the foreign catchments the count falls from 129 a month in the 1990s to 23 since 2011. Over these catchments CHIRPS is therefore close to a satellite product anchored by Nepal's gauges.

Where gauge-based grids exist, CHIRPS agrees with them within districts, and least along the Himalayan arc (Figure A.11). District totals correlate 0.66 with IMD within district (median gap 14 percent), and sub-basin totals correlate 0.70 with GPCC in Indian and 0.61 in foreign sub-basins (gaps 11 and 22 percent).

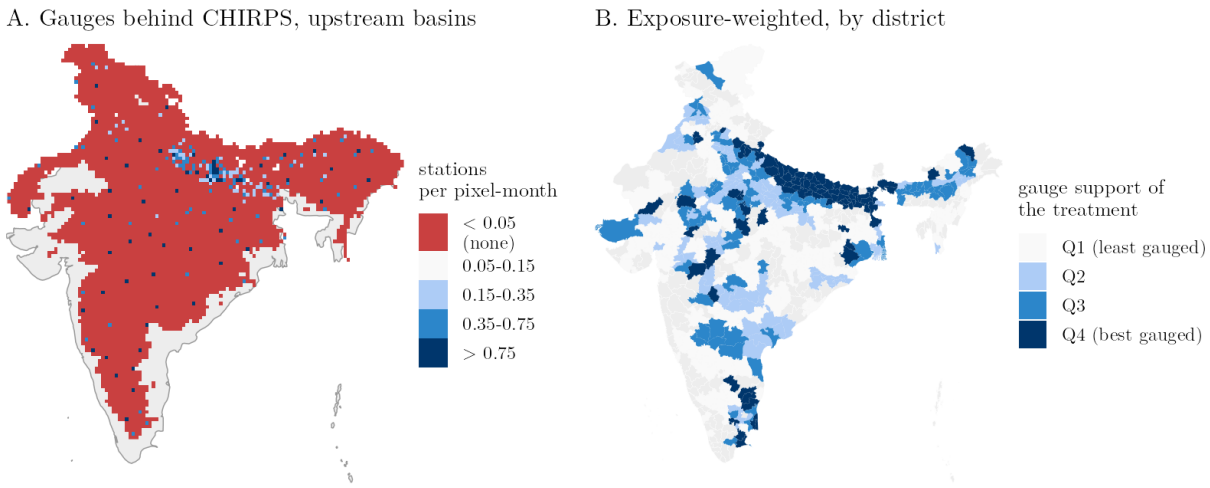


Figure A.9: The gauges behind the treatment

Notes: Panel A maps CHIRPS v3's own station-density diagnostic — the number of gauges the product blended, per 0.25-degree pixel — averaged over monsoon months (June–September) 1990–2025, masked to the upstream basins of the catchments after the neighbour exclusion. Red marks pixels averaging fewer than one blended station per twenty pixel-months. Panel B aggregates the same density to districts with the river-distance weights that build the upstream treatment; quantile bins, darker better gauged. Grey districts have no upstream catchment in the construction.

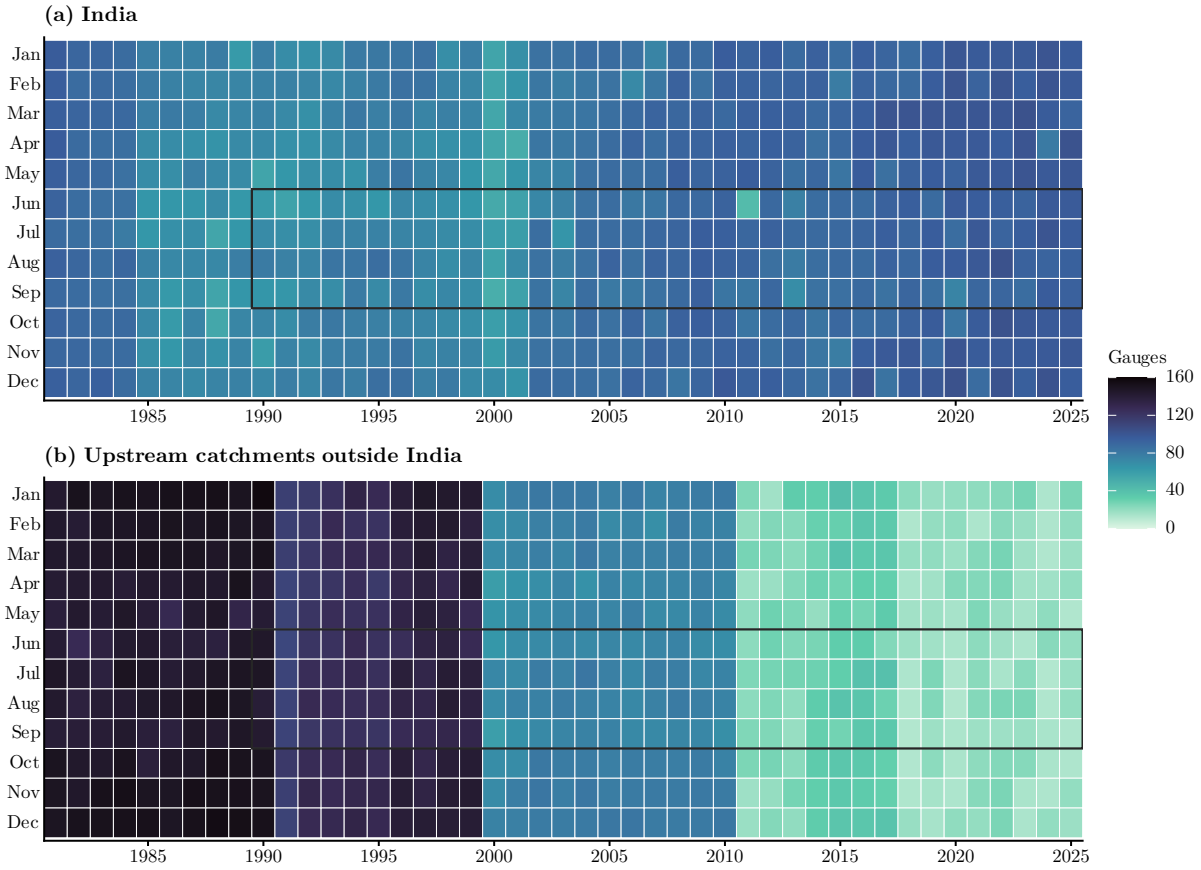


Figure A.10: The number of gauges behind CHIRPS v3, by year and month

Notes: Each cell is the number of rain gauges CHIRPS v3 blended in that month, from the product's station-density diagnostic (per 0.25-degree pixel, summed over the area; a pixel on the boundary counts by the share of it inside). Panel (a) covers India; panel (b) the upstream catchments of the paper's districts that lie outside India, in Nepal, China, Bhutan and Bangladesh. The box marks the monsoon months (June–September) of the paper's window, 1990–2025.

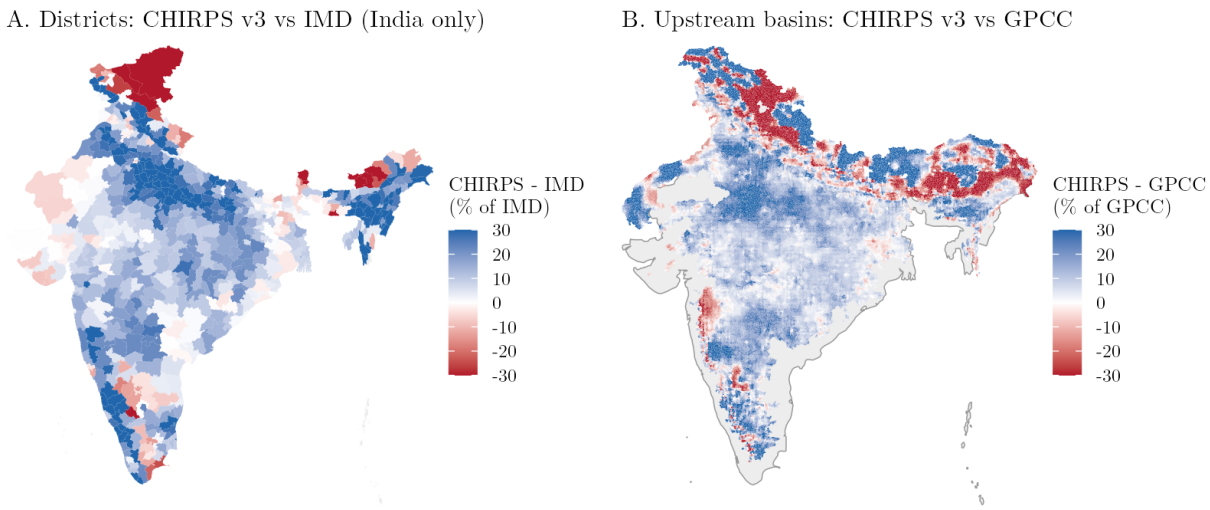


Figure A.11: CHIRPS v3 against the gauge-based grids

Notes: Panel A: the district's mean June–September CHIRPS v3 total (the panel's own local-rainfall measure) against the same total from the IMD 0.25-degree gauge-based grid, 1990–2025, as a percentage of the IMD total; blue means CHIRPS reads wetter. Panel B: the same comparison at basin level against GPCC v2022, the gauge-based product with global coverage, over the upstream basins of the treatment, 1990–2020. Gaps beyond ±30 percent are censored to the scale's ends.

Poor measurement does not drive the result. Table A.4 splits districts into terciles of the gauge support of their treatment. The least-gauged tercile carries 1.550, significant at one percent, against 1.499 in the best-gauged, significant at ten percent, and the difference is not detectable ($p = 0.13$). Classical measurement error would attenuate the estimate where gauges are scarce, so a large estimate in the least-gauged tercile argues against a measurement artifact.

Dropping sub-basins is a weaker test, and it leaves the estimate in place. Indian sub-basins alone give 0.92 (standard error 0.37, 200 districts), and dropping sub-basins above 2,500 m gives 0.93 and above 1,500 m 1.22 (standard error 0.36). The result lives neither in the badly measured high catchment nor in the foreign sub-basins. Such restrictions also change the estimand as well as the measurement, since snowmelt-fed sub-basins deliver water on a different schedule, and dropping the least-gauged tercile keeps a positive but imprecise 1.119 (standard error 0.646): they answer a power question, which is why the tercile split carries the test.

Table A.4: The headline estimate by gauge support of the treatment

	Full sample	Drop least-gauged third	Least gauged (T1)	Middle (T2)	Best gauged (T3)
log Upstream Precip. $(t-1)$	1.460***	1.119*	1.550***	0.611	1.499*
	(0.283)	(0.646)	(0.538)	(0.717)	(0.820)
Observations	7,560	4,824	2,736	2,592	2,232
Number of Districts	210	134	76	72	62
Mean of dependent variable	1.00	0.86	1.24	1.10	0.59

Notes: Poisson fixed-effects on the specification of Table 2, column 4, 1990–2025. Districts are split into terciles of the gauge support of the treatment: CHIRPS v3's own monthly station counts (Figure A.9), averaged over each district's upstream basins with the river-distance weights that build the exposure. The formal test is the interaction, which lets the treatment slope differ by tercile while holding every other coefficient common: the least-gauged tercile differs from the best-gauged by 1.05 ($p = 0.13$) and the middle by 0.80 ($p = 0.29$). Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Appendix B. Inference and Identification

This appendix checks identification first and inference second. Section B.1 tests whether exposure to the shocks, rather than the shocks themselves, drives the estimate, and Section B.6 asks whether upstream rainfall measures weather moving along the river corridor, or trade arriving by road, rather than water. The sections between them address inference. Districts along the same river share upstream sub-basins, so their treatments are correlated by construction: rain over Nepal alone enters the treatment of 57 districts and is at least half of it for 26 (Figure B.1). Standard errors that treat districts as independent then overstate precision, and the number of independent shocks behind the estimate is far smaller than the number of districts (Adão et al., 2019). I address this in three ways. Section B.2 counts the effective number of shocks and checks that no single district or state drives the estimate. Section B.4 lets the errors correlate across districts up to whole river systems, under spatial-HAC kernels and under clusters of districts that share upstream water. Section B.5 drops the asymptotic approximation and runs randomization tests that reassign the district–catchment map and the year labels.

B.1. Expected Exposure

- **Could exposure to the shocks, rather than the shocks, drive the estimate?** No: recentering on expected exposure leaves it at 1.390, significant at one percent.

Rainfall shocks can be as good as random while exposure to them is not. A district with a large, wet catchment receives more upstream rain in every year, and if such districts differ in conflict for other reasons, the treatment is confounded. Borusyak and Hull (2023) show that this confounder is expected exposure μ_i, the treatment a district would receive on average over draws of the shocks. I address it by recentering: the treatment is replaced by its deviation from μ_i, estimated as the district's mean treatment over the sample years. When climatology is stable, μ_i is constant over time and district effects already absorb it, so recentering cannot move the estimate. It reproduces the baseline 1.460 (1.460), which checks the construction rather than the assumption. The real test lets climatology drift by estimating μ_i separately for each decade. Expected exposure then carries 3.0 percent of the treatment's variance, and the estimate falls to 1.390 (standard error 0.393, $p = 0.000$), still significant at one percent (Table B.1, panel B).

B.2. The Number of Shocks

- **How many independent shocks does the estimate draw on?** 37.4 effective shocks across major sub-catchments, above the benchmark of twenty; 7.20 if each river system counts as one.
- **Does one district or state drive it?** No: dropping one state at a time keeps it between 1.32 and 1.57 (Figure E.11).

Shift–share estimates are consistent as the number of independent shocks grows, not the number of districts (Borusyak et al., 2022), so what matters is how many shocks the treatment effectively draws on. I measure this as the inverse Herfindahl index $1/\sum_b \bar{s}_b^2$, the effective sample size that Borusyak et al. (2022, p. 204) recommend reporting. Here $\bar{s}_b$ is basin b's weight in the treatment averaged over districts (mapped in Figure B.2), and the index equals the number of basins when all weigh equally. It is 2,087 across 18,109 level-10 sub-basins, 37.4 across 82 major sub-catchments and 7.20 across 21 macro basins. Level-10 sub-basins within one sub-catchment share the same

monsoon, so their shocks are not independent, and the clustered version of the condition (Borusyak et al., 2022, Assumption 5) counts shocks at the level at which they are. I take that level to be the major sub-catchment: it is the level at which the exposure-robust standard errors of Section B.4 treat the shocks as the sampling unit, and the count there, 37.4, exceeds the twenty equal-sized shocks at which the simulations of Borusyak et al. (2022) find the asymptotic approximation adequate. Treating each whole river system as one shock is the conservative extreme, with 7.20 effective shocks; the standard errors of Section B.4 allow correlation up to that level as a stress test, and the estimate clears one percent there too.

No single district or state drives the estimate. Re-estimated with one district removed, it stays between 1.37 and 1.55; with one state removed, between 1.32 and 1.57, against 1.46 in the full sample (Figure E.11). Table B.1, panel A, collects the shock counts.

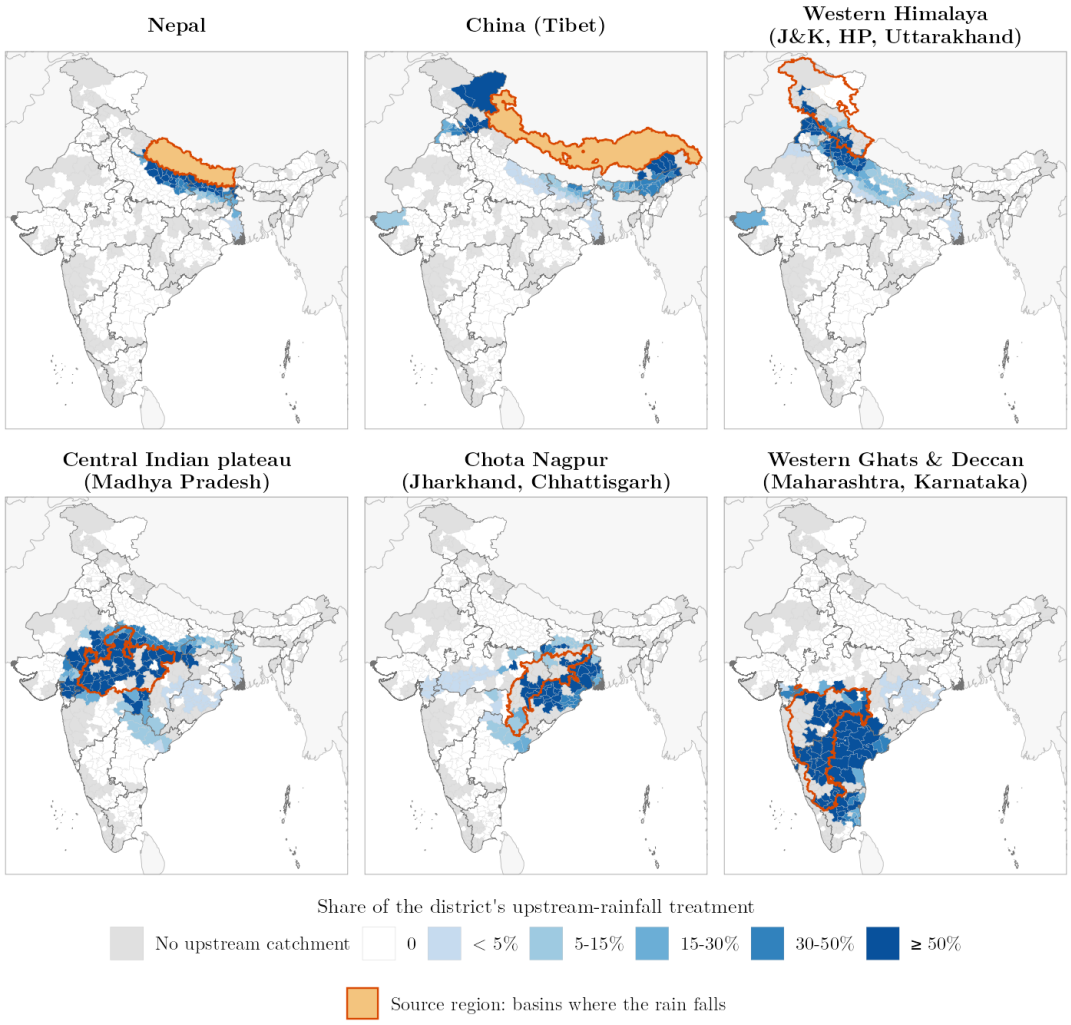


Figure B.1: Where each region's rain enters the treatment

Notes: Each panel takes one source region (outlined) and shades every district i by the share of its upstream-rainfall weight that falls in that region, $\sum_{b \in R} w_{ib}$. Basins are assigned to a region by their representative point: an Indian state, or the country outside India. Grey districts have no upstream catchment under the neighbour exclusion; white districts draw nothing from the region.

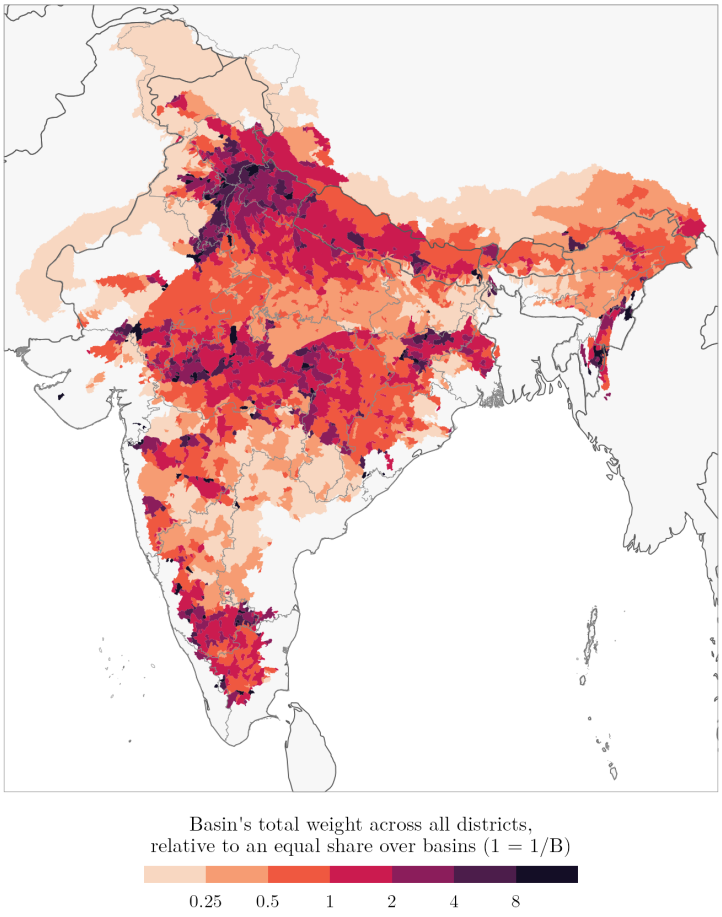


Figure B.2: Each basin's share of total exposure

Notes: Each basin b is shaded by $s_b = \frac{1}{N}\sum_i w_{ib}$, its exposure share summed over the 425 districts with an upstream catchment, relative to an equal split over the 22,237 basins that enter any district's treatment ($\sum_b s_b = 1$; 1 is an equal share). Weights as in Figure A.5.

Table B.1: Shift-share design diagnostics

A. How many independent shocks?	Units	Effective number	
HydroBASINS level-10 sub-basins	18,109	2,087	
PFAF level-5 sub-catchments	82	37.4	
PFAF level-4 macro basins	21	7.20	
largest single macro-basin share	—	0.32	
B. Recentering on expected exposure	Coefficient	Std. error	Mean of dep. var.
Baseline	1.460***	(0.235)	1.00
Recentered, time-invariant μ_i	1.460***	(0.235)	1.00
Recentered, μ_i by decade	1.390***	(0.393)	1.00
μ_i share of exposure variance	3.0%	—	

Notes: All panels are computed on the estimation sample of Table 2, column 4 — 7,560 district-years over 210 districts, 1990–2025. Panel A gives $1/\sum_b \bar{s}_b^2$ over average exposure shares: the number of *independent* shocks the design has at each level of the basin hierarchy. Panel B recenters the treatment on expected exposure μ_i. With μ_i time-invariant, district effects absorb it exactly and the estimate cannot move. Standard errors clustered by district. ** 0.05, *** 0.01.

B.3. The Spatial Specification

- **Does the estimate depend on the spatial terms the specification leaves out?** No. Added back one at a time or together, they keep it between 1.18 and 1.84, significant at one percent (Table B.2).

The preferred model of Harari and La Ferrara (2018) carries a cell's own conflict in the previous year, its neighbours' conflict in the same year and its neighbours' weather. The estimating equation keeps neighbours' weather, replaces neighbours' same-year conflict with its one-year lag, and omits three terms: the own lag, neighbours' same-year conflict and neighbours' upstream rainfall. Table B.2 adds each back in turn (columns 2–4) and all three together (column 5). The upstream coefficient stays between 1.18 and 1.84, and is 1.42 (standard error 0.36) with all three, against 1.46 in the headline.

The own lag is omitted for a property of the estimator. The fixed-effects Poisson removes each district's effect by conditioning on its total count over the panel; last year's count is part of that total, so the term carries the district effect and the estimator becomes inconsistent. Neighbours' lagged conflict is built from other districts' counts and has no such problem. Adding the own lag anyway gives 1.41, and in the linear model of Panel B, where the term is admissible, the coefficient does not move.

Neighbours' same-year conflict is omitted for simultaneity. Violence in a district and violence in its neighbours feed into each other, so the term needs an instrument or a maximum-likelihood system for all districts at once. The maximum-likelihood estimator of Harari and La Ferrara (2018) is built for a linear model of conflict incidence, and the instrument they use as a check, second-order neighbours' weather, is far too weak here: in Panel B the first-stage F is 0.32, and the instrumented estimate is uninformative. Entered directly in the Poisson model, the term lowers the coefficient to 1.18. Some fall is expected, since neighbours' same-year conflict itself rises with the district's upstream rainfall (0.044, standard error 0.020), so conditioning on it holds part of the response fixed.

Neighbours' upstream rainfall is omitted for collinearity. After fixed effects it correlates 0.91 with the district's own. Adding it raises the own coefficient to 1.84 while the neighbours' lag takes -0.79: the regression cannot divide one effect between two nearly identical series. Table B.3 and Figure B.3 show how the estimate moves as the spatial controls that are kept enter one by one.

B.4. Standard Errors under Spatial Dependence

- **Is the estimate significant if errors are allowed to be correlated across space?** Yes, at one percent under all eight alternatives (Figure B.4).

Districts that share upstream water receive correlated treatments, and nearby districts may share unobserved shocks as well. If their errors are correlated, standard errors clustered by district are too small. I re-estimate the standard error of the headline coefficient (Table 2, column 4, 1990–2025) under eight alternatives, each allowing a different form of dependence:

- *Clustering by state*: arbitrary correlation among districts of the same state, which share administration and policing.
- *Clustering by macro basin*, at PFAF levels 4 and 5: arbitrary correlation among districts of the same river system, however far apart.

Table B.2: The spatial specification: omitted terms added one at a time and together

	As estimated	Adding one omitted term			All three
		Own lagged conflict	Neighbours' same-year conflict	Neighbours' upstream rainfall	
	(1)	(2)	(3)	(4)	(5)
Panel A. Event count, fixed-effects Poisson; neighbours' same-year conflict entered directly					
log Upstream Precip. $(t-1)$	1.460***	1.410***	1.181***	1.841***	1.422***
	(0.283)	(0.423)	(0.347)	(0.302)	(0.356)
log Upstream Precip. (t)	-0.086	-0.381	-0.003	-0.053	0.008
	(0.487)	(0.423)	(0.426)	(0.464)	(0.348)
log Own Conflict $(t-1)$		0.724***			0.692***
		(0.063)			(0.056)
log Neighbours' Conflict (t)			1.178***		1.088***
			(0.157)		(0.141)
log Neighbours' Upstream Precip. $(t-1)$				-0.785***	-0.627***
				(0.192)	(0.111)
log Neighbours' Upstream Precip. (t)				0.094	-0.247
				(0.293)	(0.189)
Observations	7,560	7,560	7,560	7,488	7,488
Districts	210	210	210	208	208
Mean of dependent variable	1.00	1.00	1.00	1.01	1.01
Panel B. log(1 + *count*)*, OLS; neighbours' same-year conflict instrumented*					
log Upstream Precip. $(t-1)$	0.066**	0.066***	0.136	0.090**	0.102**
	(0.027)	(0.024)	(0.198)	(0.040)	(0.042)
log Upstream Precip. (t)	0.006	-0.007	0.021	-0.027	-0.062
	(0.028)	(0.028)	(0.071)	(0.039)	(0.082)
log Own Conflict $(t-1)$		0.479***			0.527***
		(0.044)			(0.157)
log Neighbours' Conflict (t)			-1.566		-0.764
			(4.087)		(2.280)
log Neighbours' Upstream Precip. $(t-1)$				-0.032	-0.003
				(0.044)	(0.129)
log Neighbours' Upstream Precip. (t)				0.049	0.087
				(0.045)	(0.145)
First-stage F, neighbours' same-year conflict			0.32		0.54
Observations	15,300	15,300	15,300	15,156	15,156
Districts	425	425	425	421	421
Upstream temperature, local weather	✓	✓	✓	✓	✓
Neighbours' conflict $(t-1)$ and weather	✓	✓	✓	✓	✓
District & year FE	✓	✓	✓	✓	✓

Notes: Column 1 of Panel A is the headline (Table 2, column 4). Columns 2–4 each add one term the headline omits, and column 5 adds all three. Own lagged conflict is the district's conflict in the previous year, log(1 + count); in the fixed-effects Poisson this term is inconsistent, so Panel A column 2 is a check rather than an estimate. Neighbours' same-year conflict is log(1+count) averaged over contiguous districts. In Panel A it enters directly. In Panel B it is instrumented with second-order neighbours' local rainfall and temperature, the Kelejian–Prucha instruments; the first-stage F is reported (cluster-robust Wald). Neighbours' upstream rainfall is the contiguity average of $\log R^{UP}$ over neighbours with an upstream catchment, current and lagged; its lag correlates 0.905 with the district's own after fixed effects (Panel A, column 4 sample), so the split between the two is poorly identified. Two districts have no neighbour with an upstream catchment and leave columns 4 and 5. Panel B covers every district, including those with no recorded events, which the Poisson drops. Window 1990–2025; Conley (150 km) standard errors in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table B.3: Spatial diffusion and the upstream treatment

	Incidence (LPM)				asinh(events)
	Contiguity	River W	Both	Downstream	River W
	(1)	(2)	(3)	(4)	(5)
Upstream rain $(t-1)$	0.022*	0.032**	0.003	0.024*	0.083***
	(0.011)	(0.012)	(0.024)	(0.012)	(0.021)
Own conflict $(t-1)$	0.238***	0.274***	0.238***	0.275***	0.511***
	(0.018)	(0.019)	(0.018)	(0.019)	(0.039)
Neighbours' conflict $(t-1)$, contiguity	0.321***		0.294***		
	(0.029)		(0.033)		
Upstream districts' conflict $(t-1)$, river W		0.239***	0.055*		0.148***
		(0.029)	(0.031)		(0.035)
Downstream districts' rain (t)				-0.006	
				(0.012)	
Mean of dependent variable	0.09	0.09	0.09	0.09	
Panel B: event counts (Poisson), 210 districts					
	Baseline	+ river conflict lag	+ contiguity too		
Upstream rain $(t-1)$	0.676**	0.817***	0.708***		
	(0.321)	(0.307)	(0.217)		
Mean of dependent variable	1.00	1.00	1.00		
Panel C: the flood arm — routed flood-stage days (Poisson), 193 districts					
	Baseline	+ river lag	+ queen lag	+ upstr. flood days	
Flood-stage days (t)	0.023***	0.021***	0.024***	0.006	
	(0.006)	(0.005)	(0.005)	(0.008)	
Flood-stage days $(t-1)$	0.025**	0.021**	0.017***	0.019***	
	(0.010)	(0.009)	(0.006)	(0.007)	
Upstream districts' flood days (t)				0.128***	
				(0.043)	
Mean of dependent variable	1.48	1.48	1.48	1.48	

Notes: Time-space recursive models with Durbin-type exogenous lags (Section B.3); 1990–2025; district and year fixed effects; standard errors clustered by district in parentheses. These are NOT dynamic spatial Durbin models in the sense of Harari and La Ferrara (2018): the contemporaneous spatial lag $\rho(Wy)_t$ is absent throughout. Columns (1)–(4) are linear probability models of conflict incidence on all districts; column (5) uses asinh of the event count. Column (1) is the contiguity (queen) network and also holds neighbours' weather fixed; columns (2)–(5) use the directed river network, with (3) entering both conflict lags jointly and (4) the downstream direction. Panel B estimates the count model by fixed-effects Poisson with the own conflict lag excluded (strict exogeneity) on a frame without the upstream temperature controls, so its baseline sits below the published headline and comparisons are internal to the panel. Panel C repeats the exercise for the routed flood-stage days of the flood arm, local weather held throughout and the published no-controls form reproduced first; its final column adds upstream districts' own flood days, whose post-fixed-effects correlation with own days is 0.49. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

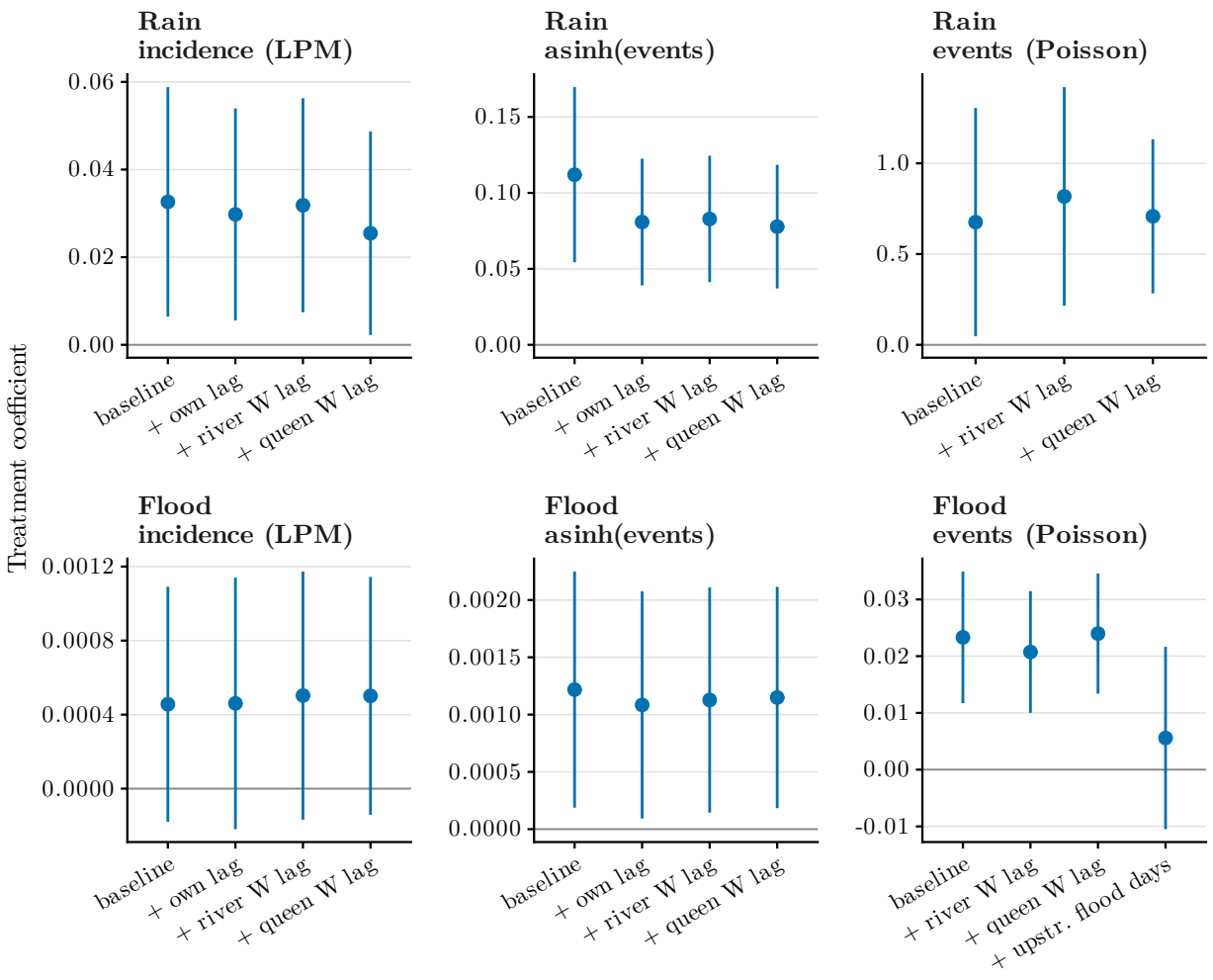


Figure B.3: The stability ladders: treatment coefficients as spatial controls enter, with 95 percent confidence intervals (district-clustered)

Notes: Top row: lagged upstream rainfall. Bottom row: contemporaneous routed flood-stage days. Left and centre columns are the linear models of Table B.3; the right column is the Poisson count model, where for rainfall the coefficient rises under diffusion controls and for flood days the final rung adds upstream districts' own flood days. Poisson ladders omit the own-conflict lag throughout (strict exogeneity); the rainfall Poisson frame lacks the upstream temperature controls, so its baseline is not the published headline.

- *Clustering by catchment group*: arbitrary correlation among the 85 groups of districts whose upstream-weight vectors have cosine similarity above 0.50 (Figure E.8).
- *Conley spatial HAC*, at cutoffs from 25 to 500 kilometres: correlation that decays with geographic distance, regardless of state or basin boundaries.
- *Network HAC*, with cosine or Jaccard kernels: correlation that rises with the upstream basins two districts share, regardless of distance.
- *Exposure-robust* standard errors at PFAF level 5 (Adão et al., 2019): the basin shocks rather than the districts are the sampling unit, the inference counterpart of the shock count in Section B.2.

The estimate clears one percent under every one of them: the smallest t-statistic is 4.97, against a critical value of 2.576 (Figure B.4).

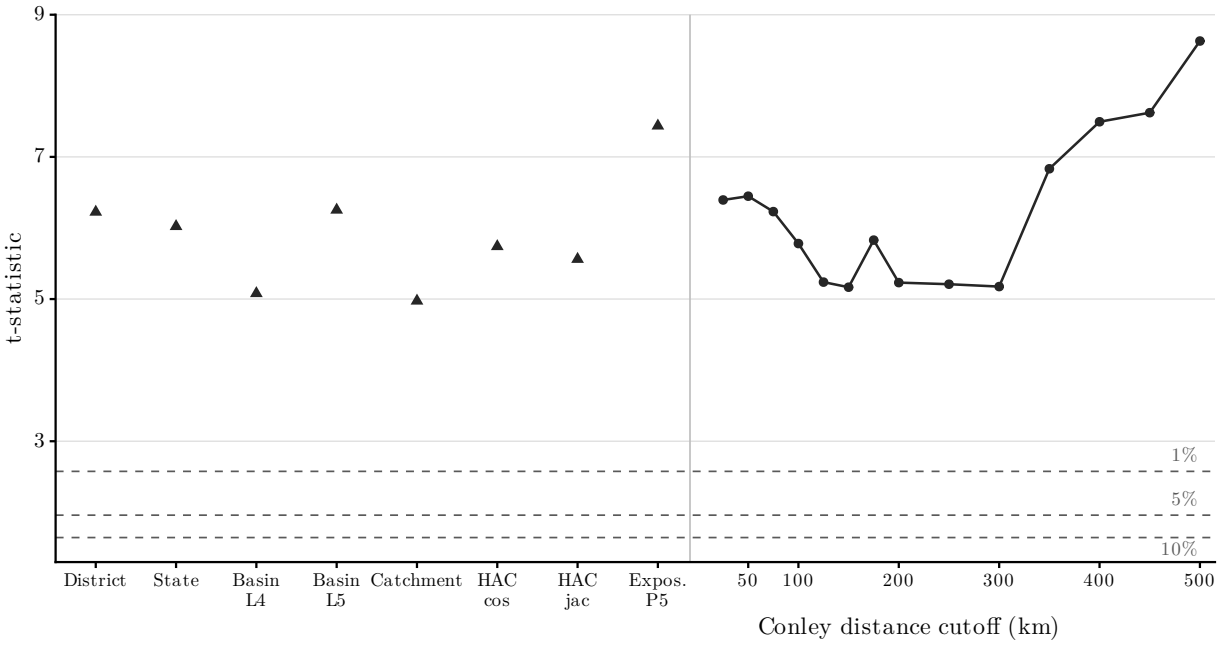


Figure B.4: The headline test statistic under alternative variance estimators

Notes: Every point is the same regression — the specification of Table 2, column 4, estimated on 1990–2025 — re-priced under a different variance estimator, so the figure varies the estimator and nothing else. The point estimate is 1.46 throughout and t is that estimate divided by the standard error in question. The connected curve gives Conley spatial-HAC statistics at distance cutoffs from 25 to 500 kilometres; the grid stops at 500 because beyond it the kernel encompasses most of the country and no longer describes local dependence. The curve is not monotone in the cutoff, and at some cutoffs the variance matrix must be repaired before use; its level is informative and its wiggles are not. The eight triangles at the left have no distance cutoff and are shown at their own labelled positions. Five cluster the residuals — on districts, states, PFAF level-4 and level-5 macro basins, and the catchment grouping built from the treatment itself in this section. The last three do not cluster at all: 'HAC cos' and 'HAC jac' are spatial-HAC estimators whose kernel is the cosine, respectively Jaccard, similarity of districts' upstream-weight vectors, so two districts fed by the same basins are treated as dependent however far apart they lie; 'Expos. P5' is the exposure-robust estimator of Adão et al. (2019), which treats the PFAF level-5 basin shocks rather than the districts as the sampling unit. The exposure-robust estimator computed at the finest basin level is deliberately omitted: with more than a thousand basin shocks its asymptotics do not bite and it returns a standard error of 0.060, far tighter than anything else here, which would imply a precision the design does not have. Dashed lines mark the two-sided normal critical values at the 10, 5 and 1 percent levels.

- **Does the district's own catchment matter, or would rain on any nearby catchment do?** Its own catchment: no reassignment within the state reaches the actual estimate ($p = 0.0002$).
- **Does the timing matter, or would rain in any year do?** The timing: re-dating the shocks breaks the link ($p = 0.0002$).
- **Does the catchment matter wherever its boundary is drawn?** Yes, at every buffer radius from 0 to 100 km ($p \leq .005$).
- **Does it matter where the dams are?** Yes: random dam labels do not reproduce the gap between free and regulated water ($p = 0.0002$).

Randomization inference asks whether the estimate could arise from the structure of the data alone. Each test re-estimates the same regression after reassigning one part of the data at random, holding the rest fixed, and compares the actual estimate with the distribution of the reassigned ones; the exact p-value is the share of draws at least as extreme. The paper reports four such tests, each with 5,000 draws.

- *The district–catchment map, within states* (Figure E.9; Figure E.10, panel (a)). *Randomized:* which catchment's rainfall series each district receives, among the districts of its state. *Held fixed:* each catchment's full time series, each district's conflict and local weather, and the panel. *Tests:* whether a district's own catchment predicts its conflict better than a plausible neighbour's. No permutation reaches the actual estimate of 1.46 ($p = 0.0002$); the draws centre on -0.087, with a 95th percentile of 0.31.
- *The year labels* (Figure E.10, panels (b) and (d)). *Randomized:* the years of the weather block — upstream rainfall and temperature and the district's own weather — with one permutation applied to every district. *Held fixed:* the map, the conflict series and the spatial terms built from it, and each year's cross-section of shocks, so that districts sharing sub-basins keep their correlated shocks. *Tests:* whether conflict follows the timing of the shocks. The actual estimate lies outside the null on the coefficient ($p = 0.0040$), the studentized statistic ($p = 0.0002$, against an actual t of 6.22) and the score statistic implied by the Poisson likelihood ($p = 0.0098$).
- *The year labels, within decades* (Figure E.10, panel (c)). As the previous test, but each year moves only within its decade, which preserves trends and relaxes the exchangeability the free permutation assumes. *Tests:* the same question without relying on trends being exchangeable ($p = 0.0002$ studentized, $p = 0.0008$ score).
- *The district–catchment map at each buffer radius* (Figure A.3, panels (f)–(g)). *Randomized:* as in the first test, at each radius from 0 to 100 kilometres, under both distance measures. *Held fixed:* as in the first test, on a fixed set of districts. *Tests:* whether the true map beats a plausible neighbour's however the catchment boundary is drawn. It does at every buffer radius ($p \leq .005$; Section A.2).

The two kinds of test answer different questions, and neither subsumes the other. The map tests shuffle space and hold time; the year-label tests shuffle time and hold space. Permuting the map within states makes it demanding: a counterfeit map still hands each district rainfall correlated with its own, so the test asks not whether upstream rainfall matters but whether *this* district's catchment matters more than a plausible neighbour's.

B.6. River Water, Corridor Weather or Road-Borne Trade?

- **Could upstream rainfall measure weather moving along the river corridor?** Not the district's own weather, the basins the neighbour exclusion removes, or the monsoon regime; on flood days the catchment beats the basins across its own divide.
- **Could it arrive through trade by road?** No: trading partners' rainfall leaves the coefficient unchanged, and at equal road travel time only upstream rain raises conflict.
- **And the downstream coefficient?** It is rainfall geography (Table E.13).

The map test of Section B.5 shows that a district's own catchment predicts its conflict better than a plausible neighbour's. Two alternatives to river water remain. Upstream rainfall could measure weather moving along the river corridor towards the district, since monsoon storms travel up the same lowlands the rivers drain. Or upstream harvests could reach the district by road. The tests below hold the geography fixed and vary only the river connection.

Seasonal rainfall totals cannot rule out corridor weather. I build two twins of each catchment that share its geography but drain elsewhere (Figure E.12): a matched twin, which replaces each catchment sub-basin by one at the same distance and bearing from the district, and a ridge twin, the sub-basins just across the catchment's watershed divide. Within district and year, their seasonal rainfall correlates 0.94 and 0.95 with the catchment's, too closely for a comparison between them to rank either (Table E.10).

Counted in flood days, the catchment beats the far side of its own divide. Rebuilt as the number of monsoon days on which the three-day rainfall total exceeds its two-year level — the rainfall a river turns into floods — the treatment separates from its twins, with correlations of 0.62 and 0.47. Beside the ridge twin the catchment keeps 0.078 while the twin falls to 0.020 (difference $p = 0.03$); the matched twin stays statistically tied (Table B.4).

Neither the district's own weather nor the monsoon regime carries the coefficient. Controlling for the district's heavy-rain days, its largest five-day total and its monthly rainfall leaves the coefficient at 1.57 (Table E.11). Rainfall on the basins the neighbour exclusion removes carries 0.40 (1.08) while upstream rainfall keeps 1.35. The part of upstream rainfall explained by the all-India anomaly, El Niño and the monsoon's active and break spells holds only 3% of its variation; the remainder carries 1.50, the full coefficient (Table E.12).

The downstream coefficient, by contrast, is rainfall geography. A twin built the same way from sub-basins the district's water never reaches reproduces it: -1.35 and -0.92 at t and $t-1$, against -1.35 and -0.93 for the real downstream rainfall (Table E.13).

Trade does not carry the upstream coefficient either. I measure the rainfall of the district's trading partners: districts connected to it by road but not by water, weighted by road travel time (Figure E.13). Holding it fixed leaves the upstream coefficient at 1.466, against 1.460; across eight partner definitions, three road networks and the railway, including the early-1990s Digital Chart of the World, it stays between 1.36 and 1.65 (Table B.5). The measure is not a dead proxy: partners' rainfall lowers the district's crop price index by -0.641 (0.196). Roads also run both ways while rivers do not: at equal road travel time, upstream rain raises conflict and downstream rain does not ($p < 0.001$; Figure B.5), and the effect does not grow where villages are better connected to roads (Table E.14).

One corridor account survives these tests: on seasonal totals, and even on flood days, the matched twin stays statistically tied with the catchment.

Table B.4: Sideways twins on flood-type statistics of upstream rainfall

	Days above the two-year level			log maximum three-day total		
	(1) Catchment	(2) + twin	(3) Twin replaces catchment	(4) Catchment	(5) + twin	(6) Twin replaces catchment
Panel A. Bearing-and-distance-matched twin						
Catchment statistic (t)	0.026	0.043*		-0.002	0.181	
	(0.022)	(0.022)		(0.171)	(0.297)	
Catchment statistic ($t-1$)	0.086***	0.060**		0.718***	0.613**	
	(0.023)	(0.024)		(0.214)	(0.289)	
Twin statistic (t)		-0.022	-0.005		-0.257	-0.111
		(0.018)	(0.015)		(0.304)	(0.166)
Twin statistic ($t-1$)		0.042***	0.063***		0.157	0.671***
		(0.013)	(0.015)		(0.273)	(0.206)
Catchment ($t-1$) − twin ($t-1$)		0.018 (0.031)			0.455 (0.513)	
Within-FE corr., twin and catchment ($t-1$)		0.62			0.85	
Observations	7,524	7,524	7,524	7,524	7,524	7,524
Number of Districts	209	209	209	209	209	209
Mean of dependent variable	1.00	1.00	1.00	1.00	1.00	1.00
Panel B. Ridge twin						
Catchment statistic (t)	0.026	0.039		-0.002	-0.082	
	(0.022)	(0.025)		(0.171)	(0.279)	
Catchment statistic ($t-1$)	0.086***	0.078***		0.715***	0.563**	
	(0.023)	(0.023)		(0.214)	(0.234)	
Twin statistic (t)		-0.025	-0.010		0.121	0.073
		(0.023)	(0.019)		(0.328)	(0.201)
Twin statistic ($t-1$)		0.020*	0.045***		0.201	0.686***
		(0.011)	(0.017)		(0.271)	(0.232)
Catchment ($t-1$) − twin ($t-1$)		0.057 (0.027)			0.362 (0.443)	
Within-FE corr., twin and catchment ($t-1$)		0.47			0.85	
Observations	7,560	7,560	7,560	7,560	7,560	7,560
Number of Districts	210	210	210	210	210	210
Mean of dependent variable	1.00	1.00	1.00	1.00	1.00	1.00

Notes: The outcome is the count of UCDP events located to a district or better, estimated by Poisson with district and year fixed effects on the Table 2 column 4 sample (1990–2025), with that column's full control set unless stated. Conley spatial-HAC standard errors (150 km) in parentheses; ***, **, and * indicate significance at 1%, 5% and 10%. The upstream treatment is rebuilt from catchment-weighted daily CHIRPS v3 basin rainfall with the paper's river-distance weights and replaces log upstream rainfall at t and $t-1$; every other control of Table 2 column 4 stays. Columns 1–3: the number of June–September days on which the centred three-day total exceeds the district's two-year level (the median of the catchment's annual maxima over 1989–2025, one threshold per district applied to catchment and twin alike). Columns 4–6: the log of the season's maximum centred three-day total. The bearing-matched twin replaces each basin of the district's upstream catchment by a level-10 basin outside the catchment and the basins the neighbour exclusion removes at the same great-circle distance (±15%) and bearing (±20°) from the district's population centre, carrying the original basin's river-distance weight; the ridge twin is every basin sharing a boundary with the catchment but outside it and those removed basins, weighted by inverse straight-line distance (Figure E.12). Columns 1 and 4 use the catchment statistic alone, columns 2 and 5 add the twin's, columns 3 and 6 use the twin's alone.

Table B.5: Upstream rainfall and trading partners' rainfall over the road and rail networks

	Baseline	+ Partners' rainfall, by road or rail				Partners only
	(1)	(2)	(3)	(4)	(5)	(6)
log Upstream Precip. $(t-1)$	1.460***	1.466***	1.387***	1.475***	1.483***	
	(0.283)	(0.282)	(0.272)	(0.282)	(0.282)	
log Partners' Precip., network-weighted $(t-1)$		-1.021	-1.536**	-1.051	-1.387	-0.423
		(1.314)	(0.737)	(1.240)	(1.261)	(1.495)
Network		OSM roads, today	OSM rail, today	OSM highways	DCW roads, 1990s	OSM roads, today
Partners' rainfall: corr. with treatment		0.10	0.09	0.10	0.10	0.10
Observations	7,560	7,524	6,876	7,560	7,560	7,524
Number of Districts	210	209	191	210	210	209
Mean of dependent variable	1.00	1.00	1.04	1.00	1.00	1.00
Estimation	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓	✓	✓

Notes: The outcome and specification are those of column 4 of Table 2; every column also includes the current-year terms of the rainfall variables shown. Partners' rainfall is the log of the average monsoon (June–September) rainfall of other districts, each weighted by $1/(1+h)$ where h is road travel time in hours, over partners with no water link to the district: it excludes the district, its neighbours, and every district with at least 5% of its area in the district's upstream, downstream or own sub-basins. Road networks: OpenStreetMap today (motorway to secondary roads); OpenStreetMap trunk and primary roads only (the national and state highway network, without post-2000 expressways); the Digital Chart of the World, compiled in the early 1990s. Column 3 uses the railway instead: OpenStreetMap mainline track routed at 40 km/h, with access to the nearest track at 30 km/h, so that h is rail travel time. Corr. with treatment: correlation of lagged partners' rainfall with lagged upstream rainfall after fixed effects and the column's controls. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

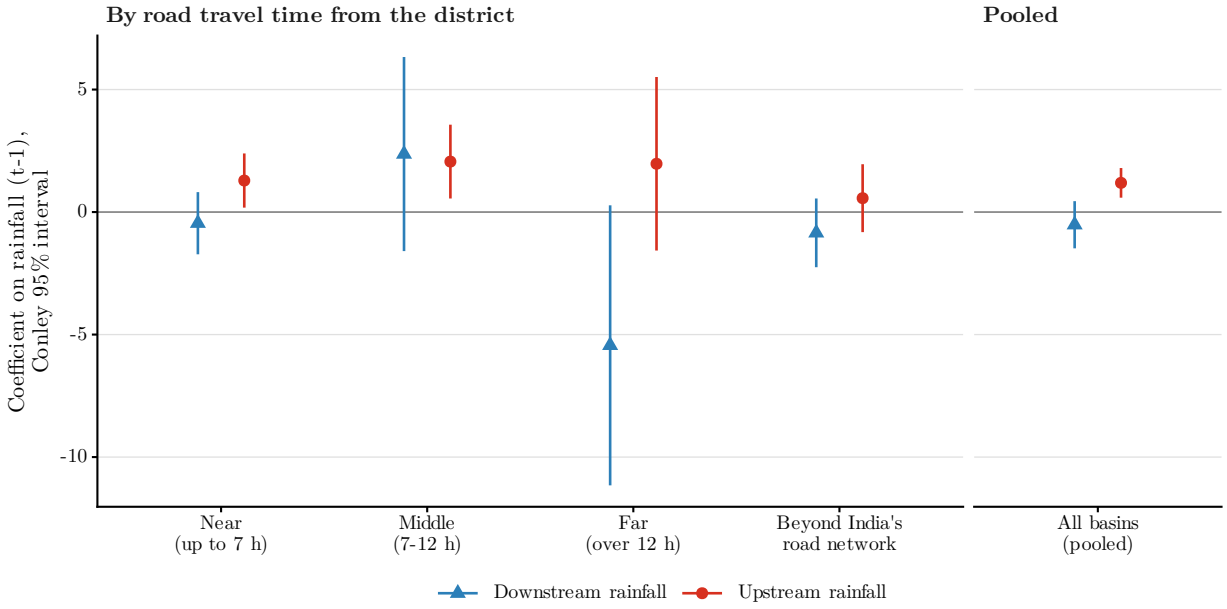


Figure B.5: Upstream and downstream rainfall at equal road travel time

Notes: One Poisson regression per panel, with the controls of column 4 of Table 2 and the published upstream rainfall replaced by the components plotted. Each sub-basin upstream or downstream of a district is assigned the road travel time from the district to the Indian districts it overlaps, and grouped into bands that are terciles of basin area over both directions; mean travel times are 4.9, 9.5 and 17.4 hours upstream and 4.8, 9.4 and 15.8 hours downstream. Both directions are weighted alike, by basin area over the district's combined upstream and downstream area. Basins with no Indian territory (Nepal, Tibet, Bangladesh, Pakistan) have no road time and are shown separately. Equality of upstream and downstream coefficients: jointly over the three bands, $p < 0.001$; pooled, $p < 0.001$. Intervals use Conley spatial-HAC standard errors (150 km).

Appendix C. Routed Water: CaMa-Flood and Its Validation

C.1. The CaMa-Flood Model

- **Why route the water?** Rainfall is a proxy; flooding also depends on soil, snow and the river network.
- **What is the routed measure?** Monsoon days on which upstream-origin discharge at the district's river exceeds its two-year level.

Upstream rainfall is a proxy for the water that reaches the district. Its weights (Section A.3) are statistical choices, not hydrology, and the same monsoon can flood a district or not depending on how wet the soil already is, how much falls as snow, and how the river network carries and stores the water (Figure 4). To measure what the river delivers, I route runoff — rain that the land surface has already passed through soil, snow and evaporation — through CaMa-Flood (Yamazaki et al., 2011, 2013), a global river routing model, at six-arc-minute resolution over South Asia for 1989–2025.

CaMa-Flood moves water along a fixed river network. The land is divided into unit catchments, each draining into the next one downstream, and each holds a river channel with a floodplain beside it, described by the channel's width, bank height and length and by a floodplain elevation profile taken from high-resolution topography. Every time step, runoff enters the channels, the water stored in each unit catchment is updated by mass balance, and outflow to the next catchment is computed from the slope of the water surface between the two. When storage exceeds the channel's capacity, water spills onto the floodplain, which slows and spreads the flood wave instead of passing it downstream unchanged.

Formally, the storage V_c (m^3) of unit catchment c evolves by

$$V_c^{t+\Delta t} \;=\; V_c^{t} \;+\; \sum_{k \in u(c)} Q_k^{t}\,\Delta t \;-\; Q_c^{t}\,\Delta t \;+\; A_c R_c^{t}\,\Delta t,$$

where $u(c)$ are the unit catchments draining directly into c, Q is outflow ($\mathrm{m}^3\,\mathrm{s}^{-1}$), A_c is the catchment's area and R_c its runoff ($\mathrm{m\,s}^{-1}$) (Yamazaki et al., 2011, eq. 9). Storage up to the channel's capacity stays in the channel; beyond it, the floodplain profile splits storage between channel and floodplain at a common water level, which gives the water depth and the flooded area (Yamazaki et al., 2011, eqs. 1–2). Outflow follows the local inertial approximation to the shallow-water equations (Bates et al., 2010),

$$Q_c^{t+\Delta t} \;=\; \frac{Q_c^{t} + \Delta t\, g\, A\, S_c}{1 + \Delta t\, g\, n^2 |Q_c^{t}| / \left(H^{4/3} A\right)},$$

where S_c is the water-surface slope to the downstream catchment, A the flow cross-section, H the hydraulic radius, n Manning's roughness and g gravity (Yamazaki et al., 2013, eq. 6). Where a river divides, bifurcation channels link unit catchments that the river network does not connect, and water flows through them toward whichever side has the lower water surface (Yamazaki et al., 2014b); $u(c)$ then includes catchments linked to c through them. These are the river splits of Section A.3.

The configuration is the model's South Asia set-up. The river map is built from MERIT Hydro (Yamazaki et al., 2019); channel widths come from satellite observation (Yamazaki et al., 2014a), filled by an empirical relation with discharge where none exists, and bank heights from

that relation. Manning's n is 0.03 in channels and 0.10 on floodplains, the water level at river mouths is fixed, and there are no reservoirs, barrages, canals, or levees. Runoff is daily ERA5-Land (Muñoz-Sabater et al., 2021), distributed to unit catchments by area. The simulation runs in three chunks (1988–1999, 2000–2018 and 2018–2025), each started from empty storage with a one-year spin-up that is discarded; re-runs from stored states reproduce the baseline bit for bit.

The measure keeps only water that arrives from upstream. For district d, a masked run removes the runoff of every unit catchment within 50 kilometres of the district (the 50-kilometre mask),

$$\tilde{q}_{kt}^{(d)} \;=\; A_k R_{kt}\, \mathbf{1}\big[\, k \notin M_{g(d)} \big]\,,$$

where $M_{g(d)}$ is the union of the 50-kilometre masks of the districts that share d's run: districts whose masks barely overlap are simulated together, in 34 runs, each district losing at most 5 percent of its upstream input to the others' masks. Simulated discharge at river cell c is then a deterministic, nonlinear function of the history of masked runoff upstream of c and of Θ, the time-invariant river and floodplain geometry and parameters of the configuration above,

$$Q_{ct}^{(d)} \;=\; \mathcal{F}_c\Big(\{\tilde{q}_{ks}^{(d)}\}_{k\in\mathcal{U}(c),\, s\le t}\,;\, \Theta\Big)\,,$$

where $\mathcal{U}(c)$ is every catchment upstream of c, including those reaching it through bifurcation channels. With local runoff masked, upstream runoff is its only time-varying input. A district's flood-stage days in year y count the monsoon days on which discharge at its river cell with the largest upstream area, c_d, exceeds that cell's two-year return level,

$$\text{FloodDays}_{dy} \;=\; \sum_{t\,\in\,\text{June–September of } y} \mathbf{1}\Big[\,\bar{Q}_{c_d t}^{(d)} > Q_{c_d}^{\text{RP2}}\Big]\,,$$

where $\bar{Q}$ is the centred three-day mean of daily outflow and Q_c^{RP2} is a Gumbel fit, by L-moments, to the 37 annual maxima of the unmasked baseline at cell c, 1989–2025. Upstream rainfall generates these days mechanically: a log-point of upstream monsoon rain adds 6.2 flood-stage days (standard error 2.28; Table C.1, column 1). Table 3 in Section V races the rainfall total against the June–September discharge total at the same cell; Table C.1 races it against these routed days.

The mask's footprint does not drive the measure. Rebuilding the ensemble with the mask drawn at the paper's neighbour-exclusion footprint instead of 50 km gives instruments correlating 0.94 within district and a flood-stage result of 0.024 (50 km) against 0.021 (neighbour exclusion) on the common sample. The 50-kilometre construction is the better identified of the two: excluding districts whose mask removes under 1 percent of upstream input leaves its estimate at 0.020 while the administrative variant loses significance, and the one neighbour-exclusion run that failed its volume-conservation gate did so for exactly this reason and is held out.

The routed measure's catchment is the model's own, not $U(i)$. Figure C.1 draws both for Dhubri: the drainage area that CaMa-Flood traces on its flow directions to the district's river cell (4,323 unit catchments, 469 thousand km^2) and the HydroBASINS catchment of Section A.2 (488 thousand km^2). The two share 94 percent of the latter's area and 98 percent of the former's; the rest lies where the two hydrographies draw the divide differently. Panel (b) sets the two exclusions side by side. The 50-kilometre mask removes the runoff of 124 unit catchments inside the drainage area, 2.9 percent of its area and 1.1 percent of the input the river cell receives; the neighbour

exclusion removes 111 sub-basins, 3.1 percent of $U(i)$. Figure C.2 shows what the masked run delivers at the cell: daily upstream-origin discharge in the wettest and driest June–September of the window, whose totals differ by a factor of 1.57. The unmasked baseline for the same years lies on top of the masked series, as it should for a district whose river drains most of the Brahmaputra basin above it.

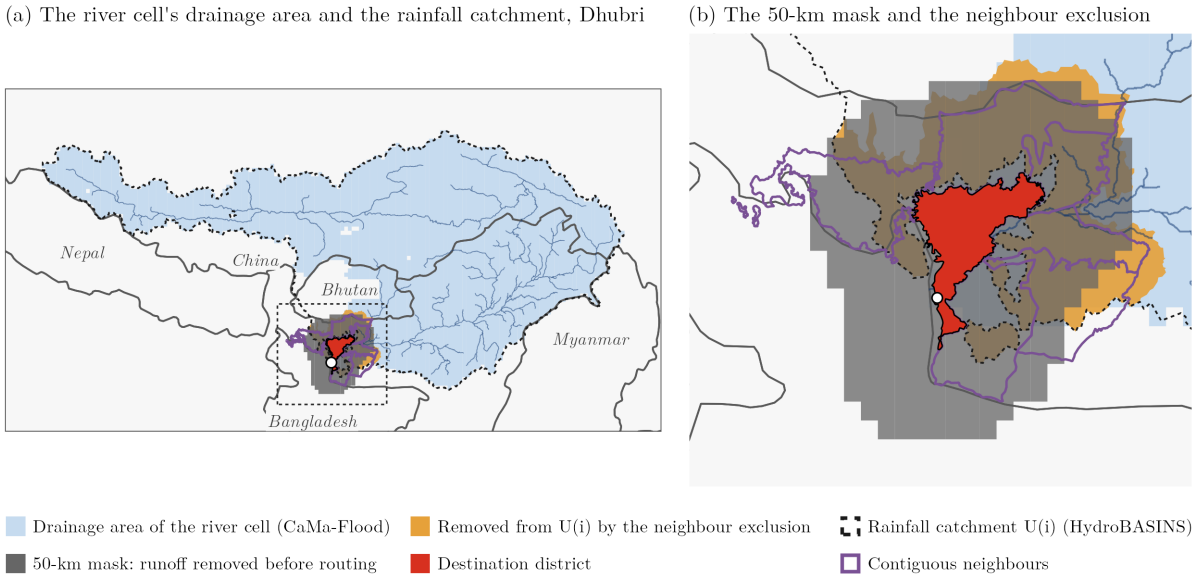


Figure C.1: The routed catchment and the rainfall catchment, for one district

Notes: Panel (a): the drainage area of Dhubri's river cell (white point; the cell with the largest upstream area in the district) on CaMa-Flood's six-arc-minute flow directions, 4,323 unit catchments, with the HydroBASINS catchment $U(i)$ of Figure 2 as the dashed outline; rivers of Strahler order five and above. Panel (b) enlarges the dashed box: the 277 unit catchments within 50 kilometres of the district, whose runoff the masked run sets to zero (grey), over the 111 sub-basins the neighbour exclusion removes from $U(i)$ (amber); the purple outlines are the district's 6 contiguous neighbours.

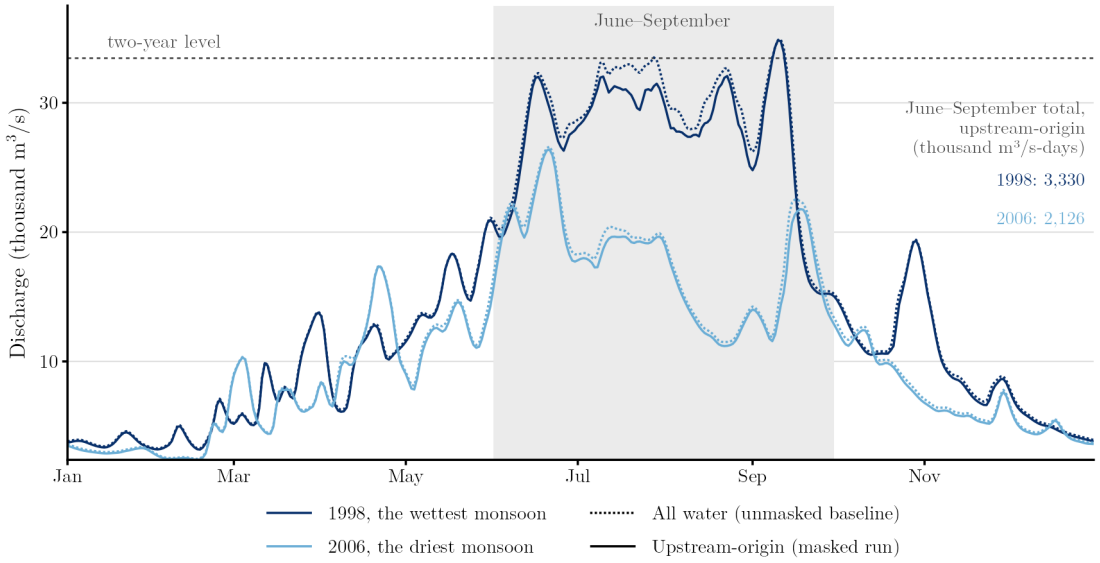


Figure C.2: Daily upstream-origin discharge at the river cell

Notes: Daily discharge at Dhubri's river cell, in thousand m^3/s, for the June–September with the largest (1998) and the smallest (2006) upstream-origin total in 1990–2025. Solid lines: the masked run, upstream origin only; dotted: the unmasked baseline. The shaded band is the window whose total, in logs, is the routed regressor of Table 3; the dashed line is the cell's two-year return level (33 thousand m^3/s), above which a day counts as a flood-stage day (Section C.3).

C.2. Validation

- **Is the simulated discharge credible?** Yes: it matches an independent routing model (median $r = 0.90$) and satellite-mapped floods.

Against an independent routing model.— Against VIC-RRM — an independent model with different physics and routing, calibrated on Bangladesh gauge observations — CaMa's daily discharge agrees closely: $r = 0.96$ at the Ganges checkpoint, median $r = 0.90$ across 365 district gauge cells, and median $\kappa = 0.48$ on the margin the design actually uses, which days exceed a cell's own return level. The models share ERA5-Land forcing, so this validates routing rather than meteorology. The resulting ordering is respected throughout: discharge is validated twice over, inundated extent is biased in level but valid in rank, and modelled flood *duration* is not used.

Validation against satellite flooding.—Over the 147 catalogued GFD events, the model's inundation extent correlates with observed flooded fractions at 0.41 (Spearman, district level), reaching 0.51 with a probability of detection of 0.90 at unit-catchment resolution on major events. The model is naturalized — no reservoirs or levees — so infrastructure-failure floods like the 2008 Kosi embankment breach are invisible to it by construction: the routed instrument captures rain-driven floods only, precisely the exogeneity property the design wants.

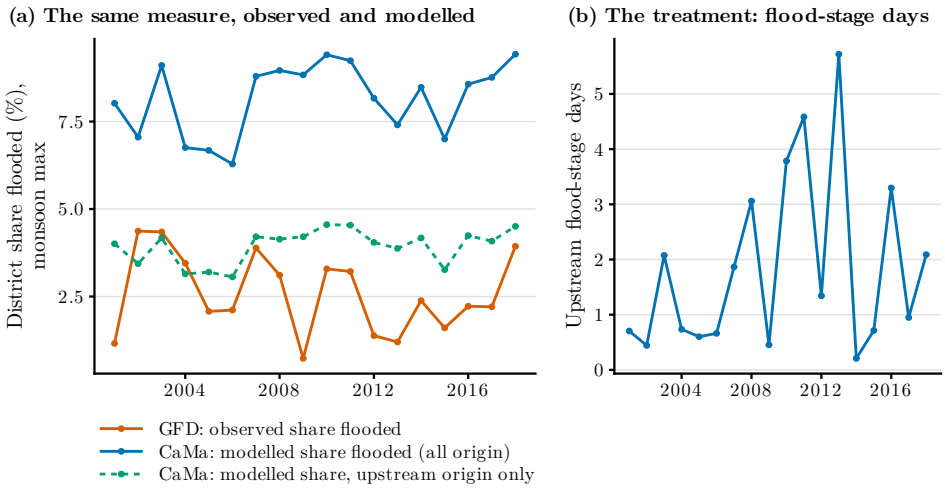


Figure C.3: Routed and observed flooding over time

Notes: Yearly means across the districts carrying both sources (387 districts with a CaMa river cell, an upstream signal and no bifurcation flag). Panel (a) is in levels — the monsoon-maximum percentage of district area flooded — for GFD (satellite-observed, any origin), the CaMa baseline run (modelled, any origin), and the CaMa masked run (modelled, upstream origin only). Panel (b) is the paper's treatment: days the three-day-mean upstream-origin discharge exceeds the two-year return level, averaged over the same districts. GFD district-years lost to cloud cover, and districts without CaMa floodplain cells, drop from their series' means.

C.3. The Flood-Stage Day Count

The day count is the hazard version of Table 3's discharge series: the June–September days on which the centred three-day mean of upstream-origin discharge at the district's river cell exceeds the cell's own two-year return level. Table C.1 enters it beside the rainfall total, as the main text enters the volume. It carries a within-year effect the volume does not — 0.020 per day in the year the river stands high, 0.018 with the spatial block — and its lagged term is halved and loses significance once the spatial block enters, while it absorbs none of the lagged rainfall coefficient.

Table C.1: Upstream rainfall and routed flood-stage days

	First link	Routed water		Both	
	(1)	(2)	(3)	(4)	(5)
log Upstream Precip. (t)	6.244***			-0.083	-0.112
	(2.277)			(0.411)	(0.507)
log Upstream Precip. ($t-1$)	0.249			0.874**	1.469***
	(1.191)			(0.396)	(0.307)
Upstream Flood-Stage Days (t)		0.020**	0.018**	0.020**	0.018**
		(0.008)	(0.008)	(0.009)	(0.008)
Upstream Flood-Stage Days ($t-1$)		0.028***	0.015**	0.025**	0.011
		(0.010)	(0.007)	(0.011)	(0.007)
Observations	6,336	6,336	6,336	6,336	6,336
Number of Districts	176	176	176	176	176
Mean of dependent variable	2.32	1.10	1.10	1.10	1.10
Estimation	OLS	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓	✓
Spatial controls			✓		✓

Notes: The outcome is the count of UCDP events located to a district or better, estimated by Poisson on 1990–2025 on the districts carrying a CaMa river cell, a usable upstream signal and no bifurcation flag. All five columns run on ONE sample. Column 1 is the first link: flood-stage days regressed by OLS on upstream rainfall and the controls, so it reads days per log point of rainfall. Every column carries the same control set as Table 2: upstream temperature, local rainfall and local temperature, current and lagged. Columns 3 and 5 add the spatial block — neighbours' lagged conflict and neighbours' current and lagged local rainfall and temperature, averaged under row-normalised queen contiguity. Flood-stage days count the days on which the centred three-day mean of upstream-origin discharge exceeded the cell's fitted two-year return level in June–September. Local rainfall and temperature enter every column and are not shown. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Within district the count correlates only 0.18 with the rainfall treatment, against 0.47 for the seasonal volume, which is why the main text carries the volume: a threshold count keeps the extreme tail of the discharge and discards its level.

C.4. Inundated Extent and the Flood Catalogue

- **Do flooded area and depth matter for conflict?** No: they predict deaths, not conflict.
- **Why not use the Global Flood Database?** Its events come from a news catalogue that detects floods non-randomly.

Extent and depth.—Inundated *area* and flood *depth* — the margins an optical flood map measures — carry no effect and absorb none of the flood-stage coefficient. This is not an artifact of measurement scale: rebuilt like-for-like, as days on which the district's flooded fraction exceeded its *own* return level, the footprint remains a tight zero, alone and entered beside discharge on identical observations. Depth so normalized does move conflict, but depth-days and discharge-days correlate at 0.94 within district — two readings of the same river — so it corroborates rather than competes. Nor is weighting the explanation: rebuilt weighting each pixel by population or kharif harvested area, the footprint still carries nothing.

The footprint measures are also validated on their own terms: on the Dartmouth catalogue's impact fields across 144 usable India events, a log point of flooded area raises recorded deaths by 0.32 and displacement by 0.73. The measures that best predict who dies predict nothing about

conflict, while routed discharge days continue to — humanitarian severity and conflict relevance are different quantities.

The footprint's null is adaptation, not irrelevance: under abnormal inundation land moves *into* flood-tolerant rice and *out* of waterlogging-sensitive cash crops — the flood-tolerance margin of Emerick et al. (2016). Spread water is absorbed by land use; the channel-force margin, dominated by the first days at flood stage, is the margin the portfolio cannot insure, and the conflict response concentrates in the paddy-floodplain systems whose agriculture engages river water, not where water-vulnerable crops dominate. One methodological note: the flood channel operates chiefly within the year, the agricultural-damage channel into the following one, so testing the flood channel against the wrong lag will fail.

Modeled flooding versus the flood catalogue.—The natural alternative to a routing model is the Global Flood Database (Tellman et al., 2021), and the reason to decline it is measurement (Patel, 2024). Every one of its 147 India events descends from the news-based Dartmouth catalogue, and the error is testable in-sample: small floods are missing non-randomly, detection depends on who lives downstream, and detection varies with cloud cover — which tracks the storm itself. The regression the catalogue invites shows the consequence: a catalogued-event indicator predicts *fewer* recorded conflict events, but the GFD's own mapped extent carries nothing, a CaMa-built indicator matched to the catalogue's base rate is significantly *positive*, and the negative appears identically in two independent news-derived conflict datasets — attention displacement within a shared news pipeline, not pacification. The design therefore uses continuous physical quantities throughout and retains the GFD in the validation role below.

C.5. The Flood Threshold

- **Does the flood threshold define the result?** No: refitting it on the full record moves nothing.

Does the flood threshold define the result?.—The return level is what makes a day a flood day. The routed panel fits the levels on the full 37-year routing record; the 2000–2018 panel it replaced used only 19 annual maxima. On the 2000–2018 sample, where both exist, the longer record raises the two-year level by a median 4.5% and cuts mean flood-stage days from 2.22 to 1.87, while the two series still correlate at 0.973. Put side by side in the same specification, nothing moves: routed flood days read 0.0277 against 0.0276, upstream rainfall 0.781 against 0.788. The ordering Section V draws on does not depend on where the threshold was placed.

C.6. The Shape of the Flood Response and the Origin of the Water

- **Do floods explain why conflict falls in dry upstream years?** No. Holding flood-stage days fixed leaves the dry-year decline in place (Figure C.4), and so does counting smaller floods.
- **Does the origin of the flood water matter?** Separating flood days by origin leaves the upstream-origin response unchanged (Table C.2).

Floods above the two-year level do not explain why conflict falls in dry catchment years as steadily as it rises in wet ones (Figure C.4). The response across deciles of lagged upstream rainfall is close to linear, and dry years could still act through floods if dry monsoons produced them. They produce fewer: relative to the middle deciles, the four driest lower the same season's upstream-origin flood-stage days by 0.45 to 0.80 days and the three wettest raise them by 1.5 to 3.0. Holding

flood-stage days at t and $t-1$ fixed leaves the conflict profile where it was: the driest decile's coefficient is -0.56 without the flood terms and -0.56 with them, the wettest decile's 0.49 and 0.48, while each flood-stage day carries 0.019 (0.013 counting all water). Smaller floods do not carry the dry-year response either. A flood-stage day requires discharge above the river cell's two-year level, so I lower the bar to the cell's 90th or 75th percentile of daily discharge over 1989–2025 and count the monsoon days above it, in total or only below the two-year level. Dry monsoons prevent these smaller floods too: relative to the middle deciles, the four driest have 1.7 fewer upstream-origin days above the 90th percentile and 3.1 fewer above the 75th. Holding them fixed at t and $t-1$ moves the four driest deciles' average coefficient from -0.37 to between -0.37 and -0.34, on the upstream-origin and the full run alike, a shrinkage of at most 8%.

Separating flood days by the origin of the water leaves the upstream-origin response unchanged (Table C.2). The routed-water race of Table 3 counts flood-stage days in a run with the 50-kilometre mask, which removes the district's own and nearby runoff. The full run counts more, and the difference is the flood-stage days the district's own runoff adds: zero in 64% of district-years and 1.6 on average, against 2.0 upstream-origin days. Added to column 6 of Table 3, they leave the upstream-origin days at 0.017 against 0.018 and upstream rainfall at 1.48. The district's own floods carry a response of similar size, estimated less precisely: 0.009 ($p = 0.12$) at t and 0.012 ($p = 0.08$) at $t-1$.

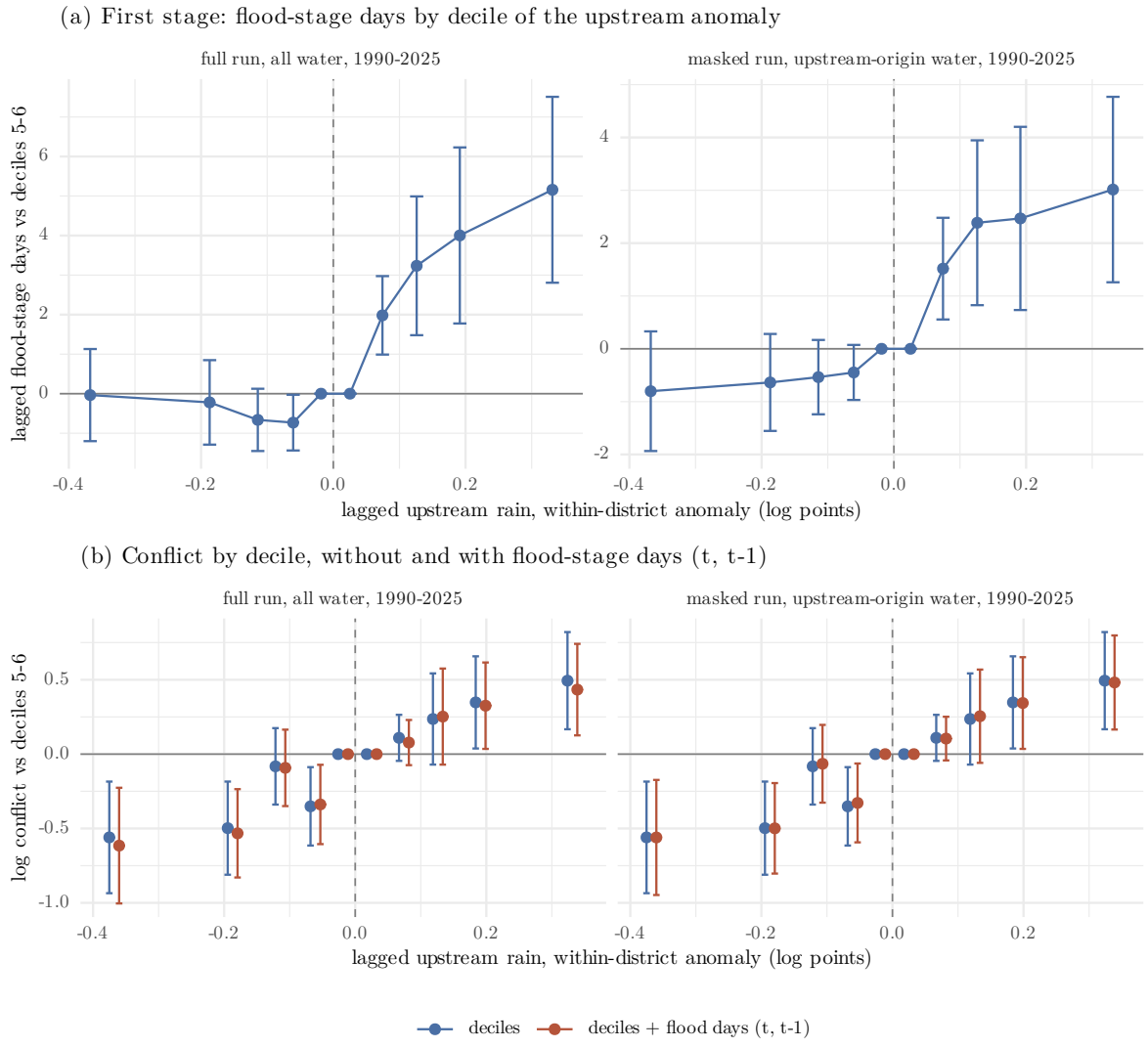


Figure C.4: Flood-stage days and conflict by decile of the lagged upstream rainfall anomaly

Notes: The horizontal axis is the within-district anomaly of log lagged upstream rainfall, cut into deciles on the Table 2 column 4 estimation rows; points sit at each decile's mean anomaly, with deciles 5–6 as the reference, and bars are Conley 95% intervals. The sample is the districts carrying a CaMa river cell, a usable upstream signal and no bifurcation flag, 1990–2025: 6,336 district-years on 176 districts. Panel (a) is the first stage: lagged flood-stage days (JJAS days on which the centred three-day discharge at the district's gauge cell exceeds its two-year level) on the decile dummies with the column 4 controls and district and year effects, OLS; left, the full naturalized run (all water); right, the run with the 50-kilometre mask, which removes the district's own and nearby runoff (upstream-origin water, the measure of Table 3). Panel (b) is the column 4 Poisson regression of conflict on the same decile dummies on each sample, without (blue) and with (red) flood-stage days at t and $t-1$; the flood-day coefficients are 0.013 and 0.011 (full run, t and $t-1$) and 0.019 and 0.008 (upstream-origin water).

Table C.2: Flood-stage days by origin of the water

	(1) Table 3 column 6	(2) + local-origin days	(3) + local-origin days, floored
log Upstream Precip. (t)	-0.112	-0.194	-0.200
	(0.507)	(0.522)	(0.522)
log Upstream Precip. ($t-1$)	1.469***	1.484***	1.482***
	(0.307)	(0.306)	(0.306)
Upstream-origin flood-stage days (t)	0.018**	0.017**	0.017**
	(0.008)	(0.008)	(0.008)
Upstream-origin flood-stage days ($t-1$)	0.011	0.010	0.010
	(0.007)	(0.007)	(0.007)
Local-origin flood-stage days (t)		0.009	0.009
		(0.006)	(0.006)
Local-origin flood-stage days ($t-1$)		0.012*	0.013*
		(0.007)	(0.007)
Local-origin days floored at zero			✓
Observations	6,336	6,336	6,336
Number of Districts	176	176	176
Mean of dependent variable	1.10	1.10	1.10

Notes: Column 1 is Table 3 column 6. Upstream-origin flood-stage days are JJAS days on which the centred three-day discharge at the district's gauge cell exceeds its two-year level in the run with the 50-kilometre mask, which removes the district's own and nearby runoff (the measure of Table 3); local-origin days are the days on which it does so in the full naturalized run, minus the upstream-origin days, at the same cell and threshold. Column 3 floors the difference at zero. Conley spatial-HAC standard errors (150 km) in parentheses; ***, **, and * indicate significance at 1%, 5% and 10%.

Appendix D. Mechanism

This appendix collects the evidence on how the upstream shock reaches conflict: when the response appears (Appendix D.1), what happens to agriculture, prices and labour (Appendix D.2), and what public employment does (Appendix D.3).

D.1. The Timing of the Response

- **When does conflict respond to upstream monsoon rain?** From the turn of the year, peaking in the autumn after the monsoon (Figure D.1).
- **Does the spatial block change the timing?** No. With neighbours' conflict lagged a year, as in the tables, both rungs give the same response.
- **Which kind of violence responds?** Battles between the state and armed groups; attacks on civilians do not (Figure D.2).

The main text dates the treatment to a year; this subsection asks when within the following two years the response appears, and whether it is specific to the monsoon or merely to rainfall.

Re-indexing.—A district-month is relabelled by monsoon year v (June to May) and position $k \in \{0, \dots, 11\}$, months since June — a complete relabelling of the calendar, with $y = v + \mathbb{1}\{k \geq 7\}$. Let C_{ivk} count events UCDP places in district i in that month, located to a district and dated to a week or better.

Specification.—Let $R^M_{i,v}$ be upstream monsoon rainfall — exactly the main text's treatment — and $R^N_{i,v}$ the complementary January–May and October–December total, both one number per district-year. With T^M and T^N the matching upstream temperatures,

$$\begin{aligned}\log \mathbb{E}\big[C_{ivk}\big] = {} & \beta^M_k \log R^M_{i,v} + \beta^M_{k+12} \log R^M_{i,v-1} + \beta^N_k \log R^N_{i,v} + \beta^N_{k+12} \log R^N_{i,v-1} \\ & + \gamma^M_k T^M_{i,v} + \gamma^M_{k+12} T^M_{i,v-1} + \gamma^N_k T^N_{i,v} + \gamma^N_{k+12} T^N_{i,v-1} \\ & + \mathbf{x}'_{ivk}\boldsymbol{\theta} + (W\mathbf{z}_{ivk})'\boldsymbol{\psi} + \alpha_i + \lambda_{vk},\end{aligned} \tag{1}$$

estimated by Poisson pseudo-maximum-likelihood with standard errors clustered on district. Each coefficient vector has twenty-four elements: an observation at position k uses element k, the season that opened its own monsoon year, and element $k + 12$, the season before. This is one regression, not twenty-four — estimating it month by month would lose more than half the observations, since a Poisson drops districts with no conflict in the month being estimated.

What varies, and what does not.—The rainfall on the right-hand side is a seasonal total, so what changes across the twenty-four coefficients is which month's violence sits on the left; month-to-month movement in the estimates is not structure. The twenty-four positions all fall after the monsoon begins, so the design contains no pre-shock months and offers no placebo lead.

Controls and fixed effects.—$\mathbf{x}_{ivk}$ holds the district's own rainfall and temperature that month and the month before; $\mathbf{W}\mathbf{z}_{ivk}$ holds the queen-contiguity averages of neighbours' conflict twelve months earlier and of neighbours' own weather, rainfall in logs on both sides. The contemporaneous spatial lag is omitted for the reason Table B.3 gives, and the Poisson carries no own lagged conflict term. The time effect is $\lambda_{vk} = \delta_{vk}$, one parameter for every calendar month in the sample (418 of them): a single month-year effect absorbs anything common to India in that month — an election, a national operation, a change in reporting — where separate month-of-year and year effects absorb only the average season and the average year. It does not absorb the horizon structure, because

within a month-year cell every district sits at the same two horizons, so β_k and β_{k+12} are identified from variation across districts in the two vintages' rainfall. This is the strictest time control the design admits. Two rungs are reported, without the spatial block and with it.

Result.—Figure D.1 plots both rungs over 1990–2025, the window of Tables 2–E.3. Monsoon rainfall is flat or negative through the season itself and runs positive through the following year: summed over months seven to eighteen it gives 5.52 ($p = 0.043$) under month-year effects alone and 5.53 ($p = 0.046$) with the spatial block, the peak falling in November of the following year (0.91). Non-monsoon rainfall, which is nearly orthogonal to monsoon rainfall once the fixed effects are removed, raises no conflict over the same window. Its sum is negative and not significant in either rung: -2.89 ($p = 0.119$) and -2.49 ($p = 0.135$). The delayed rise in conflict is a monsoon phenomenon rather than a rainfall phenomenon, which is what the agricultural mechanism requires.

What the spatial block carries.—The two rungs differ in neither size nor shape: adding neighbours' conflict and weather leaves the monsoon sum at 5.53, against 5.52 without them. Neighbours' conflict enters lagged twelve months, as the tables' column 4 lags it a year. Lagged one month, it would itself respond to the same shock, since neighbours share upstream catchments, and it absorbed about two thirds of the profile.

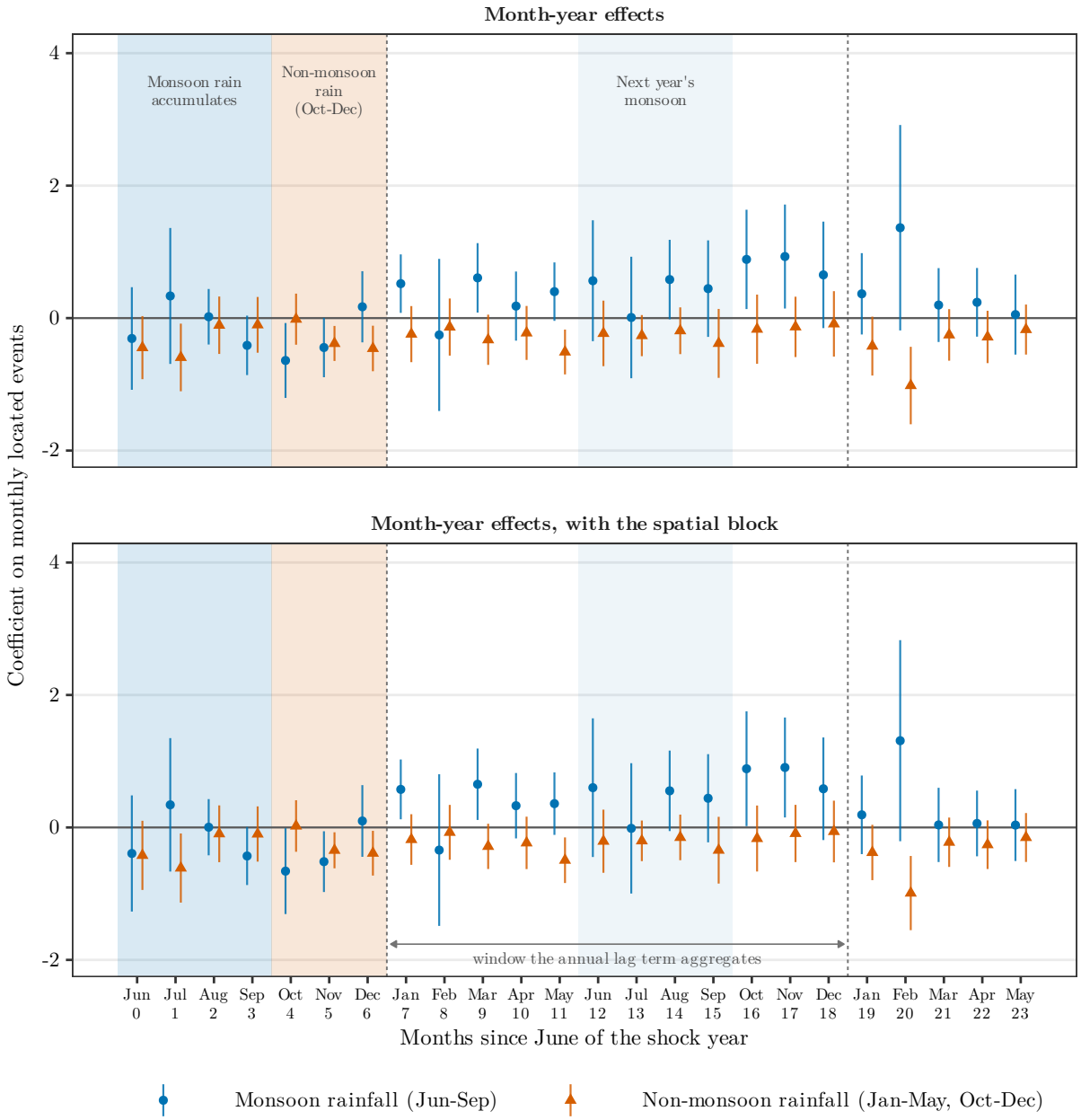


Figure D.1: Monthly horizons of the upstream-rainfall shock, by season

Notes: Each panel plots $\hat{\beta}_h^M$ and $\hat{\beta}_h^N$ from one estimate of equation (1), both seasons carried simultaneously at all twenty-four horizons, against months since June of the shock year. Points are Poisson coefficients with 95 percent cluster-by-district intervals. The panels differ only in the spatial block; both carry district and month-year effects, and 418 month-year cells span 1990–2025. Shaded bands mark the months in which each rainfall measure is accumulated. The third band is the following year's monsoon; the dashed rules enclose horizons 7–18, the window the annual lag term aggregates. Both rungs are estimated on 82,764 district-months over 198 districts. The spatial block lags neighbours' conflict twelve months, as the annual tables lag it a year.

Which violence carries it.—Figure D.2 re-estimates the spatial rung with the outcome split by UCDP type of violence. State-based violence, fighting between the Government of India and an armed group, is 66 percent of located events and carries the profile: flat through the monsoon and the harvest, then positive at every horizon of the following year, summing to 4.15 ($p = 0.002$) over horizons 7–12 and 3.42 ($p = 0.001$) over horizons 13–18. One-sided violence, armed groups' attacks on civilians, does not respond: 2.86 ($p = 0.651$) over horizons 7–18. Non-state violence, fighting between organized groups, has too few events to estimate precisely (303 events on 52 districts;

11.75, $p = 0.296$).

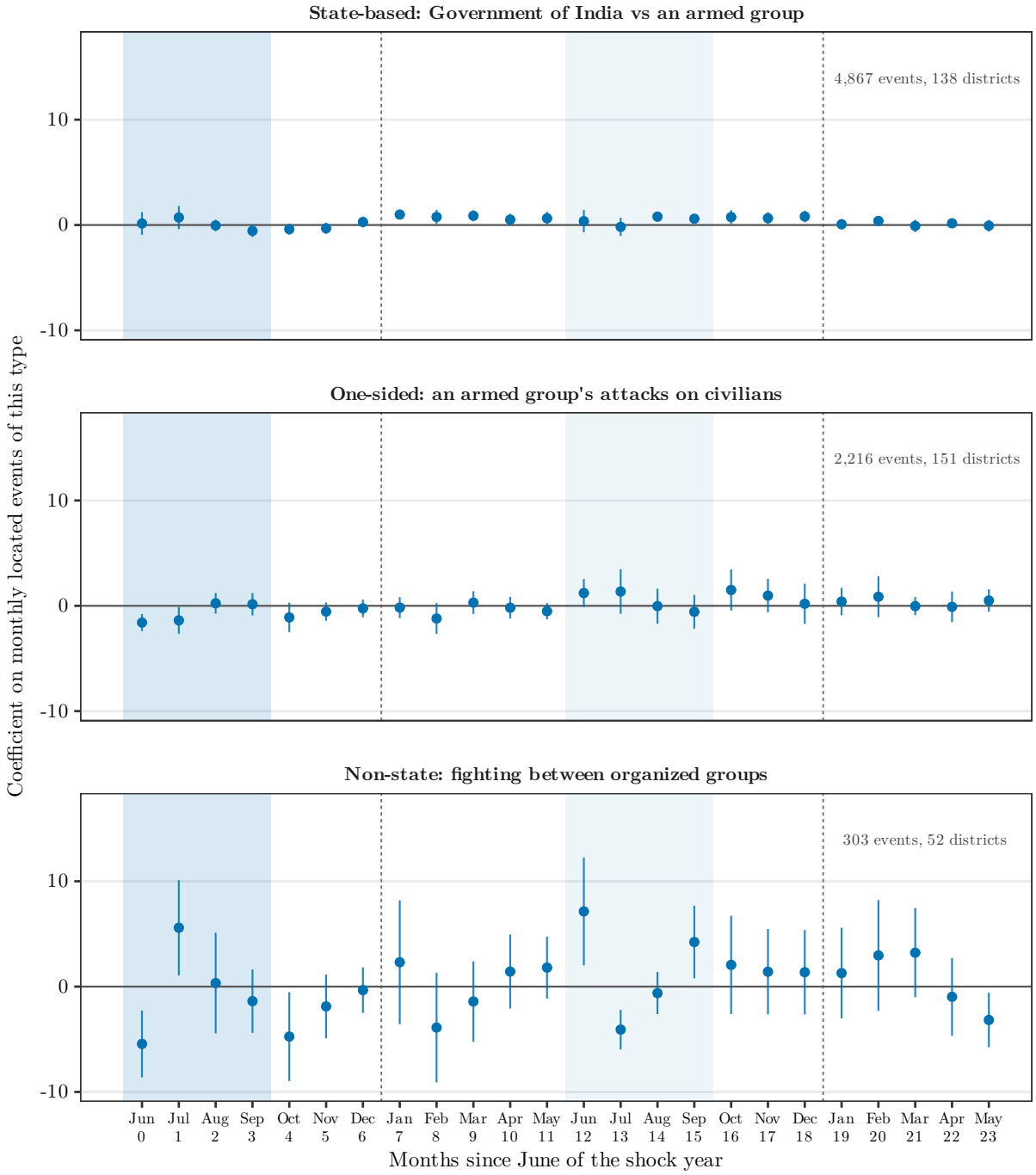


Figure D.2: Monthly horizons of the monsoon shock, by type of violence

Notes: Each panel re-estimates the spatial rung of Figure D.1 with monthly located events of one UCDP type of violence as the outcome. Monsoon horizons are shown; the non-monsoon horizons, temperatures, own and neighbours' weather, neighbours' conflict lagged twelve months, and district and month-year effects are carried as in Figure D.1. Points are Poisson coefficients with 95 percent cluster-by-district intervals; the Poisson keeps districts with at least one event of that type, so the samples differ by type. Shading and dashed rules as in Figure D.1.

- **Which tables does the main text's mechanism section read?** Table D.1 (the district level: four sources of water, one column per outcome) and Table D.2 (the household after an upstream shock).
- **What does upstream rain do to agriculture?** Floods destroy kharif planting (Tables D.5–D.7).
- **And to labour?** Households supply more labour as non-farm earnings fall, and the wage falls (Tables D.13–D.15).

Table D.1: The mechanism at the district level: four sources of water, one column per outcome

	Kharif area	Cereal prod.	Real ag. wage	Nominal wage	Origin output
	(1)	(2)	(3)	(4)	(5)
log Upstream Precip. (t)	-0.102	-0.232**	-0.081**	-0.066**	0.269***
	(0.071)	(0.106)	(0.037)	(0.031)	(0.049)
log Upstream Precip. $(t-1)$	0.140**	0.045	-0.127***	-0.092***	0.203***
	(0.061)	(0.103)	(0.041)	(0.034)	(0.053)
log Downstream Precip. (t)	0.217***	0.316***	0.054*	-0.030	-0.011
	(0.056)	(0.093)	(0.032)	(0.024)	(0.037)
log Downstream Precip. $(t-1)$	0.159***	0.193**	0.094***	0.028	0.102***
	(0.049)	(0.088)	(0.032)	(0.027)	(0.039)
Estimation	OLS	OLS	OLS	OLS	OLS
District & Year FE	✓	✓	✓	✓	✓
Observations	8,079	7,715	4,556	4,556	8,649
Districts	341	341	276	276	339

Notes: Every column is one regression carrying all four rainfall measures at once — upstream, local, neighbour and downstream, each contemporaneous and lagged — with upstream and local temperatures and their lags, district and year fixed effects, and standard errors clustered on district, as in Table D.11. Each column is estimated on its outcome's own full sample; Table D.11, columns 5–8, reports the same outcomes restricted to the conflict-estimation districts. Only the upstream and downstream terms are shown. All outcomes are logs. Kharif planted area sums hectares over one crop set used by every mechanism exhibit — the six largest kharif crops by planted area (rice, cotton, bajra, maize, groundnut, pigeonpea). Cereal production sums tonnes over the four cereals (rice, maize, bajra, jowar). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table D.2: The household after an upstream shock: income, labour allocation and consumption

	Income			Labour allocation			Welfare
	Farm income	Non-farm earnings	Any remittance	Any business	Business days	Farm days	Consumption
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
log Upstream Precip. (t)	0.305	-4.100***	0.004	-0.025	-8.544	-2.369	-0.650***
	(1.529)	(1.259)	(0.076)	(0.055)	(10.912)	(17.640)	(0.177)
log Upstream Precip. ($t-1$)	0.701	-3.424**	0.075	0.030	9.275	4.859	0.054
	(1.686)	(1.430)	(0.064)	(0.028)	(6.540)	(20.670)	(0.172)
log Downstream Precip. (t)	3.242***	1.024	0.044	-0.011	-3.067	25.730**	0.110
	(1.152)	(0.858)	(0.037)	(0.027)	(6.013)	(12.049)	(0.111)
log Downstream Precip. ($t-1$)	2.040*	-0.167	-0.044	0.011	-0.284	-9.108	-0.026
	(1.097)	(0.827)	(0.044)	(0.021)	(4.612)	(9.433)	(0.122)
Estimation	OLS	OLS	OLS	OLS	OLS	OLS	OLS
District & Wave FE	✓	✓	✓	✓	✓	✓	✓
Observations	48,386	48,386	48,386	186,361	186,361	186,361	48,337
Districts	221	221	221	221	221	221	221
Mean of dependent variable			0.11	0.07	16.57	33.08	

Notes: Every column is one regression carrying all four rainfall measures at once — upstream, local, neighbour and downstream, each contemporaneous and lagged — with upstream and local temperatures and their lags, district and survey-wave fixed effects, and standard errors clustered on district, as in Table D.11. Only the upstream and downstream terms are shown. IHDS households (columns 1–3 and 7) and individuals (columns 4–6), 2005 and 2012 waves; the treatment is the mean over each wave's reference years, lagged to the previous wave's. Income: farm income is own-account agricultural income — crop, agricultural property and animal income — accruing to households that operate land or livestock. Labour allocation: whether the person runs a non-farm business, days supplied to it counting non-participants as zero, and farm days on the same convention. Welfare: consumption is the log of household consumption. Farm and non-farm income are asinh of rupees, participation and remittance receipt are linear probabilities, days are counts; coefficients are not comparable in size across columns. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

The routed water carries the same chain on its own variation. Table D.3 puts Table 3's flood-stage days into Table D.1's specification, in place of upstream rainfall and beside it, on the districts that carry a routed series. A flood-stage day cuts kharif planted area by -0.006 log points in the year it occurs ($p = 0.05$) and lowers the real wage (-0.002), and it raises conflict in that year (0.014) and the next (0.019); with both measures in the regression neither moves, the lagged rainfall elasticity for conflict going from 1.13 to 1.08. The two measures are nearly uncorrelated within district, so each carries the full chain from water to planting, wages and conflict on variation the other does not use.

Table D.3: The mechanism's first links with the routed water

	Kharif area (1)	Cereal prod. (2)	Real ag. wage (3)	Nominal wage (4)	Conflict (5)
Panel A: Upstream rainfall (Table D.1 specification)					
log Upstream Precip. (t)	-0.058	-0.181	-0.070*	-0.056	-0.104
	(0.079)	(0.114)	(0.042)	(0.034)	(0.341)
log Upstream Precip. ($t-1$)	0.149**	0.033	-0.145***	-0.097**	1.132***
	(0.068)	(0.111)	(0.046)	(0.040)	(0.330)
Panel B: Routed flood-stage days in its place					
Upstream Flood-Stage Days (t)	-0.006**	-0.008	-0.002*	-0.001	0.014**
	(0.003)	(0.006)	(0.001)	(0.001)	(0.007)
Upstream Flood-Stage Days ($t-1$)	-0.002	-0.003	-0.001	-0.003*	0.023**
	(0.002)	(0.003)	(0.002)	(0.001)	(0.011)
Panel C: Both					
log Upstream Precip. (t)	-0.022	-0.139	-0.065	-0.057*	-0.151
	(0.075)	(0.103)	(0.042)	(0.034)	(0.339)
log Upstream Precip. ($t-1$)	0.162**	0.052	-0.145***	-0.090**	1.081***
	(0.071)	(0.113)	(0.046)	(0.039)	(0.333)
Upstream Flood-Stage Days (t)	-0.006**	-0.008	-0.002*	-0.001	0.014*
	(0.003)	(0.006)	(0.001)	(0.001)	(0.007)
Upstream Flood-Stage Days ($t-1$)	-0.002	-0.003	-0.000	-0.002*	0.019*
	(0.002)	(0.003)	(0.002)	(0.001)	(0.011)
Observations	6,910	6,663	3,862	3,862	5,256
Number of Districts	291	291	238	238	146
Mean of dependent variable					1.18
Estimation	OLS	OLS	OLS	OLS	Poisson
District & Year FE	✓	✓	✓	✓	✓

Notes: Table D.1's specification — upstream, local, neighbour and downstream rainfall at both horizons with their temperatures, district and year effects, standard errors clustered by district — on the districts that carry a usable routed series (a CaMa river cell, an upstream signal and no bifurcation flag), with the routed flood-stage days of Table 3 entering in place of upstream rainfall (Panel B) and beside it (Panel C). Columns 1–4 are the district outcomes of Table D.1; column 5 is conflict on the same districts under the same right-hand side, which is not Table 3's (that table carries the spatial block and Conley errors). Local, neighbour and downstream terms enter every column and are not shown. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table D.4 asks Table 4's question of the routed water: is its response concentrated where agriculture is? Panel A says it is not. Interacted with agricultural dependence alone, the discharge shows no gradient for conflict (-0.207 at $t-1$) or for the real wage (-0.012), where upstream rainfall on the same districts shows 0.973*** and -0.050. Panel B locates the response. In agricultural districts on large rivers, last monsoon's discharge raises conflict (0.558**; 0.628** holding the rival moderators fixed) and lowers the real wage (-0.061** in the same year, -0.065** the next) and the nominal wage (-0.053**), the signs the chain predicts; at the sample's average river the dependence gradient is absent (-0.278 for conflict). River size is standardised within this sample, every district of which has a river, so the lower-order terms read at its own average river. The simulated discharge at a small river cell is a poor measure of the water a district receives, and the chain appearing where the river is large is what a noisy measure predicts; large drainage area also marks plains districts, so the panel locates the response rather than explaining it.

Table D.4: Is the routed water's response concentrated where agriculture is?

	Conflict events		Real ag. wage		Nominal wage	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Agricultural dependence						
log Routed Discharge, June–September (t)	-0.365**	-0.385**	-0.019	-0.030*	-0.020	-0.027**
	(0.177)	(0.172)	(0.014)	(0.016)	(0.012)	(0.013)
× Agric. dependence	0.304	-0.131	-0.015	-0.032*	-0.007	-0.017
	(0.302)	(0.392)	(0.015)	(0.017)	(0.011)	(0.012)
log Routed Discharge, June–September ($t-1$)	0.304*	0.258	0.018	0.016	0.006	0.005
	(0.165)	(0.163)	(0.015)	(0.016)	(0.014)	(0.016)
× Agric. dependence	-0.207	-0.163	-0.012	-0.017	-0.013	-0.020
	(0.296)	(0.385)	(0.014)	(0.014)	(0.012)	(0.013)
Panel B: Agricultural dependence and the size of the river						
log Routed Discharge, June–September (t)	-0.325*	-0.322**	-0.009	-0.024	-0.014	-0.022
	(0.183)	(0.163)	(0.017)	(0.020)	(0.014)	(0.015)
× Agric. dependence	0.196	-0.119	-0.026	-0.041**	-0.014	-0.022*
	(0.295)	(0.396)	(0.016)	(0.017)	(0.012)	(0.013)
× River size	0.077	0.076	0.025	0.018	0.015	0.013
	(0.150)	(0.160)	(0.017)	(0.017)	(0.011)	(0.011)
× Agric. dependence × River size	0.300	0.152	-0.061**	-0.059*	-0.032*	-0.032
	(0.252)	(0.273)	(0.030)	(0.030)	(0.020)	(0.020)
log Routed Discharge, June–September ($t-1$)	0.262	0.229	0.033*	0.030	0.018	0.016
	(0.182)	(0.173)	(0.017)	(0.019)	(0.014)	(0.016)
× Agric. dependence	-0.278	-0.204	-0.025	-0.028*	-0.023*	-0.028**
	(0.314)	(0.408)	(0.015)	(0.014)	(0.013)	(0.013)
× River size	-0.318*	-0.355*	0.036*	0.032*	0.028*	0.024
	(0.169)	(0.183)	(0.018)	(0.018)	(0.017)	(0.016)
× Agric. dependence × River size	0.558**	0.628**	-0.065**	-0.059*	-0.053**	-0.048*
	(0.262)	(0.269)	(0.031)	(0.031)	(0.026)	(0.027)
Observations	5,508	5,472	4,003	3,980	4,003	3,980
Number of Districts	153	152	247	246	247	246
Mean of dependent variable	1.16	1.16				
Estimation	Poisson	Poisson	OLS	OLS	OLS	OLS
District & Year FE	✓	✓	✓	✓	✓	✓
Rival moderators × discharge		✓		✓		✓

Notes: Table 4 with routed discharge — the June–September total of upstream-origin discharge simulated by CaMa-Flood at the district's river cell, in logs, the regressor of Table 3 — in place of upstream rainfall, on the districts that carry a routed series. Every column interacts current and lagged discharge and downstream rainfall with agricultural dependence: 2005 Economic Census non-agricultural employment per head, reversed so that higher means more dependent on agriculture. The upstream rainfall terms are dropped; every other control is Table 4's. Columns 1–2 are Poisson on located UCDP events, columns 3–6 OLS on the male field-labour wage, deflated by the local paddy price (3–4) and nominal (5–6). Columns 2, 4 and 6 add the same interactions with Table 4's three rival moderators. Panel B adds interactions with river size, the log upstream drainage area at the district's river cell, standardised over the districts in this sample — every one of which has a river, so the scale is not Table 4's, which includes districts without one. Local weather enters uninteracted, and the downstream terms are not shown. Standard errors clustered by district in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table D.5: Kharif crop margins: extensive and intensive

	Rice	Cotton	Bajra	Maize	Groundnut	Pigeonpea
	(1)	(2)	(3)	(4)	(5)	(6)
Area, extensive: any planting (0/1)						
log Upstream Precip. (t)	-0.060***	-0.090***	-0.058**	-0.135***	-0.066**	-0.055**
	(0.014)	(0.027)	(0.025)	(0.024)	(0.028)	(0.026)
log Downstream Precip. (t)	0.072***	-0.011	0.007	0.081***	0.014	0.017
	(0.016)	(0.026)	(0.022)	(0.024)	(0.031)	(0.026)
Observations	8,300	8,300	8,300	8,300	8,300	8,300
Mean of dependent variable	0.77	0.41	0.60	0.84	0.60	0.80
Area, intensive: log hectares, growing district-years						
log Upstream Precip. (t)	0.035	0.336**	-0.049	-0.376***	0.015	-0.256**
	(0.076)	(0.131)	(0.123)	(0.087)	(0.130)	(0.100)
log Downstream Precip. (t)	0.086	0.028	0.042	0.057	0.433***	0.103
	(0.063)	(0.105)	(0.114)	(0.091)	(0.117)	(0.111)
Observations	6,389	3,406	4,944	6,955	4,951	6,648
Production, extensive: any output (0/1)						
log Upstream Precip. (t)	-0.063***	-0.080***	-0.064***	-0.164***	-0.073***	-0.051**
	(0.014)	(0.026)	(0.025)	(0.029)	(0.028)	(0.025)
log Downstream Precip. (t)	0.070***	-0.020	0.002	0.090***	0.017	0.012
	(0.016)	(0.026)	(0.022)	(0.027)	(0.032)	(0.025)
Observations	8,300	8,300	8,300	8,300	8,300	8,300
Mean of dependent variable	0.77	0.40	0.59	0.82	0.59	0.79
Production, intensive: log tonnes, producing district-years						
log Upstream Precip. (t)	0.297***	0.665***	0.012	-0.273***	0.092	-0.158
	(0.089)	(0.183)	(0.133)	(0.098)	(0.143)	(0.106)
log Downstream Precip. (t)	0.051	-0.003	0.093	0.115	0.402***	0.018
	(0.070)	(0.149)	(0.153)	(0.103)	(0.111)	(0.123)
Observations	6,381	3,282	4,900	6,773	4,888	6,563

Notes: Every cell is one regression on the specification of Table D.11's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Extensive rows are linear probability models on an indicator for any planting (or any output) among district-years where the crop is observed; intensive rows are logs on the district-years that grow (produce) it. Columns are the six largest crops by the season's planted area, grown in at least forty districts. The season is the monsoon's own: rain falls on the standing crop. ***, **, and * denote 1%, 5%, and 10%.

Table D.6: Rabi crop margins: extensive and intensive

	Wheat (1)	Chickpea (2)	Rape & mustard (3)	Jowar (4)	Maize (5)	Barley (6)
Area, extensive: any planting (0/1)						
log Upstream Precip. (t)	0.004	-0.012	-0.012	0.020	0.007	0.002
	(0.013)	(0.021)	(0.024)	(0.017)	(0.023)	(0.019)
log Downstream Precip. (t)	0.003	-0.031	0.057***	-0.057***	-0.037*	0.007
	(0.012)	(0.025)	(0.020)	(0.020)	(0.022)	(0.016)
Observations	8,248	8,248	8,248	8,248	8,248	8,248
Mean of dependent variable	0.94	0.87	0.85	0.14	0.37	0.56
Area, intensive: log hectares, growing district-years						
log Upstream Precip. (t)	0.291***	-0.131	-0.037	0.091	-0.038	-0.044
	(0.071)	(0.103)	(0.086)	(0.206)	(0.235)	(0.084)
log Downstream Precip. (t)	-0.087	0.105	-0.069	-0.123	0.558***	0.073
	(0.057)	(0.096)	(0.078)	(0.200)	(0.212)	(0.095)
Observations	7,759	7,157	7,042	1,173	3,014	4,593
Production, extensive: any output (0/1)						
log Upstream Precip. (t)	0.000	-0.012	-0.001	0.021	0.008	0.011
	(0.012)	(0.022)	(0.022)	(0.017)	(0.023)	(0.019)
log Downstream Precip. (t)	0.003	-0.042*	0.047**	-0.058***	-0.037*	-0.004
	(0.012)	(0.024)	(0.021)	(0.020)	(0.022)	(0.016)
Observations	8,248	8,248	8,248	8,248	8,248	8,248
Mean of dependent variable	0.94	0.85	0.85	0.14	0.36	0.54
Production, intensive: log tonnes, producing district-years						
log Upstream Precip. (t)	0.287***	-0.129	-0.038	0.025	-0.065	0.006
	(0.085)	(0.110)	(0.094)	(0.195)	(0.239)	(0.089)
log Downstream Precip. (t)	-0.198***	0.070	-0.112	-0.125	0.710***	0.031
	(0.065)	(0.104)	(0.083)	(0.200)	(0.219)	(0.110)
Observations	7,753	7,020	6,994	1,169	3,006	4,492

Notes: Every cell is one regression on the specification of Table D.11's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Extensive rows are linear probability models on an indicator for any planting (or any output) among district-years where the crop is observed; intensive rows are logs on the district-years that grow (produce) it. Columns are the six largest crops by the season's planted area, grown in at least forty districts. The monsoon precedes rabi sowing, so the contemporaneous term is rain on the soil moisture the winter crop is sown into. ***, **, and * denote 1%, 5%, and 10%.

Table D.7: Kharif crop prices and the value of the harvest

	Rice	Cotton	Bajra	Maize	Groundnut	Pigeonpea
	(1)	(2)	(3)	(4)	(5)	(6)
Farm-harvest price, annual (log)						
log Upstream Precip. (t)	0.004	0.055	-0.042	-0.043*	-0.064**	0.013
	(0.025)	(0.038)	(0.030)	(0.025)	(0.030)	(0.026)
log Downstream Precip. (t)	-0.076***	0.043	0.006	0.018	0.039	0.027
	(0.020)	(0.045)	(0.023)	(0.021)	(0.029)	(0.027)
Observations	5,013	1,734	3,429	4,740	3,794	3,912
Revenue: price × production (log)						
log Upstream Precip. (t)	0.385***	0.168	-0.007	-0.310**	-0.027	-0.007
	(0.109)	(0.409)	(0.209)	(0.127)	(0.175)	(0.167)
log Downstream Precip. (t)	-0.148*	-0.048	0.175	0.268*	0.296**	0.117
	(0.080)	(0.215)	(0.185)	(0.136)	(0.133)	(0.225)
Observations	2,489	887	1,975	2,641	2,133	2,535

Notes: Every cell is one regression on the specification of Table D.11's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Prices are the DLD farm-harvest series, which is annual — it is not reported by season — so the revenue rows pair the annual price with kharif production: roughly the value of the harvest, and gross revenue rather than profit, since input costs are not observed. Because $\log(pq) = \log p + \log q$, the revenue coefficient equals the sum of the price and production responses on its own sample; it will not equal the sum of the rows printed above it or in Table D.5, which are estimated on the larger samples their own margins allow. Both blocks condition on producing district-years with a price quote, so they are intensive margins. ***, **, and * denote 1%, 5%, and 10%.

Table D.8: Kharif crop margins on the conflict-estimation sample

	Rice (1)	Cotton (2)	Bajra (3)	Maize (4)	Groundnut (5)	Pigeonpea (6)
Area, extensive: any planting (0/1)						
log Upstream Precip. (t)	-0.035*	-0.037	-0.028	-0.137***	-0.048	-0.018
	(0.019)	(0.037)	(0.035)	(0.039)	(0.043)	(0.036)
log Downstream Precip. (t)	0.079***	-0.072**	-0.023	0.105***	-0.027	-0.000
	(0.022)	(0.030)	(0.030)	(0.036)	(0.039)	(0.036)
Observations	3,841	3,841	3,841	3,841	3,841	3,841
Mean of dependent variable	0.67	0.42	0.49	0.78	0.48	0.80
Area, intensive: log hectares, growing district-years						
log Upstream Precip. (t)	-0.008	0.216	-0.130	-0.442***	-0.084	-0.249**
	(0.094)	(0.194)	(0.194)	(0.144)	(0.208)	(0.124)
log Downstream Precip. (t)	0.282***	-0.044	0.083	0.128	0.431***	0.069
	(0.080)	(0.138)	(0.147)	(0.122)	(0.163)	(0.123)
Observations	2,563	1,595	1,901	2,985	1,836	3,072
Production, extensive: any output (0/1)						
log Upstream Precip. (t)	-0.042**	-0.028	-0.032	-0.159***	-0.041	-0.023
	(0.018)	(0.038)	(0.036)	(0.043)	(0.046)	(0.035)
log Downstream Precip. (t)	0.078***	-0.076**	-0.029	0.098**	-0.021	-0.003
	(0.023)	(0.030)	(0.031)	(0.040)	(0.040)	(0.034)
Observations	3,841	3,841	3,841	3,841	3,841	3,841
Mean of dependent variable	0.67	0.40	0.49	0.75	0.47	0.79
Production, intensive: log tonnes, producing district-years						
log Upstream Precip. (t)	0.392***	0.860***	0.018	-0.174	0.161	-0.083
	(0.122)	(0.280)	(0.197)	(0.177)	(0.207)	(0.153)
log Downstream Precip. (t)	0.282***	-0.015	0.106	0.208	0.458***	0.054
	(0.090)	(0.214)	(0.188)	(0.140)	(0.155)	(0.142)
Observations	2,559	1,544	1,890	2,883	1,803	3,048

Notes: Every cell is one regression on the specification of Table D.11's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Extensive rows are linear probability models on an indicator for any planting (or any output) among district-years where the crop is observed; intensive rows are logs on the district-years that grow (produce) it. Columns are the six largest crops by the season's planted area, grown in at least forty districts. The sample is restricted to the districts the conflict Poisson retains — at least one located UCDP event over 1990–2025; columns are kept identical to the full-sample table for comparability. ***, **, and * denote 1%, 5%, and 10%.

Table D.9: Rabi crop margins on the conflict-estimation sample

	Wheat (1)	Chickpea (2)	Rape & mustard (3)	Jowar (4)	Maize (5)	Barley (6)
Area, extensive: any planting (0/1)						
log Upstream Precip. (t)	-0.040**	-0.037	-0.047*	-0.007	-0.030	-0.011
	(0.019)	(0.033)	(0.028)	(0.022)	(0.038)	(0.033)
log Downstream Precip. (t)	0.026**	-0.060*	0.034	-0.026	-0.027	0.006
	(0.013)	(0.031)	(0.022)	(0.025)	(0.026)	(0.023)
Observations	3,903	3,903	3,903	3,903	3,903	3,903
Mean of dependent variable	0.97	0.86	0.91	0.08	0.37	0.54
Area, intensive: log hectares, growing district-years						
log Upstream Precip. (t)	0.293**	-0.245	-0.345***	0.151	-0.291	0.072
	(0.122)	(0.153)	(0.113)	(0.581)	(0.362)	(0.133)
log Downstream Precip. (t)	-0.169**	0.130	-0.050	0.153	1.160***	-0.008
	(0.075)	(0.107)	(0.079)	(0.458)	(0.355)	(0.112)
Observations	3,785	3,373	3,564	299	1,451	2,107
Production, extensive: any output (0/1)						
log Upstream Precip. (t)	-0.040**	-0.038	-0.053**	-0.007	-0.028	-0.001
	(0.019)	(0.033)	(0.027)	(0.022)	(0.038)	(0.032)
log Downstream Precip. (t)	0.026**	-0.066**	0.029	-0.026	-0.026	-0.008
	(0.013)	(0.032)	(0.022)	(0.025)	(0.026)	(0.024)
Observations	3,903	3,903	3,903	3,903	3,903	3,903
Mean of dependent variable	0.97	0.85	0.91	0.08	0.37	0.53
Production, intensive: log tonnes, producing district-years						
log Upstream Precip. (t)	0.206	-0.282*	-0.416***	0.104	-0.216	0.091
	(0.140)	(0.167)	(0.112)	(0.509)	(0.359)	(0.141)
log Downstream Precip. (t)	-0.261***	0.111	-0.134	0.331	1.252***	-0.031
	(0.085)	(0.112)	(0.085)	(0.434)	(0.361)	(0.132)
Observations	3,785	3,330	3,557	298	1,448	2,073

Notes: Every cell is one regression on the specification of Table D.11's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Extensive rows are linear probability models on an indicator for any planting (or any output) among district-years where the crop is observed; intensive rows are logs on the district-years that grow (produce) it. Columns are the six largest crops by the season's planted area, grown in at least forty districts. The sample is restricted to the districts the conflict Poisson retains — at least one located UCDP event over 1990–2025; columns are kept identical to the full-sample table for comparability. Rabi jowar falls to 33 growing districts here, below the forty-district floor the full-sample selection imposes. ***, **, and * denote 1%, 5%, and 10%.

Table D.10: Kharif crop prices and the value of the harvest on the conflict-estimation sample

	Rice (1)	Cotton (2)	Bajra (3)	Maize (4)	Groundnut (5)	Pigeonpea (6)
Farm-harvest price, annual (log)						
log Upstream Precip. (t)	0.026	0.046	-0.033	-0.043	-0.045	0.106***
	(0.036)	(0.055)	(0.049)	(0.034)	(0.039)	(0.034)
log Downstream Precip. (t)	-0.080***	0.110*	-0.006	-0.004	0.047	0.026
	(0.023)	(0.064)	(0.038)	(0.027)	(0.041)	(0.043)
Observations	2,794	896	1,414	2,377	1,645	1,871
Revenue: price × production (log)						
log Upstream Precip. (t)	0.349**	0.222	-0.258	-0.175	-0.060	0.113
	(0.139)	(0.552)	(0.282)	(0.202)	(0.326)	(0.229)
log Downstream Precip. (t)	0.002	-0.132	0.331	0.273*	0.500***	0.487*
	(0.095)	(0.316)	(0.274)	(0.160)	(0.181)	(0.289)
Observations	1,043	473	689	1,083	743	1,172

Notes: Every cell is one regression on the specification of Table D.11's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Prices are the DLD farm-harvest series, which is annual — it is not reported by season — so the revenue rows pair the annual price with kharif production: roughly the value of the harvest, and gross revenue rather than profit, since input costs are not observed. Because $\log(pq) = \log p + \log q$, the revenue coefficient equals the sum of the price and production responses on its own sample; it will not equal the sum of the rows printed above it or in Table D.5, which are estimated on the larger samples their own margins allow. Both blocks condition on producing district-years with a price quote, so they are intensive margins. ***, **, and * denote 1%, 5%, and 10%. The sample is restricted to the districts the conflict Poisson retains — at least one located UCDP event over 1990–2025; columns are kept identical to the full-sample table for comparability.

Table D.11: Four sources of water in one regression

	Each outcome's own sample				Conflict-estimation districts				Mean of dep. var.	
	Upstream (1)	Local (2)	Neighbour (3)	Downstream (4)	Upstream (5)	Local (6)	Neighbour (7)	Downstream (8)	Own sample	Conflict districts
A. Conflict										
Conflict events (Poisson) $(t-1)$	1.144***	0.622	-1.413	-0.977***	1.144***	0.622	-1.413	-0.977***	1.06	1.06
	(0.324)	(0.910)	(1.260)	(0.282)	(0.324)	(0.910)	(1.260)	(0.282)		
B. District agriculture, prices and wages										
Kharif planted area (t)	-0.102	0.033	0.181	0.217***	-0.343***	0.063	0.372	0.337***		
	(0.071)	(0.168)	(0.206)	(0.056)	(0.128)	(0.268)	(0.343)	(0.087)		
Fixed-weight quantity index $(t-1)$	0.159*	-0.144	0.147	0.118**	0.006	-0.035	0.022	0.174**		
	(0.081)	(0.194)	(0.226)	(0.054)	(0.143)	(0.328)	(0.385)	(0.085)		
Implicit price index $(t-1)$	-0.057**	0.036	-0.047	0.029	-0.061**	0.061	-0.068	-0.009		
	(0.022)	(0.051)	(0.056)	(0.021)	(0.026)	(0.076)	(0.085)	(0.029)		
Fixed-weight price index (t)	-0.012	0.019	-0.042	0.001	-0.007	0.050	-0.050	-0.001		
	(0.021)	(0.045)	(0.058)	(0.016)	(0.031)	(0.071)	(0.094)	(0.021)		
Upstream districts' output (t)	0.269***	-0.010	0.099	-0.011	0.189***	0.009	0.172	-0.094**		
	(0.049)	(0.120)	(0.145)	(0.037)	(0.067)	(0.156)	(0.184)	(0.046)		
Real agricultural wage $(t-1)$	-0.127***	-0.004	0.026	0.094***	-0.108**	-0.045	0.135	0.087**		
	(0.041)	(0.077)	(0.100)	(0.032)	(0.054)	(0.095)	(0.127)	(0.038)		
C. IHDS households										
Farm income (asinh) (t)	0.305	-3.365*	2.620	3.242***	4.144	-4.808**	0.265	2.903**		
	(1.529)	(1.715)	(2.068)	(1.152)	(2.645)	(1.925)	(2.044)	(1.312)		
Non-farm income (asinh) $(t-1)$	-3.424**	1.472	1.183	-0.167	-5.905***	4.087	2.613	-0.472		
	(1.430)	(1.786)	(2.539)	(0.827)	(2.201)	(2.730)	(2.921)	(1.058)		
Receives any remittance $(t-1)$	0.075	0.038	-0.248*	-0.044	0.013	0.139	-0.325*	-0.015	0.11	0.10
	(0.064)	(0.118)	(0.131)	(0.044)	(0.104)	(0.155)	(0.187)	(0.070)		
Consumption (log) (t)	-0.650***	0.559**	-0.077	0.110	0.213	-0.316	0.202	-0.289*		
	(0.177)	(0.217)	(0.247)	(0.111)	(0.396)	(0.339)	(0.280)	(0.163)		
Districts, conflict row			177				177			
District & Year (or Wave) FE			✓				✓			

Notes: Every row is one regression carrying all four rainfall measures at once, each entered contemporaneously and lagged, with upstream and local temperatures and their lags, district and year (or survey-wave) fixed effects, and standard errors clustered on district. Each row reports one horizon, marked against its label: $(t-1)$ for the lagged term and (t) for the contemporaneous one. Crop quantities and the prices they set are read contemporaneously; conflict, wages and the slower household margins are read at $t-1$. Upstream is the river-distance-weighted catchment draining into the district excluding the district and its neighbours; downstream mirrors it exactly, on the catchment the district drains into; neighbour is the row-normalised queen-contiguity average of local rainfall. Columns 1–4 estimate each outcome on whatever districts it supports; columns 5–8 restrict every outcome to the 177 districts the conflict Poisson retains. Local and neighbour rainfall correlate 0.964 after fixed effects and carry variance inflation factors of 14.0 and 16.4, so neither is separately identified; upstream and downstream carry 2.50 and 2.26.

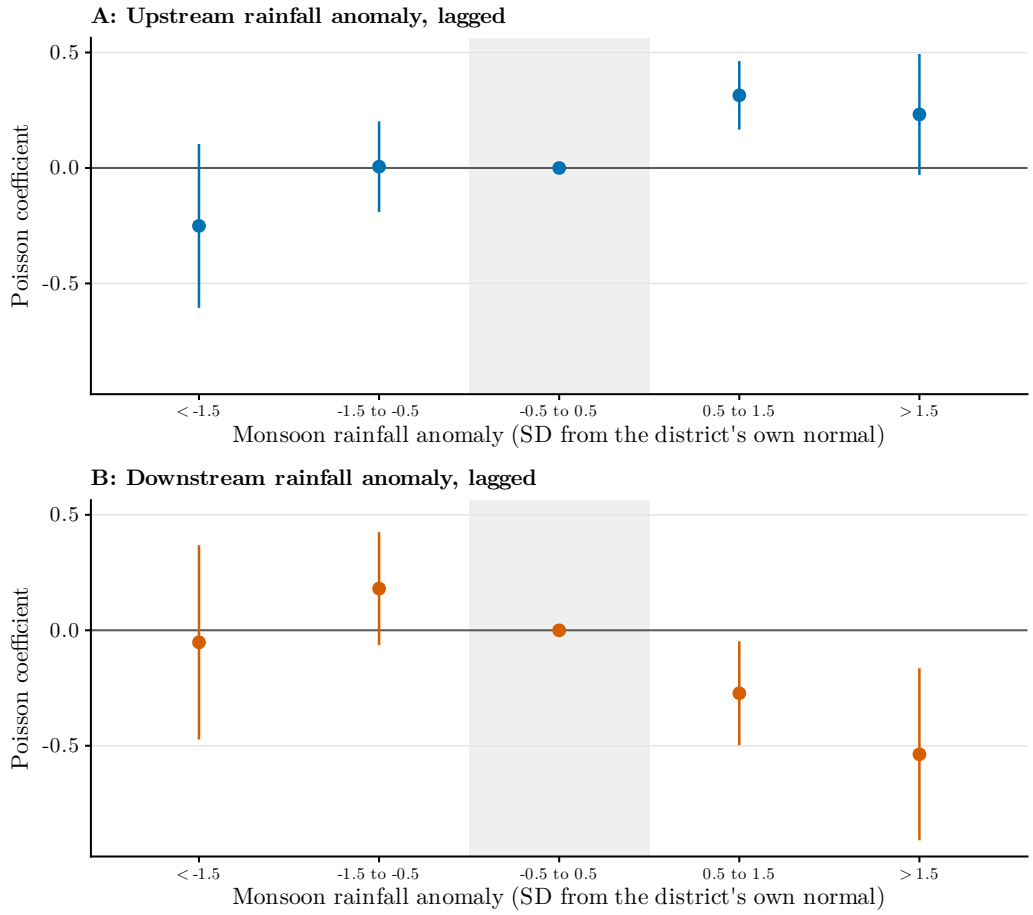


Figure D.3: The dose–response in upstream and downstream rainfall anomalies

Notes: One fixed-effects Poisson regression carrying both sources as five-bin functions of their own standardised monsoon anomaly, with the middle bin (−0.5 to 0.5 standard deviations, shaded) omitted, so every coefficient reads against a normal year. Both terms are lagged, as in the headline. Controls are the local rainfall anomaly and the upstream, downstream and local temperature anomalies, each current and lagged, with district and year effects; standard errors are clustered on district and bars are 95 percent intervals. 6,372 district-years over 177 districts.

Table D.12: Conflict and the mechanism outcomes in rainfall anomalies

	Conflict events (1)	Planted area (2)	Crops planted (3)	Quantity index (4)	Implicit price (5)	Real wage (6)	Nominal wage (7)
Upstream anomaly, contemporaneous (t)							
< -1.5 SD	0.231	-0.084	0.126**	-0.163**	0.016	-0.042	-0.073***
	(0.197)	(0.056)	(0.058)	(0.064)	(0.015)	(0.027)	(0.023)
-1.5 to -0.5	-0.118	0.097***	0.094***	0.061**	0.017**	-0.003	-0.012
	(0.118)	(0.026)	(0.029)	(0.029)	(0.007)	(0.011)	(0.009)
0.5 to 1.5	0.059	0.018	-0.060**	0.041	0.004	-0.033**	-0.008
	(0.138)	(0.031)	(0.027)	(0.033)	(0.008)	(0.015)	(0.014)
> 1.5 SD	0.217	0.045	-0.113***	-0.006	0.033**	-0.058**	-0.035
	(0.189)	(0.045)	(0.039)	(0.046)	(0.013)	(0.027)	(0.021)
Upstream anomaly, lagged ($t-1$)							
< -1.5 SD	-0.289*	-0.012	-0.011	0.013	0.023	0.027	-0.020
	(0.162)	(0.071)	(0.053)	(0.073)	(0.014)	(0.028)	(0.022)
-1.5 to -0.5	-0.048	0.032	0.036	0.004	0.017**	0.002	-0.009
	(0.105)	(0.024)	(0.026)	(0.028)	(0.007)	(0.012)	(0.010)
0.5 to 1.5	0.368***	0.069**	-0.018	0.050	-0.017**	-0.025	-0.010
	(0.095)	(0.029)	(0.025)	(0.032)	(0.008)	(0.016)	(0.013)
> 1.5 SD	0.229	0.109**	-0.115**	0.063	0.018	-0.056**	-0.049**
	(0.154)	(0.047)	(0.049)	(0.051)	(0.012)	(0.027)	(0.023)
Downstream anomaly, contemporaneous (t)							
< -1.5 SD	-0.296	-0.158***	-0.151**	-0.123***	0.020	-0.024	0.017
	(0.201)	(0.042)	(0.065)	(0.044)	(0.015)	(0.024)	(0.020)
-1.5 to -0.5	0.307***	-0.112***	-0.079***	-0.128***	0.031***	-0.018	0.008
	(0.106)	(0.022)	(0.028)	(0.027)	(0.007)	(0.014)	(0.012)
0.5 to 1.5	-0.064	0.037*	0.044**	0.062***	0.002	0.002	-0.006
	(0.111)	(0.022)	(0.022)	(0.023)	(0.007)	(0.013)	(0.010)
> 1.5 SD	-0.221*	0.192***	0.136***	0.195***	0.030**	0.008	-0.004
	(0.127)	(0.053)	(0.041)	(0.053)	(0.012)	(0.023)	(0.018)
Downstream anomaly, lagged ($t-1$)							
< -1.5 SD	-0.043	0.033	-0.085	0.048	-0.019	-0.030	-0.006
	(0.216)	(0.041)	(0.055)	(0.048)	(0.014)	(0.025)	(0.022)
-1.5 to -0.5	0.199*	-0.051**	-0.029	-0.038	0.016**	-0.021	-0.006
	(0.112)	(0.023)	(0.026)	(0.027)	(0.007)	(0.015)	(0.013)
0.5 to 1.5	-0.291**	0.031	-0.048**	0.045*	0.022***	0.000	0.006
	(0.141)	(0.020)	(0.024)	(0.024)	(0.008)	(0.014)	(0.011)
> 1.5 SD	-0.475***	0.177***	0.070	0.147**	0.022	0.028	0.017
	(0.169)	(0.052)	(0.052)	(0.057)	(0.013)	(0.025)	(0.021)
Estimation	Poisson	OLS	OLS	OLS	OLS	OLS	OLS
Observations	6,372	8,079	8,300	8,079	5,806	3,723	3,723
Districts	177	341	342	341	263	276	276
Mean of dependent variable	1.06		4.01				

Notes: Every column is one regression in which upstream, downstream and local monsoon rainfall enter as unrestricted bin functions of each basin-set's own standardised anomaly — the deviation from the district's long-run mean divided by its own standard deviation — at both horizons jointly, with contemporaneous and lagged temperature anomalies as controls and district and year fixed effects; standard errors are clustered on district. The normal bin (-0.5 to 0.5 SD) is omitted, so every coefficient reads against a normal year; local rainfall's bins are controls and are not shown. Mechanism outcomes use the six-crop constructions of Table D.11 on each outcome's own sample, 1997–2022, by OLS; conflict is the fixed-effects Poisson on located events over 1990–2025. Crops planted counts how many of the six kharif crops have positive area, on district-years observing all six. ***, **, and * denote 1%, 5%, and 10%.

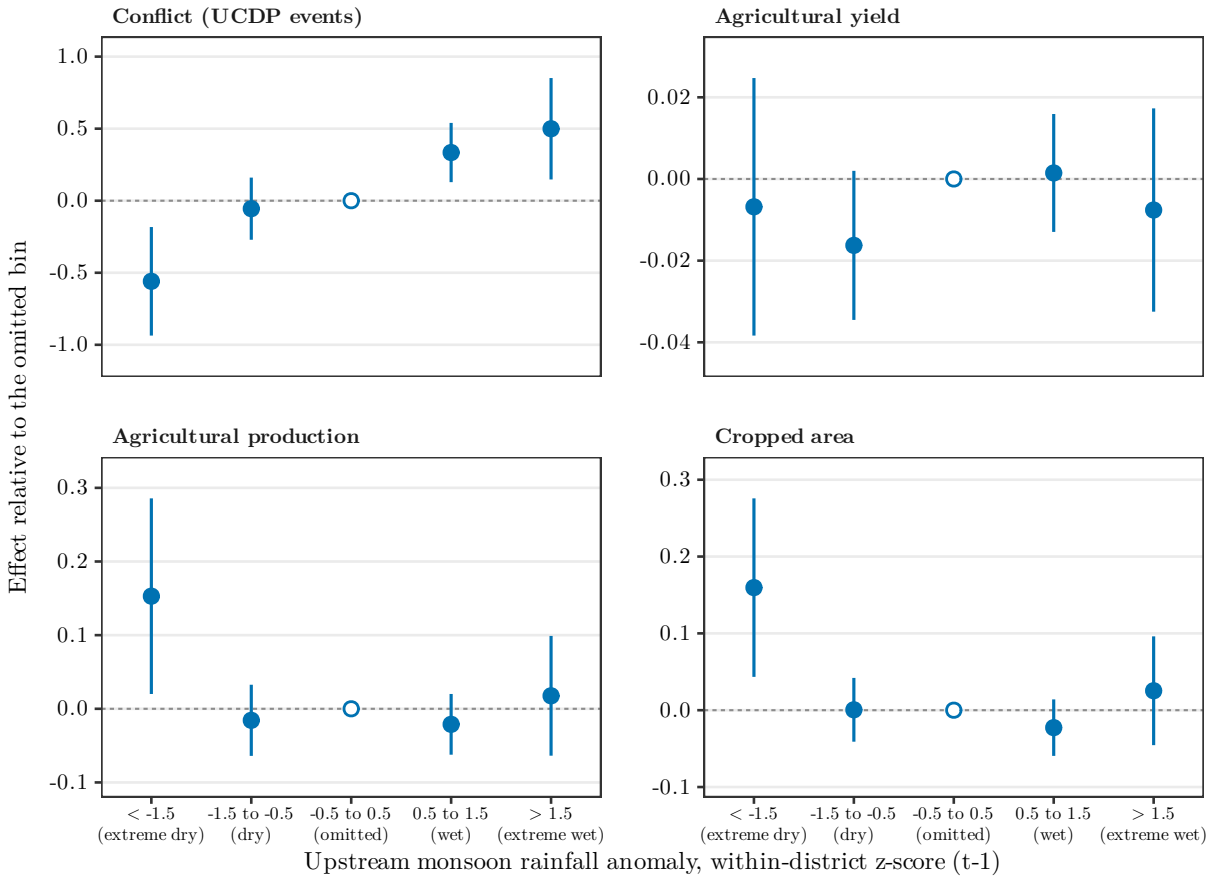


Figure D.4: Conflict and agriculture across the rainfall anomaly

Notes: Points are coefficients on bins of the lagged upstream rainfall anomaly, relative to the omitted −0.5 to 0.5 bin, with 95 percent cluster-by-district intervals. The anomaly is a within-district z-score computed on each district's own annual history. The conflict panel is Poisson on located events over 1990–2025 under the specification of Table 2, column 4 — the same sample of 8,280 district-years and 230 districts as the headline. The agricultural panels are OLS on log outcomes over the years those series cover.

Table D.13: The wage response is a multi-year adjustment

	Real wage (1)	Nominal wage (2)
log Upstream Precip. $(t-1)$	-0.065	-0.050*
	(0.042)	(0.031)
log Upstream Precip. $(t-2)$	-0.046**	-0.012
	(0.023)	(0.023)
log Upstream Precip. $(t-3)$	-0.035	-0.037
	(0.028)	(0.024)
log Upstream Precip. $(t-4)$	0.034	-0.010
	(0.030)	(0.025)
log Upstream Precip. $(t-5)$	-0.012	-0.020
	(0.029)	(0.024)
Cumulative, $\sum_{k=1}^{5}$	-0.123	-0.129
	(0.103)	(0.090)
Observations	3,872	3,872
Districts	276	276

Notes: Five annual lags of log upstream monsoon rainfall entered together, with local, neighbour and downstream rainfall and the temperature controls as in Table D.2, district and year fixed effects, and standard errors clustered on district. The cumulative row is the sum of the five lags with the standard error of that sum. ***, **, and * denote 1%, 5%, and 10%.

Table D.16: The four sources on the Economic Census, three rounds

	Upstream (1)	Local (2)	Neighbour (3)	Downstream (4)
Establishments	0.256*	-0.459	-0.070	0.124
	(0.134)	(0.344)	(0.463)	(0.174)
Non-farm employment	0.222	-0.115	-0.057	0.168
	(0.136)	(0.357)	(0.496)	(0.186)
Manufacturing employment	-0.140	-0.550	0.897	-0.577**
	(0.210)	(0.628)	(0.844)	(0.265)
Services employment	0.305**	0.011	-0.395	0.346**
	(0.130)	(0.303)	(0.435)	(0.170)
Observations		1,080		
Districts		360		
District & Round FE		✓		

Notes: The specification of Table D.11 on the Economic Census. Each round's treatment is the mean over the years between it and the previous round, and the lagged term is the mean over the window before that. The 1990 round is excluded. The panel is balanced across the three remaining rounds.

Table D.14: Labour supply and earnings after an upstream shock: NSS agricultural margins, by fieldwork quarter

	Any ag. wage work (1)	Days, ag. wage work (2)	Ag. wage earnings (3)	All wage earnings (4)
All households				n = 547,606
log Upstream Precip. (t)	+0.0160**	+0.1012**	+0.1143**	-0.0591
	(0.0076)	(0.0456)	(0.0476)	(0.0819)
log Upstream Precip. ($t-1$)	+0.0061	+0.0182	+0.0828	+0.0842
	(0.0091)	(0.0545)	(0.0562)	(0.1098)
Mean of dependent variable	0.05	0.26		
Interviewed Jul–Sep, during the monsoon				n = 137,591
log Upstream Precip. (t)	+0.0133	+0.1026	+0.0996	-0.1628
	(0.0136)	(0.0772)	(0.0870)	(0.1367)
log Upstream Precip. ($t-1$)	-0.0109	-0.0885	-0.0424	-0.1817
	(0.0156)	(0.0910)	(0.0978)	(0.1692)
Mean of dependent variable	0.05	0.29		
Interviewed Oct–Dec, kharif harvest (+2 months)				n = 137,619
log Upstream Precip. (t)	+0.0153	+0.0818	+0.1040	+0.0198
	(0.0130)	(0.0794)	(0.0818)	(0.1361)
log Upstream Precip. ($t-1$)	+0.0025	-0.0344	+0.0683	+0.3293*
	(0.0162)	(0.0965)	(0.1021)	(0.1701)
Mean of dependent variable	0.05	0.28		
Interviewed Jan–Mar (+5 months)				n = 135,937
log Upstream Precip. (t)	+0.0070	+0.0526	+0.0572	-0.2774*
	(0.0121)	(0.0726)	(0.0798)	(0.1452)
log Upstream Precip. ($t-1$)	+0.0029	+0.0163	+0.0395	-0.2410
	(0.0154)	(0.0942)	(0.0993)	(0.1784)
Mean of dependent variable	0.05	0.25		
Interviewed Apr–Jun, following year (+8 months)				n = 136,451
log Upstream Precip. (t)	+0.0283***	+0.1671***	+0.1969***	+0.1559
	(0.0103)	(0.0642)	(0.0664)	(0.1453)
log Upstream Precip. ($t-1$)	+0.0295**	+0.1776**	+0.2655***	+0.4037**
	(0.0133)	(0.0811)	(0.0864)	(0.1740)
Mean of dependent variable	0.04	0.23		

Notes: NSS Employment–Unemployment rounds 61, 66 and 68 (2004–05, 2009–10, 2011–12), working-age individuals, on the four sources of water — upstream, local, neighbouring and downstream rainfall — with temperatures, each at the contemporaneous and lagged monsoon, district and round fixed effects, individual age, sex and education, and standard errors clustered on district. Each round follows exactly one monsoon — 2004, 2009 and 2011 — which is the contemporaneous term. THE REFERENCE PERIOD IS ONE WEEK: days run 0–7. Earnings are wage and salary receipts for that same week in asinh, which keeps the structural zeros but prices the jump from zero in rupees, so the coefficients read as sign and significance rather than as elasticities. NSS allocates households to fieldwork quarters BY DESIGN, so the panels compare households drawn to be alike rather than households that selected their timing. The quarters sit at different distances from the SAME monsoon, and the panel headings give that distance: July–September is surveyed during it and April–June eight months after it ended, in the following calendar year — every panel is post-treatment. ***, **, and * denote 1%, 5%, and 10%.

Table D.15: The non-farm margin after an upstream shock: NSS, by fieldwork quarter

	Casual non-farm			Self-employment		Regular salaried
	Any work	Days	Earnings	Any	Days	Earnings
	(1)	(2)	(3)	(4)	(5)	(6)
All households						n = 547,606
log Upstream Precip. (t)	-0.0155	-0.0884	-0.1172*	+0.0215	+0.2485*	-0.0600
	(0.0100)	(0.0663)	(0.0670)	(0.0226)	(0.1472)	(0.0706)
log Upstream Precip. ($t-1$)	-0.0298***	-0.1855**	-0.2033**	+0.0330	+0.2474	+0.1959**
	(0.0115)	(0.0736)	(0.0803)	(0.0239)	(0.1549)	(0.0783)
Mean of dependent variable	0.06	0.36		0.33	2.07	
Interviewed Jul–Sep, during the monsoon						n = 137,591
log Upstream Precip. (t)	-0.0294*	-0.1320	-0.2241**	+0.0118	+0.1660	-0.0561
	(0.0162)	(0.1024)	(0.1013)	(0.0318)	(0.2081)	(0.1059)
log Upstream Precip. ($t-1$)	-0.0142	-0.0844	-0.0812	+0.0642*	+0.5585**	-0.0697
	(0.0140)	(0.0874)	(0.0973)	(0.0375)	(0.2338)	(0.1222)
Mean of dependent variable	0.05	0.32		0.35	2.14	
Interviewed Oct–Dec, kharif harvest (+2 months)						n = 137,619
log Upstream Precip. (t)	-0.0127	-0.0900	-0.0875	+0.0169	+0.2248	+0.0058
	(0.0120)	(0.0755)	(0.0845)	(0.0312)	(0.2052)	(0.1012)
log Upstream Precip. ($t-1$)	-0.0337**	-0.1813*	-0.2085*	+0.0067	+0.0527	+0.4547***
	(0.0154)	(0.0946)	(0.1069)	(0.0320)	(0.2095)	(0.1374)
Mean of dependent variable	0.06	0.35		0.33	2.09	
Interviewed Jan–Mar (+5 months)						n = 135,937
log Upstream Precip. (t)	-0.0254*	-0.1556	-0.1920*	+0.0603**	+0.4901***	-0.1438
	(0.0150)	(0.0982)	(0.1046)	(0.0288)	(0.1737)	(0.1138)
log Upstream Precip. ($t-1$)	-0.0455***	-0.2740**	-0.3339***	+0.0671**	+0.4614**	+0.0489
	(0.0170)	(0.1097)	(0.1188)	(0.0340)	(0.2061)	(0.1218)
Mean of dependent variable	0.06	0.39		0.33	2.05	
Interviewed Apr–Jun, following year (+8 months)						n = 136,451
log Upstream Precip. (t)	+0.0050	+0.0196	+0.0322	+0.0010	+0.1448	-0.0717
	(0.0139)	(0.0900)	(0.0993)	(0.0322)	(0.2065)	(0.1240)
log Upstream Precip. ($t-1$)	-0.0265	-0.2026	-0.1942	-0.0025	-0.0634	+0.3281**
	(0.0203)	(0.1259)	(0.1422)	(0.0363)	(0.2489)	(0.1482)
Mean of dependent variable	0.06	0.39		0.31	2.00	

Notes: NSS Employment–Unemployment rounds 61, 66 and 68 (2004–05, 2009–10, 2011–12), working-age individuals, on the four sources of water — upstream, local, neighbouring and downstream rainfall — with temperatures, each at the contemporaneous and lagged monsoon, district and round fixed effects, individual age, sex and education, and standard errors clustered on district. Each round follows exactly one monsoon — 2004, 2009 and 2011 — which is the contemporaneous term. THE REFERENCE PERIOD IS ONE WEEK: days run 0–7. Earnings are wage and salary receipts for that same week in asinh, which keeps the structural zeros but prices the jump from zero in rupees, so the coefficients read as sign and significance rather than as elasticities. NSS allocates households to fieldwork quarters BY DESIGN, so the panels compare households drawn to be alike rather than households that selected their timing. The quarters sit at different distances from the SAME monsoon, and the panel headings give that distance: July–September is surveyed during it and April–June eight months after it ended, in the following calendar year — every panel is post-treatment. ***, **, and * denote 1%, 5%, and 10%. Self-employment days are unconditional and include own-cultivation, whose EARNINGS the Employment–Unemployment schedule does not collect.

Table D.17: Nightlights and the four sources of water, VIIRS

	Mean light (1)	Total light (2)
log Upstream Precip. (t)	-0.043	-0.041
	(0.050)	(0.050)
log Upstream Precip. $(t-1)$	0.072	0.070
	(0.056)	(0.057)
log Local Precip. (t)	0.067	0.059
	(0.070)	(0.069)
log Local Precip. $(t-1)$	-0.023	-0.029
	(0.066)	(0.066)
log Downstream Precip. (t)	0.240**	0.241**
	(0.096)	(0.096)
log Downstream Precip. $(t-1)$	0.251**	0.252**
	(0.098)	(0.099)
Observations	3,240	3,240
Number of Districts	360	360
Estimation	OLS	OLS
District & Year FE	✓	✓

Notes: District-year panel, 2013 – 2021 — the VIIRS record. The outcome is the log of district mean radiance in column 1 and of total radiance in column 2. Each column enters all four sources of water under the specification of Table 2: upstream, local and downstream monsoon rainfall, current and lagged, with their temperature controls and the spatial block is deliberately ABSENT here. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table D.18: Is the downstream effect a dam-release effect?

A. Downstream rainfall interacted with dam presence	Downstream rain, no dam (1)	× dam upstream (2)	Implied effect with a dam (3)	Mean of dep. var.
Conflict events (Poisson)	-1.695***	1.419**	-0.276	1.06
	(0.387)	(0.600)	—	
Kharif cereal area	0.249***	0.130	0.379	
	(0.088)	(0.116)	—	
Kharif cereal production	0.150	0.315**	0.465	
	(0.098)	(0.131)	—	
Real agricultural wage	0.075**	0.032	0.107	
	(0.036)	(0.033)	—	

B. The same, with an agricultural interaction alongside	Downstream rain (1)	× dam upstream (2)	× agriculture (3)	Mean of dep. var.
Irrigated share, 2000	-1.697***	1.412**	-0.195	1.06
	(0.405)	(0.597)	(0.313)	
Cropped-area share, 1990s	-1.660***	1.429**	-0.215	1.06
	(0.391)	(0.582)	(0.241)	
Districts	177, of which 68% have a dam			
District & Year FE	✓			

Notes: Panel A adds an interaction between downstream rainfall (both horizons) and an indicator for any irrigation dam completed by 1989 in the district's upstream catchment, to the four-source specification of Table D.11. Main effects are retained. The moderator is time-invariant, so district effects absorb its level. In Panel B both interactions are entered together, with the agricultural moderator standardised so the main term is the effect at its mean.

D.3. Public Employment

- **Does the employment guarantee absorb the shock?** No: it moderates the effect of local rain, not of upstream rain (Tables D.19 and D.20).

Table D.19: The employment guarantee and local rainfall: replication and measure swaps

	Rain, pre-NREGA	× NREGA	Rain, post-NREGA	Districts	Mean of dep. var.
His events, his rainfall	−1.402*** (0.276)	+1.298*** (0.329)	−0.103 (0.158)	239	4.55
His events, CHIRPS rainfall	−1.328*** (0.248)	+1.539*** (0.310)	+0.211 (0.161)	239	4.55
UCDP events, his rainfall	−0.632 (0.413)	+0.845* (0.493)	+0.213 (0.251)	211	1.56
UCDP events, CHIRPS rainfall	−1.477*** (0.361)	+1.698*** (0.425)	+0.220 (0.234)	211	1.56

Notes: Fetzer (2020)'s reduced form — local monsoon rainfall interacted with district NREGA availability, his sample and fixed effects — re-estimated here, then with his TRMM rainfall replaced by this paper's CHIRPS measure, his news-based event counts replaced by UCDP, and both. The moderation replicates in each: rainfall's pacifying effect before the programme, absent after. The post-programme interval bounds what can be claimed — it rules out a post effect half the pre-programme size but not a quarter of it, so the programme roughly halves the local rainfall–conflict relationship rather than eliminating it. Standard errors clustered on district.

Table D.20: NREGA and the two rainfall shocks under tightening identification

	Local rainfall		Upstream rainfall		Mean of dep. var.	
	× NREGA	Rain, post	× NREGA	Rain, post	Local	Upstream
A. Post-programme interaction, 1990–2025						
District + year	+0.511(0.394)	+0.178(0.268)	+1.552(0.388) ∗ ∗∗	+1.290(0.350) ∗ ∗∗	0.74	0.76
+ phase × year	+0.486(0.462)	+0.013(0.283)	+1.828(0.431) ∗ ∗∗	+1.239(0.326) ∗ ∗∗	0.74	0.77
+ state × year	+0.436(0.240)∗	+0.165(0.324)	+0.294(0.353)	+0.719(0.378)∗	1.12	1.20
+ state × phase × year	+0.299(0.343)	+0.093(0.315)	+0.405(0.290)	+0.644(0.450)	1.49	1.60
B. Staggered roll-out: NREGA × rain within region-year						
Full window	−0.028(0.044)		−0.077(0.073)		0.75	0.81
2004–2010	−0.057(0.034)∗		−0.081(0.057)		2.24	2.49
2006–2008, staggering live	−0.063(0.033)∗		−0.088(0.036) ∗ ∗		2.97	3.10

Notes: Panel A interacts each rainfall source (lagged, logged) with an indicator for NREGA being active in the district, on the fixed-effects Poisson of Table 2, and tightens the fixed effects one step at a time; shown are the interaction and the implied post-programme rain coefficient. Under district and year effects the upstream interaction is large — but "after NREGA" is also simply "later." Panel B removes "later" entirely: region × year effects with phase-specific rain slopes, so the interaction is identified only from roll-out timing within the same region and year. What survives is near zero for both sources — the programme neither amplifies nor absorbs either shock on this margin. The 2006–2008 window is where phases 1–3 differ within a year; its sample is small and the coefficients are correspondingly fragile. Standard errors clustered on district. The elasticity pair the text quotes (1.344 to 1.252) is the upstream response before and after within region-year, from the same specification as Panel B.

Appendix E. Additional Figures and Tables

Exhibits are grouped by the main-text section they support; each carries its own notes.

E.1. Descriptive Figures

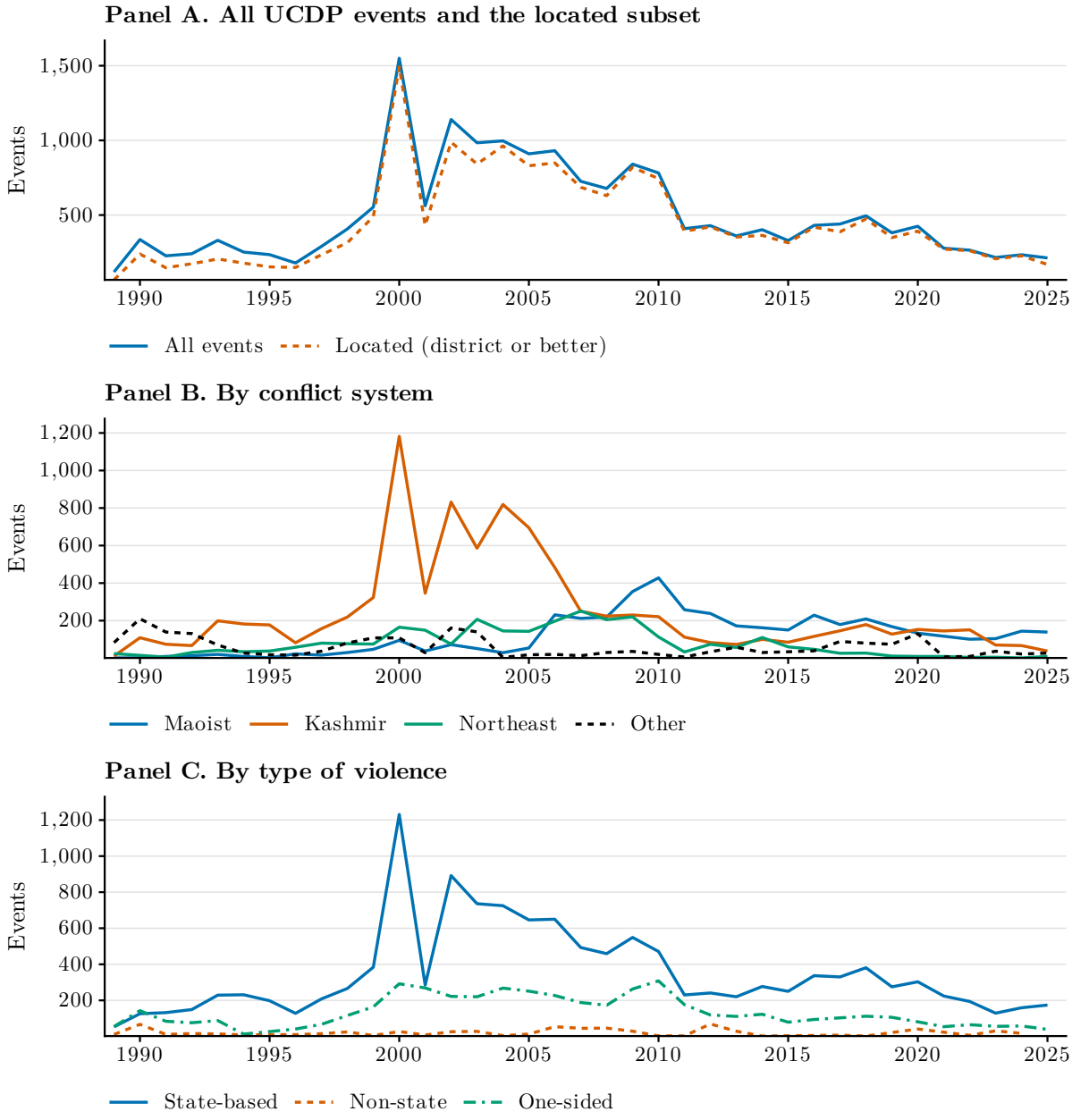


Figure E.1: UCDP conflict events in India, 1989–2025

Notes: Panel A plots all events against the subset UCDP locates to a district or better, the outcome used throughout the paper. Panel B splits all events by conflict system, assigned from the UCDP conflict name (Maoist, Kashmir, Northeast, and a residual holding India–Pakistan interstate events, communal non-state violence, and unaffiliated one-sided violence). Panel C splits them by UCDP's type of violence.

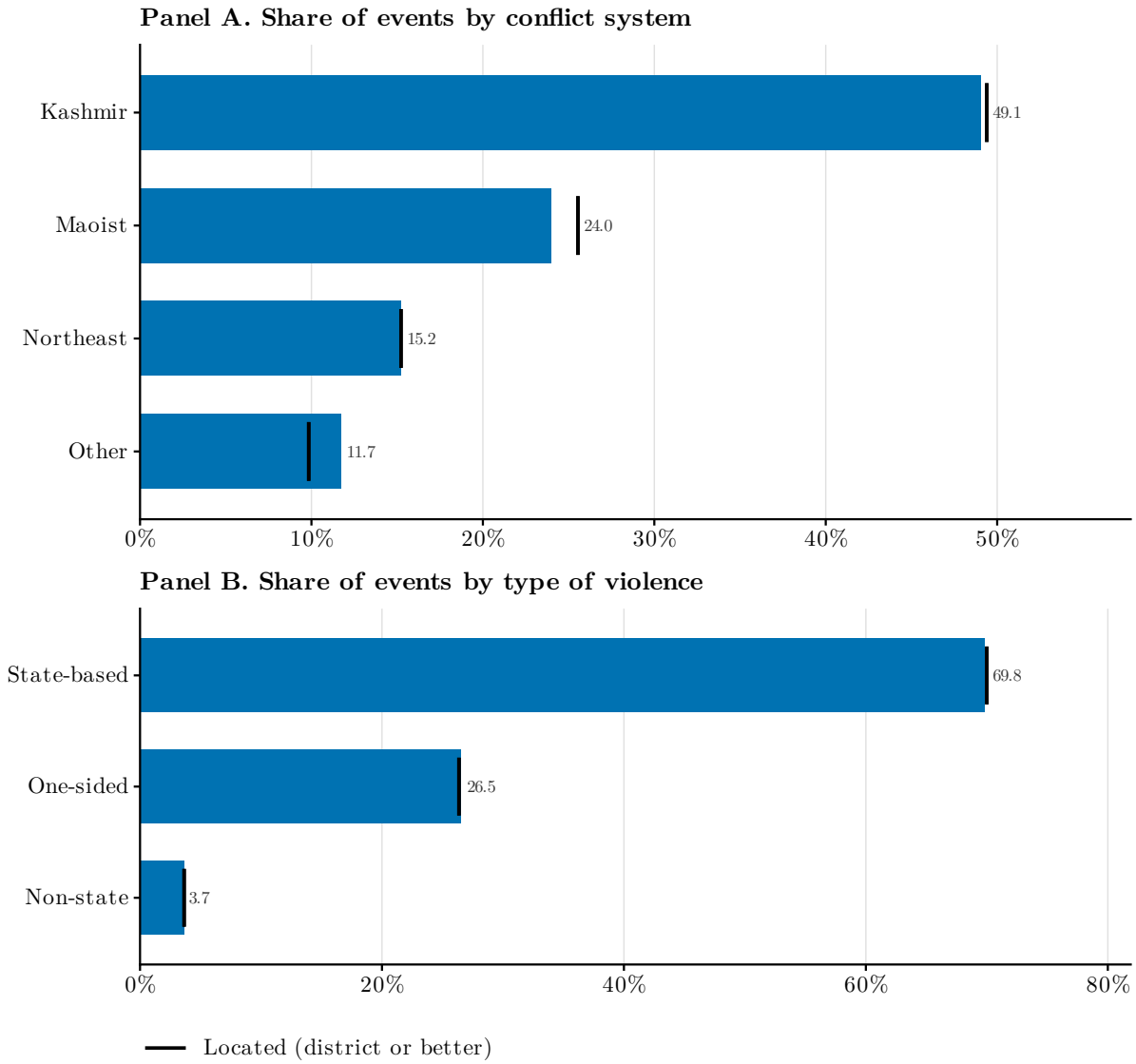


Figure E.2: Composition of UCDP events in India, 1989–2025

Notes: Bars give each category's share of all events over the full record; the tick on each bar gives its share of the located subset the estimation uses. Bars and ticks coincide to within 1.9 percentage points everywhere. Categories are as in Figure E.1.

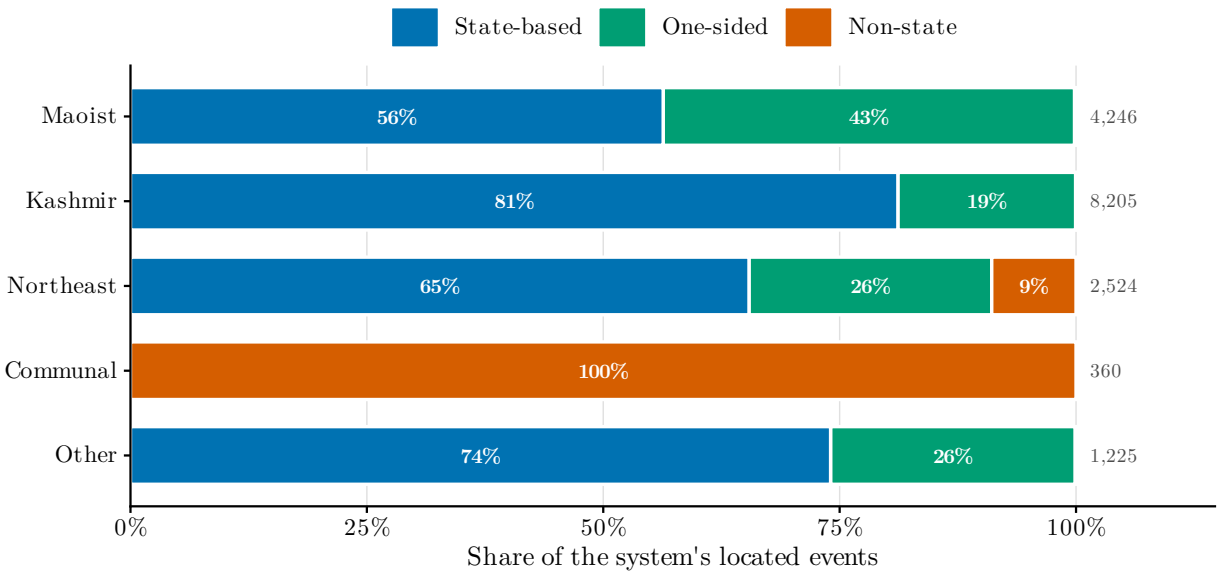


Figure E.3: Mix of violence types within each conflict system, located events, 1990–2025

Notes: Bars are each system's located events divided among UCDP's three types of violence and sum to 100 percent; the count beside each bar is the events it rests on. Among the three systems the paper estimates, Maoist violence is the most evenly divided (56.3 percent state-based against 43.5 percent one-sided), which is why its pooled count moves least: the two components respond in opposite directions and largely cancel. Kashmir is 81.2 percent state-based and the Northeast 65.4 percent. Restricting to located events changes no cell of this table by more than 2.9 percentage points, so the composition here is not an artifact of the precision rule.

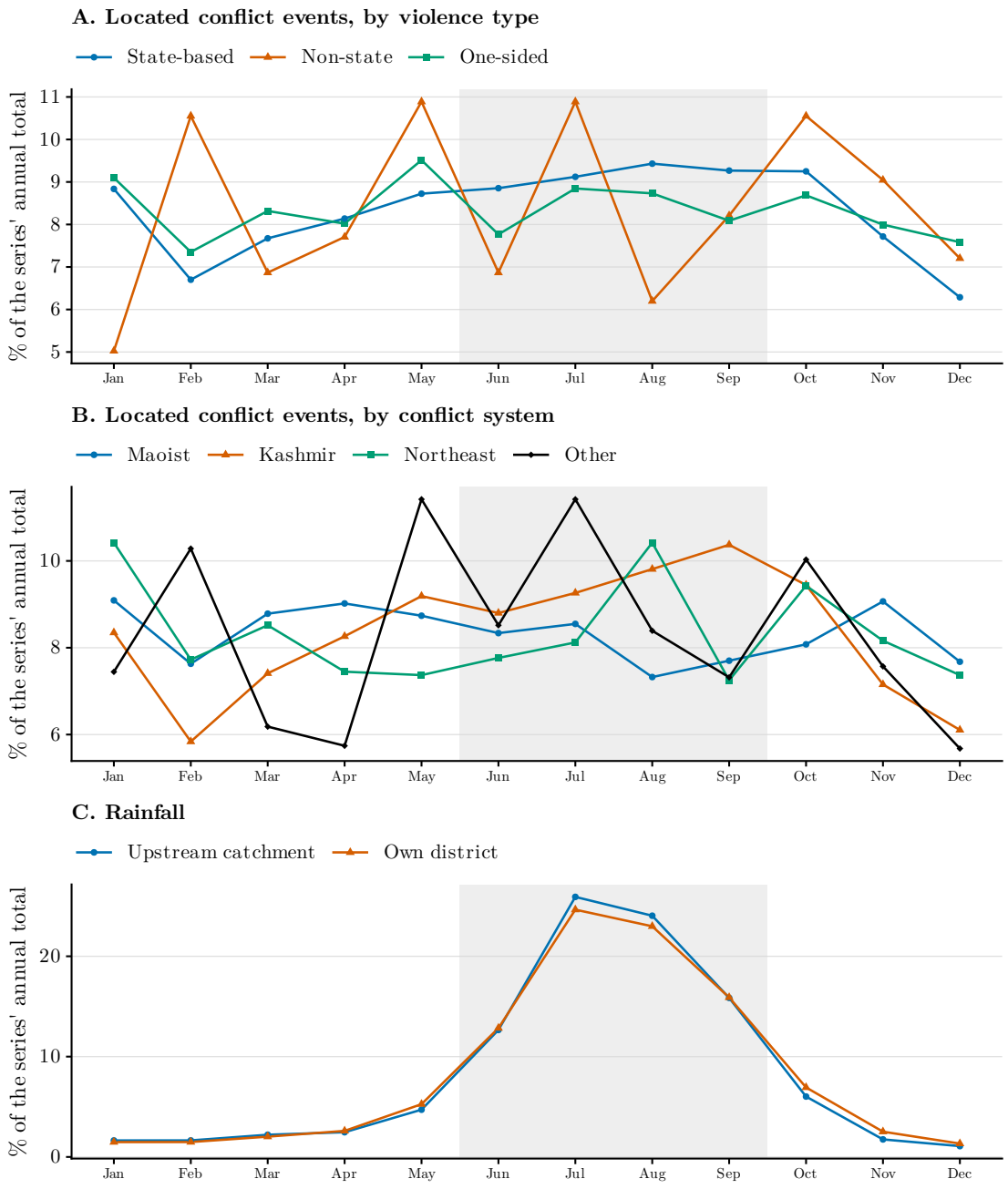


Figure E.4: Monthly composition of conflict and rainfall, 1990–2025

Notes: Each series is normalised to sum to 100 percent across the twelve calendar months, so the panels show *when* each series occurs rather than how large it is. Panels A and B use located UCDP events; violence types are UCDP's state-based, non-state and one-sided categories, and conflict systems are the paper's Maoist, Kashmir and Northeast groupings with everything else pooled into Other. Panel C rebuilds upstream rainfall from the basin-month series through the same crosswalk and the same river-distance weights that the annual treatment uses; district monthly shares are computed first and then averaged across districts. The shaded band marks June–September.

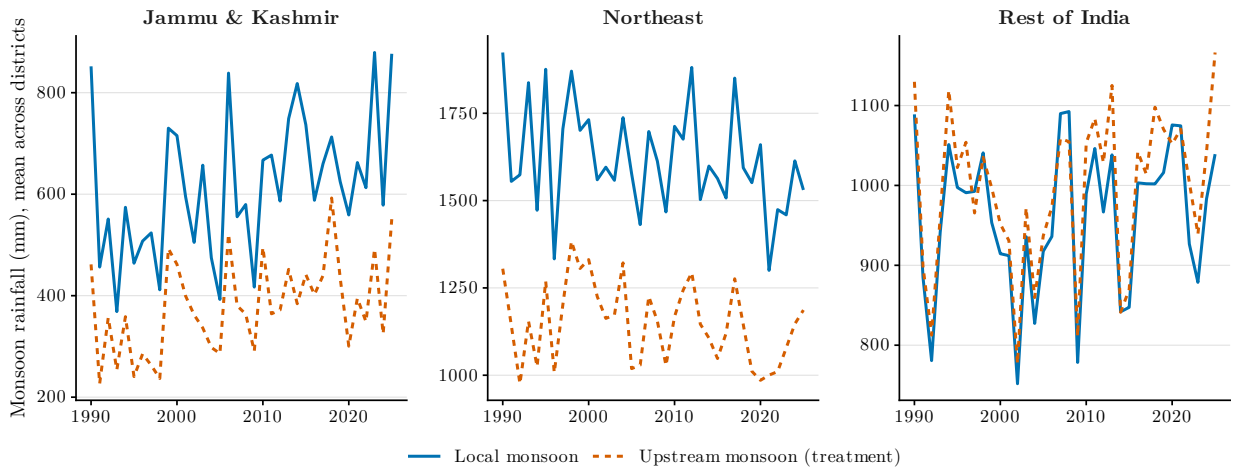


Figure E.5: Monsoon rainfall over time, local and upstream, by region

Notes: Yearly means across the estimation-sample districts of the two rainfall variables the regressions use: the district's own monsoon total and the river-distance-weighted monsoon total over its upstream catchment. Regions are assigned by state (Jammu & Kashmir; the seven Northeastern states; the rest).

E.2. Main-Result Robustness

Every column of Table 2 survives state × year fixed effects (Table E.1). They absorb anything common to a state in a year and leave 36 percent of the within-district variation in lagged upstream rainfall (60 percent once neighbours' weather is held fixed); on what remains the elasticity is 1.39 to 1.62, significant at the 1 percent level under every clustering and spatial cutoff I try and in a within-state randomization test.

The routed-water race does not survive the same test. Table E.2 repeats Table 3 with state × year effects: the lagged upstream rainfall elasticity is 1.52 to 1.77 and significant at the 1 percent level in every column, while no routed-discharge coefficient exceeds 0.27 in absolute value or comes within sight of significance (smallest $p = 0.13$). Whatever the river adds in Table 3 is common to a state in a year; the rainfall effect is not.

The response is not stationary within a conflict system. Figure E.6 re-estimates Table 2, column 4, on every 11-year window of the sample, for all conflict and for each system's own count, with the year's located events beneath. The full-window estimates are 1.21 (standard error 0.75) for Maoist violence, 0.41 (0.22) for Kashmir and 1.76 (0.60) for the Northeast (Table E.5). All conflict stays positive throughout, between 0.83 (window centred on 2015) and 2.12. Maoist violence does not respond before the 2004 merger that formed CPI (Maoist) (-0.44 in the window centred on 2000), rises to 3.80 by 1995, and returns to zero as the insurgency recedes after 2016. Kashmir responds in the insurgency's first phase (1.55 centred on 1995) and turns negative after the 2003 ceasefire, when its events thin to a few valley districts: 26 of its 26 windows rest on fewer than 20 districts and are drawn hollow. The Northeast strengthens over the period, reaching 2.21 by 2015. Windows overlap, so neighbouring estimates share most of their data and the figure reads as a trend, not as independent tests; Figure E.7 shows that 9- and 15-year windows trace the same paths.

The upstream effect is not specific to UCDP. Table E.7 estimates the paper's regression on a second corpus, the South Asia Terrorism Portal incident counts assembled by Fetzer (2020), on one sample of district-years (2000–2012), beside the UCDP outcomes aggregated to the same

Table E.1: Upstream rainfall and conflict, with state × year fixed effects

	No spatial controls	Spatial controls			
	(1)	(2)	(3)	(4)	(5)
log Upstream Precip. (t)	0.174	0.175	0.140	0.127	0.217
	(0.433)	(0.403)	(0.441)	(0.425)	(0.338)
log Upstream Precip. ($t-1$)	1.388***	1.476***	1.489***	1.562***	1.619***
	(0.335)	(0.370)	(0.404)	(0.433)	(0.416)
log Local Precip. (t)	0.021	0.041	-0.341	-0.397	-0.647
	(0.364)	(0.308)	(0.606)	(0.549)	(0.530)
log Local Precip. ($t-1$)	-0.347	-0.337	0.303	0.267	0.180
	(0.315)	(0.337)	(0.968)	(1.000)	(1.026)
log Neighbours' Conflict ($t-1$)		0.391***		0.376**	0.316
		(0.138)		(0.152)	(0.197)
log Neighbours' Local Precip. (t)			0.749	0.877	1.327
			(1.056)	(1.042)	(1.011)
log Neighbours' Local Precip. ($t-1$)			-1.110	-1.019	-0.719
			(1.371)	(1.419)	(1.381)
log Downstream Precip. (t)					-0.032
					(0.402)
log Downstream Precip. ($t-1$)					-0.801**
					(0.374)
1 SD of identifying variation: a wetter monsoon of (%)	11	11	9	9	9
Implied effect of a 1 SD wetter monsoon (%)	15	17	14	15	16
Observations	4,876	4,876	4,876	4,876	3,886
Number of Districts	210	210	210	210	177
Mean of dependent variable	1.55	1.55	1.55	1.55	1.74
Estimation	Poisson	Poisson	Poisson	Poisson	Poisson
District FE	✓	✓	✓	✓	✓
State × Year FE	✓	✓	✓	✓	✓

Notes: Table 2 with its year effects replaced by state × year effects; the outcome, sample, controls and spatial terms of each column are unchanged. Poisson drops state-years with no events, so observations fall while every district remains, and the mean of the outcome rises. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table E.2: Upstream rainfall and the water in the river, with state × year fixed effects

	No spatial controls			Spatial controls		
	(1)	(2)	(3)	(4)	(5)	(6)
log Upstream Precip. (t)	0.288		0.370	0.248		0.341
	(0.437)		(0.422)	(0.420)		(0.423)
log Upstream Precip. ($t-1$)	1.534***		1.523***	1.769***		1.724***
	(0.358)		(0.404)	(0.482)		(0.514)
log Routed Discharge, June–September (t)		-0.128	-0.149		-0.135	-0.157
		(0.212)	(0.182)		(0.193)	(0.167)
log Routed Discharge, June–September ($t-1$)		0.191	-0.002		0.269	0.072
		(0.153)	(0.180)		(0.176)	(0.192)
Observations	3,869	3,869	3,869	3,869	3,869	3,869
Number of Districts	170	170	170	170	170	170
Mean of dependent variable	1.78	1.78	1.78	1.78	1.78	1.78
Estimation	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
District FE	✓	✓	✓	✓	✓	✓
State × Year FE	✓	✓	✓	✓	✓	✓
Spatial controls				✓	✓	✓

Notes: Table 3 with its year effects replaced by state × year effects; the outcome, sample, controls and spatial block of each column are unchanged, and local rainfall and temperature enter every column and are not shown. Poisson drops state-years with no events, so observations fall while the mean of the outcome rises. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

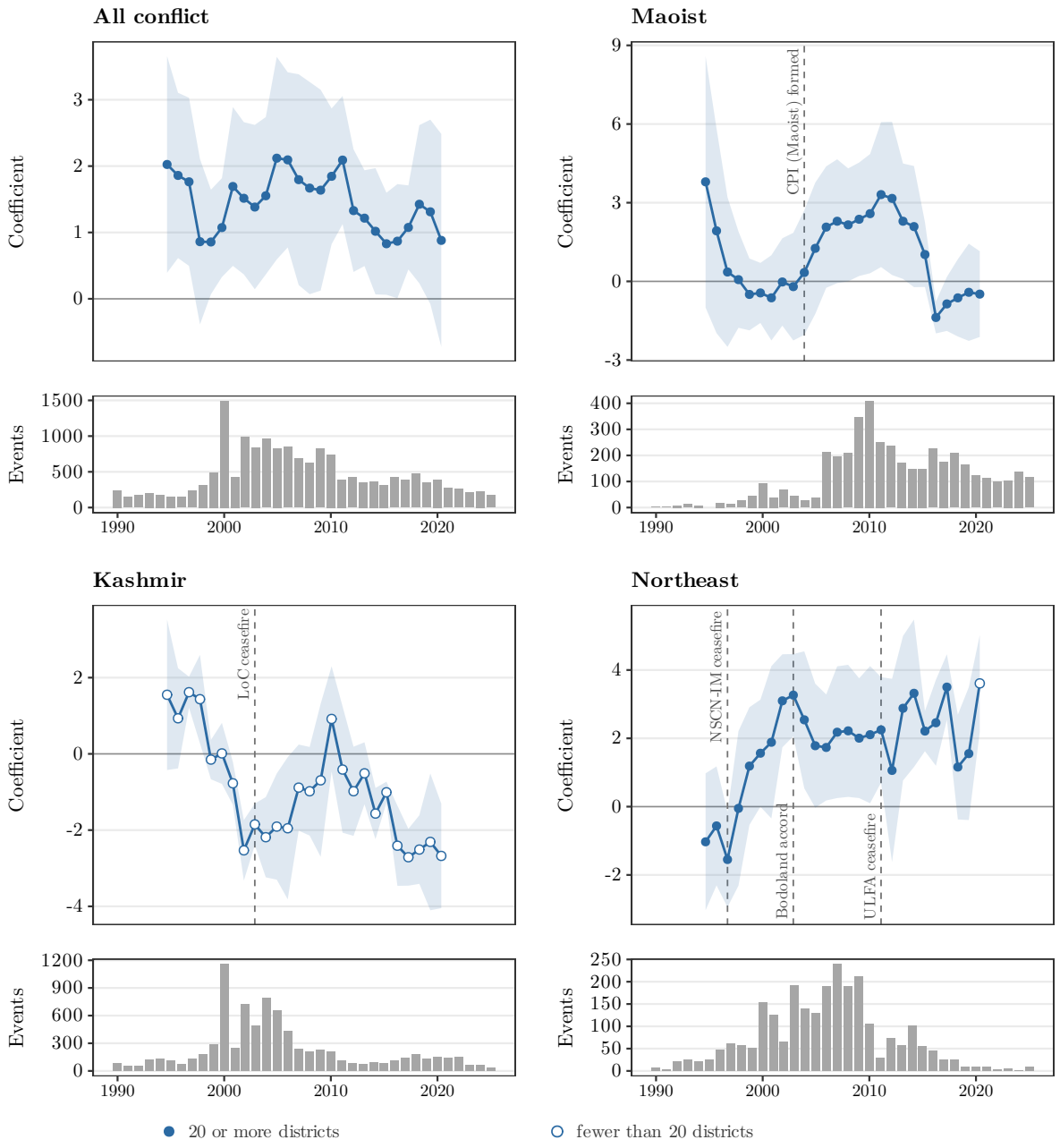


Figure E.6: Upstream rainfall and conflict over time, by conflict system

Notes: In each cell the upper panel is the coefficient on log Upstream Precip. $(t-1)$ from Table 2, column 4, estimated on each 11-year window of 1990–2025 and plotted at the window's centre year, with its 95 percent Conley (150 km) interval; the lower panel is the series' located events per year. The outcome is all located events or the events of one conflict system, assigned from the UCDP conflict name (Table A.1). The Poisson keeps only districts with events in the window, and windows resting on fewer than 20 districts are drawn hollow. Dashed lines mark externally dated events: the November 2003 India–Pakistan ceasefire along the Line of Control, the September 2004 merger forming CPI (Maoist), and for the Northeast the 1997 NSCN-IM ceasefire, the 2003 Bodoland accord and the 2011 ULFA ceasefire. Figure E.7 repeats them with 9- and 15-year windows.

Table E.3: Is the monsoon the right window?

	Monsoon	Crop-calendar alternatives		
	(1)	(2)	(3)	(4)
log Upstream Precip., monsoon $(t-1)$	1.460***		1.547***	1.479***
	(0.283)		(0.301)	(0.295)
log Upstream Precip., growing season $(t-1)$		0.748***		
		(0.290)		
log Upstream Precip., non-monsoon window $(t-1)$			-0.066	
			(0.119)	
Growing-season component orthogonal to the monsoon $(t-1)$				-0.316
				(0.270)
Observations	7,560	7,350	7,350	7,350
Number of Districts	210	210	210	210
Mean of dependent variable	1.00	1.01	1.01	1.01
Estimation	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓

Notes: Every column is the specification of Table 2, column 4, on 1990–2025; only the rainfall window changes. *Crop-calendar weights.* MIRCA2000 reports, for each district, the share of its cropped area under crops in each calendar month; these shares are time-invariant. Monthly upstream rainfall is multiplied by those shares and summed, so column 2 weights the whole year by when that district's own agriculture actually happens. *The non-monsoon term.* Columns 3 and 4 split the weights at the monsoon boundary and renormalise each half to sum to one, so the non-monsoon regressor carries *only* October–May rainfall, weighted by the share of the district's cropping falling in those months — broadly the rabi season. That window straddles two calendar years, so it is dated by monsoon year: the term dated t runs from October of year t to May of year $t+1$, and the lagged term covers the same eight months one year earlier. *Column 4.* The regressor is the residual from projecting the whole-year crop-calendar measure on monsoon rainfall within district and year — the crop-calendar content the monsoon window does not already carry. It is a generated regressor and its standard errors are not corrected for that first step. Current-period terms, weather controls and the spatial block are estimated throughout and not displayed. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

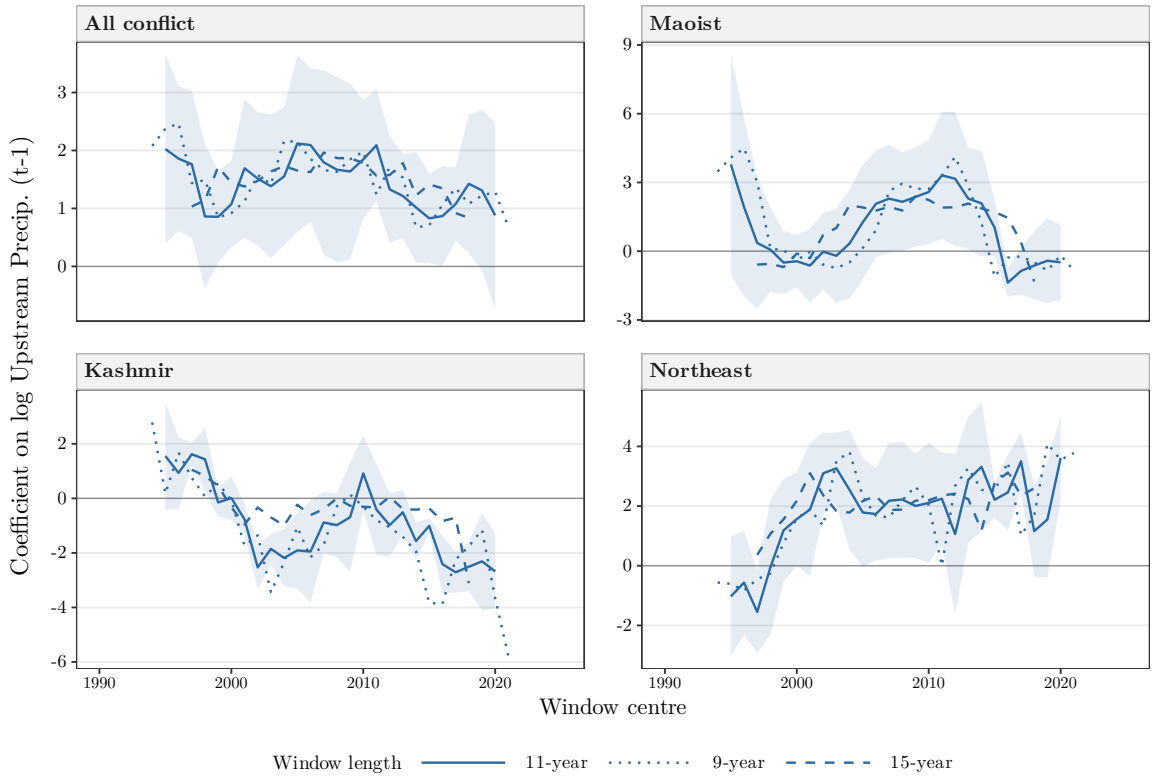


Figure E.7: Rolling windows of 9, 11 and 15 years

Notes: The series of Figure E.6 re-estimated with 9-year and 15-year windows; the shaded band is the 11-year window's 95 percent Conley interval. Specification and sample rules as in Figure E.6.

Table E.4: Upstream rainfall and conflict with the territory the census does not enumerate

	Baseline	Spatial controls			
	(1)	(2)	(3)	(4)	(5)
A. As published: territory dropped					
log Upstream Precip. $(t-1)$	0.910**	0.994***	1.351***	1.460***	1.242***
	(0.359)	(0.314)	(0.351)	(0.283)	(0.277)
Observations	7,560	7,560	7,560	7,560	6,372
Number of Districts	210	210	210	210	177
Mean of dependent variable	1.00	1.00	1.00	1.00	1.06
Located events in sample	7,564	7,564	7,564	7,564	6,764
B. Territory's events reassigned to the nearest Indian district					
log Upstream Precip. $(t-1)$	0.904**	0.996***	1.343***	1.455***	1.237***
	(0.356)	(0.308)	(0.359)	(0.285)	(0.279)
Observations	7,560	7,560	7,560	7,560	6,372
Number of Districts	210	210	210	210	177
Mean of dependent variable	1.01	1.01	1.01	1.01	1.07
Located events in sample	7,638	7,638	7,638	7,638	6,838
C. Territory kept as a district					
log Upstream Precip. $(t-1)$	1.046***	0.965***	1.452***	1.470***	1.173***
	(0.288)	(0.276)	(0.294)	(0.260)	(0.245)
Observations	7,596	7,596	7,596	7,596	6,372
Number of Districts	211	211	211	211	177
Mean of dependent variable	1.08	1.08	1.08	1.08	1.06
Located events in sample	8,218	8,218	8,218	8,218	6,764

Notes: Columns as in Table 2. The territory is the part of the former state of Jammu and Kashmir administered by Pakistan and China, which the 2001 census map draws as one unit but the census did not enumerate. Panel A is Table 2. Panel B assigns each of the territory's located events to the nearest Indian district. Panel C adds the territory as a district: its treatment is built as for every other district, and it enters the neighbour matrix. It has no downstream catchment and so does not enter column 5, which moves only through the neighbour terms of the districts bordering it. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table E.5: Which conflicts respond, cut each way

	All conflict (1)	State-based (2)	One-sided (3)	Non-state (4)
Panel A. By type of violence				
log Upstream Precip. (t)	-0.086	-0.026	-0.378	0.555
	(0.487)	(0.511)	(0.767)	(1.329)
log Upstream Precip. ($t-1$)	1.460***	1.429***	1.063*	2.491**
	(0.283)	(0.324)	(0.595)	(1.178)
log Local Precip. (t)	-0.088	0.049	-1.466	3.403**
	(0.579)	(0.593)	(0.910)	(1.679)
log Local Precip. ($t-1$)	0.507	0.881	-0.936	5.595**
	(1.025)	(1.238)	(1.156)	(2.660)
1 SD of identifying variation: a wetter monsoon of (%)	12	12	12	12
Implied effect of a 1 SD wetter monsoon (%)	18	18	13	33
Observations	7,560	5,040	5,508	1,848
Districts	210	140	153	66
Mean of dependent variable	1.00	0.98	0.42	0.19
	Maoist (5)	Kashmir (6)	Northeast (7)	Communal (8)
Panel B. By conflict system				
log Upstream Precip. (t)	-1.233*	0.752**	0.624	1.107
	(0.723)	(0.379)	(1.124)	(1.451)
log Upstream Precip. ($t-1$)	1.206	0.408*	1.758***	1.965
	(0.746)	(0.225)	(0.595)	(1.274)
log Local Precip. (t)	-1.091	-0.476	0.322	2.662
	(1.090)	(0.702)	(0.742)	(2.430)
log Local Precip. ($t-1$)	-1.545	-1.078**	-0.576	5.260
	(1.097)	(0.428)	(0.394)	(3.225)
1 SD of identifying variation: a wetter monsoon of (%)	10	16	9	12
Implied effect of a 1 SD wetter monsoon (%)	13	6	17	25
Observations	3,255	720	1,190	1,334
Districts	93	20	34	58
Mean of dependent variable	0.95	2.62	1.45	0.20
Estimation	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓

Notes: Every cell is a separate Poisson regression on 1990–2025 under the specification of Table 2, column 4: district and year fixed effects, the full weather controls, and the spatial block. Only the outcome changes. Neighbours' conflict in the spatial block is built from the column's own outcome. Types are UCDP's type of violence, systems are assigned from the UCDP conflict name. Poisson drops districts recording no events of a given kind, which is why the district counts differ down the table. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table E.6: Conflict system by type of violence

	Maoist		Kashmir		Northeast	
	State-based	One-sided	State-based	One-sided	State-based	One-sided
	(1)	(2)	(3)	(4)	(5)	(6)
log Upstream Precip. (t)	-0.954*	-1.130	0.588	1.451***	1.064	-1.515
	(0.534)	(1.181)	(0.406)	(0.475)	(1.185)	(1.212)
log Upstream Precip. ($t-1$)	0.260	2.593***	0.383*	0.662	1.230***	1.560
	(0.672)	(0.941)	(0.197)	(0.546)	(0.475)	(1.590)
log Local Precip. (t)	-0.516	-2.019	-0.081	-2.329	-0.052	0.998**
	(0.743)	(1.553)	(0.431)	(1.558)	(0.634)	(0.453)
log Local Precip. ($t-1$)	-0.238	-3.005**	-0.416	-2.277**	-0.328	-1.790
	(1.064)	(1.226)	(0.386)	(1.107)	(0.244)	(1.117)
1 SD of identifying variation: a wetter monsoon of (%)	11	10	16	17	9	9
Implied effect of a 1 SD wetter monsoon (%)	3	28	6	11	12	14
Observations	2,835	2,070	540	462	1,190	644
Districts	81	69	15	14	34	23
Mean of dependent variable	0.66	0.60	2.76	0.85	1.02	0.68
Estimation	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓	✓	✓

Notes: Each column is a separate Poisson regression on 1990–2025 under the specification of Table 2, column 4: district and year fixed effects, the full weather controls, and the spatial block. Only the outcome changes — events of one conflict system and one UCDP type of violence. Neighbours' conflict in the spatial block is built from the column's own outcome. Poisson drops districts recording no events of a given kind, which is why the district counts differ across columns; the columns are therefore not a decomposition of any single total. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

districts. SATP's count of lethal incidents responds at 0.57, against 2.27 for UCDP events on the same sample, and the gap is composition. In SATP the response sits in terrorist attacks (1.49) and in civilians killed (2.42), and is absent from attacks on security forces (-0.22) and from security-force deaths (-0.16). UCDP shows the same split on this sample: one-sided violence responds (3.61) while state-based violence does not (1.04). SATP records proportionally more security-force incidents, which do not respond, so its aggregate count dilutes an effect that both corpora show in civilian-directed violence.

The downstream coefficient does not cross to SATP. Table E.8 adds the downstream block of Table 2, column 5, to the same regressions, on the 4,276 district-years in 335 districts for which the downstream series exists. The block leaves every upstream coefficient where it was, in both corpora. UCDP keeps a contemporaneous downstream coefficient on this sample (-0.89 for all events, -1.05 for deaths, -1.22 for one-sided violence), while the lagged one loses precision. No SATP series responds to downstream rainfall: lethal incidents give -0.39 and 0.01, civilians killed -1.50 with a standard error of the same size, and the militant and security-force casualty series carry positive, imprecise coefficients. The upstream effect is a property of the violence that both corpora record; the downstream coefficient is, so far, a property of UCDP's record of it.

Table E.7: Upstream rainfall by type of violence in two conflict corpora

	Mean	log Upstream Precip. $(t-1)$	Observations
Panel A. UCDP, events located to a district or better			
All events	0.63	2.267***	1,782
		(0.629)	
State-based	0.37	1.043	1,386
		(0.635)	
One-sided	0.23	3.608***	1,166
		(0.931)	
All deaths	1.65	1.687***	1,769
		(0.607)	
in state-based events	0.95	0.367	1,380
		(0.562)	
in one-sided events	0.62	2.538**	1,156
		(1.117)	
Panel B. SATP, as assembled by Fetzer (2020)			
Lethal incidents	1.01	0.574**	2,173
		(0.286)	
Attacks on security forces	0.15	-0.219	1,253
		(0.627)	
Terrorist attacks	0.42	1.491***	1,339
		(0.467)	
Attacks on civilians	0.58	0.473	1,818
		(0.353)	
Persons killed	2.15	1.883***	2,028
		(0.690)	
Militants	0.93	1.340**	1,530
		(0.603)	
Security forces	0.72	-0.156	1,359
		(0.841)	
Civilians	1.51	2.425***	1,729
		(0.906)	

Notes: Each row is a separate Poisson regression of the outcome named on the paper's weather controls — current and lagged upstream rainfall and temperature, and current and lagged local rainfall and temperature — on one sample: the district-year panel of South Asia Terrorism Portal (SATP) incidents of Fetzer (2020), 2000–2012, 5,580 district-years in 437 of his districts, with the paper's weather variables mapped to his districts and UCDP events aggregated to them. The Poisson drops districts with no events of a row's kind, so observation counts differ by row. Fixed effects are those of Fetzer's specification: district, and year interacted with his conflict-treatment grouping. The spatial block of Table 2 is not included. UCDP deaths are the best estimate ('best') per event. SATP lethal incidents are total incidents net of those recorded as non-lethal; attack and casualty categories are SATP's own. Standard errors clustered by district in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table E.8: Downstream rainfall by type of violence in two conflict corpora

		log Upstream Precip. $(t-1)$		log Downstream Precip.		
	Mean	without downstream	with downstream	(t)	$(t-1)$	Observations
Panel A. UCDP, events located to a district or better						
All events	0.57	2.032**	2.052**	-0.888*	-1.088	1,212
		(0.900)	(0.857)	(0.461)	(0.703)	
State-based	0.36	1.054	1.099	-0.588	-0.664	912
		(0.934)	(0.863)	(0.469)	(0.715)	
One-sided	0.18	3.465***	3.201***	-1.220**	-0.994	846
		(1.139)	(1.072)	(0.617)	(0.809)	
All deaths	1.55	1.637**	1.413*	-1.048**	-0.334	1,203
		(0.772)	(0.726)	(0.478)	(0.633)	
in state-based events	0.91	1.017	0.863	0.020	0.332	907
		(0.709)	(0.580)	(0.641)	(0.799)	
in one-sided events	0.56	1.814	1.373	-0.983	-0.518	837
		(1.291)	(1.305)	(0.815)	(1.059)	
Panel B. SATP, as assembled by Fetzer (2020)						
Lethal incidents	0.91	0.845**	0.802**	-0.393	0.010	1,479
		(0.346)	(0.374)	(0.411)	(0.418)	
Attacks on security forces	0.13	0.529	0.377	0.778	-1.106	833
		(0.892)	(0.886)	(0.754)	(0.681)	
Terrorist attacks	0.36	0.863	0.851	-1.165	-0.561	911
		(0.573)	(0.596)	(0.714)	(0.519)	
Attacks on civilians	0.48	0.743*	0.786**	-0.331	-0.708	1,217
		(0.420)	(0.375)	(0.502)	(0.475)	
Persons killed	2.04	2.889***	2.950***	0.521	-0.669	1,340
		(0.823)	(0.783)	(0.455)	(0.689)	
Militants	0.89	1.461**	1.472**	0.489	0.800	1,041
		(0.731)	(0.716)	(0.788)	(0.682)	
Security forces	0.69	1.366	1.109	0.878	1.272	930
		(1.054)	(1.053)	(1.177)	(0.846)	
Civilians	1.44	2.882**	2.962***	-0.558	-1.505	1,200
		(1.125)	(1.083)	(0.585)	(1.003)	

Notes: The construction of Table E.7 — each row a separate Poisson regression of the outcome named on the paper's weather controls, on the district-year panel of Fetzer (2020), 2000–2012, with his district and year-by-treatment-group fixed effects — restricted to the 4,276 district-years in 335 districts for which the downstream series exists. Columns 2 and 3 report the upstream coefficient without and with the downstream block of Table 2, column 5 (current and lagged downstream rainfall and temperature); columns 4 and 5 are the downstream rainfall coefficients from the latter regression. The Poisson drops districts with no events of a row's kind, so observation counts differ by row. Standard errors clustered by district in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

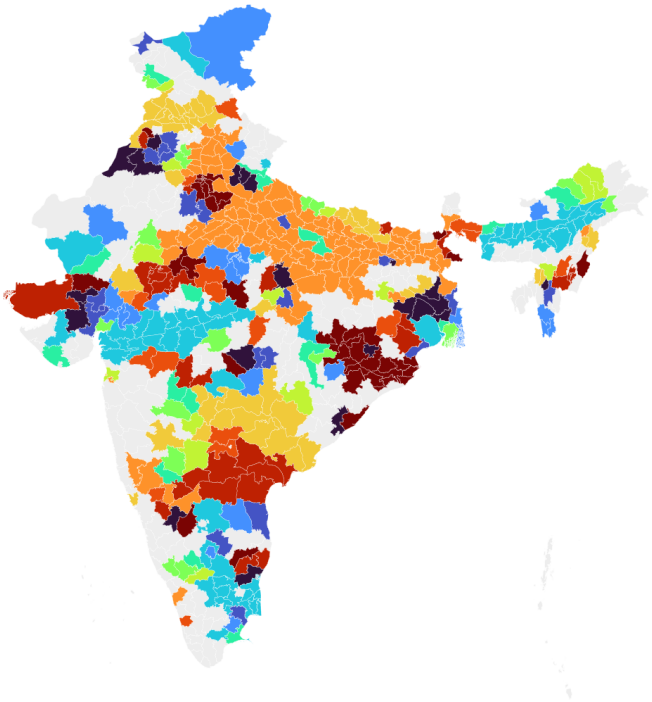


Figure E.8: Districts grouped by shared upstream catchment

Notes: Connected components of the district-by-district cosine similarity in upstream exposure, thresholded at 0.50. Exposure is the river-distance weight vector over HydroBASINS level-10 units that builds the treatment, so two districts are joined when more than 0.50 of their upstream water is shared, directly or through a chain of such links. Districts drawn in flat grey have no upstream catchment in the construction. The partition is used in Section B as a clustering level for standard errors, not as an estimation sample.

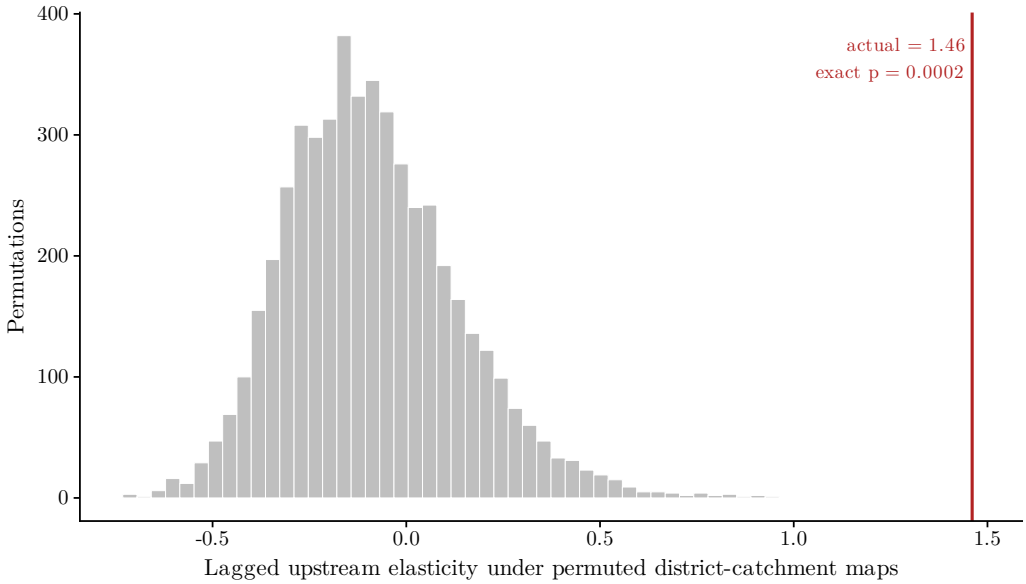


Figure E.9: Randomization inference

Notes: Distribution of the lagged upstream elasticity across 5,000 within-state permutations of the district–catchment assignment; the red line is the estimate under the true assignment. Section B.5 describes the test.

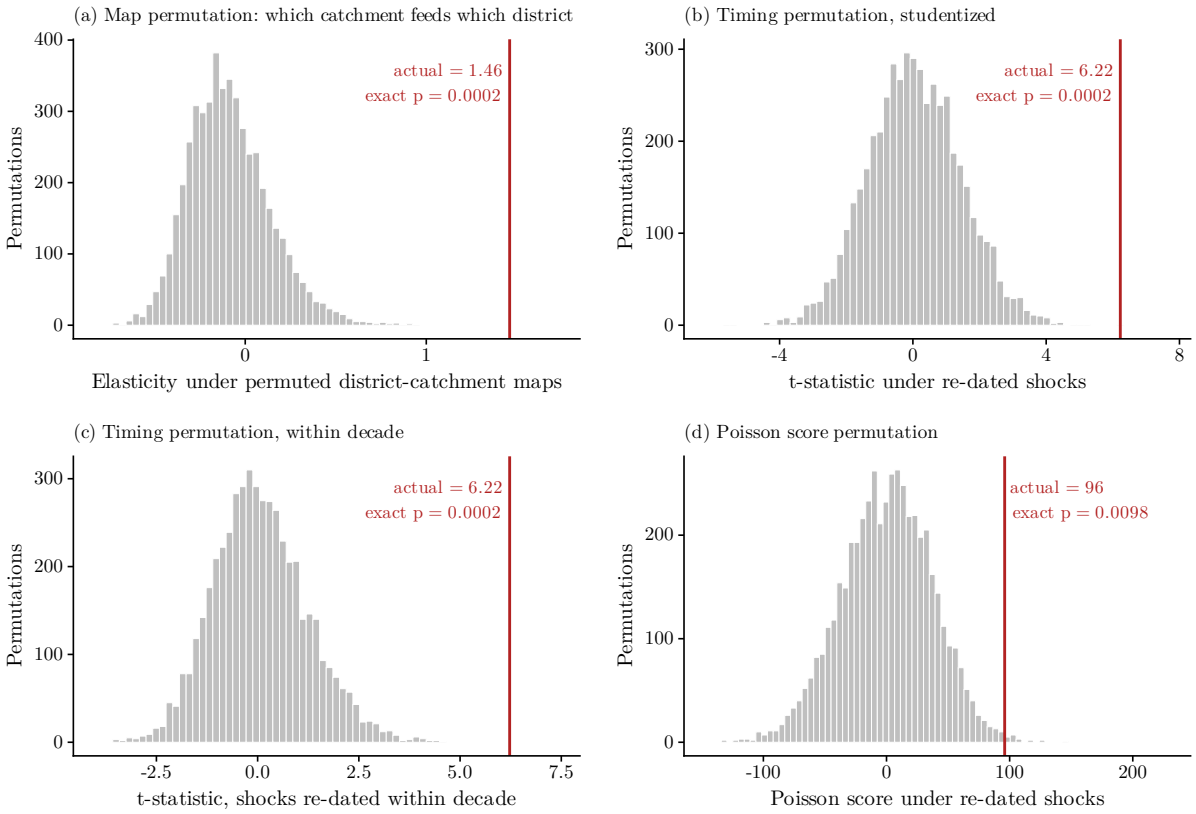


Figure E.10: Four randomization tests of the same estimate

Notes: Section B.5 describes each test. Every panel re-estimates the specification of Table 2, column 4 on 1990–2025 under 5,000 permutations; the grey histogram is the null distribution, the vertical rule the observed statistic. Panel (a) permutes the district–catchment map within states, holding time fixed. Panels (b)–(d) permute the year labels of the whole cross-section jointly, holding the map fixed: one permutation per draw applied to every district, so the spatial dependence of the shock is preserved and only the shock–outcome alignment breaks. The outcome and the spatial terms built from it are never re-dated. Panel (c) restricts each permutation to stay within its decade, which preserves trend and relaxes the exchangeability the free permutation assumes. Panel (d) uses the score statistic implied by the Poisson likelihood. Permuting the upstream rainfall term alone rather than the whole forcing block gives $p = 0.0006$ on the studentized statistic; that narrower version leaves the treatment's own temperature control and the local weather at their true dates, so the regressors are no longer mutually aligned.

Table E.9: Flooding at migrant origins and conflict at destinations

Origin definition	FloodMig (t)	Cluster SE	Perm. p (coef.)	Perm. p (t)
Origin state (D-02)	-0.033	(0.024)	0.396	0.395
Origin district (D-11)	0.002	(0.013)	0.950	0.949
Hybrid	0.007	(0.017)	0.813	0.822

Notes: Poisson fixed-effects, 2002–2025, the destination's own routed flood days and local weather included in every column. FloodMig is the migration-share-weighted mean of origin flood days, with shares from the 2001 Census. D-02 gives origin states only; D-11 gives origin districts within a state. No definition produces an effect surviving the year-block permutation.

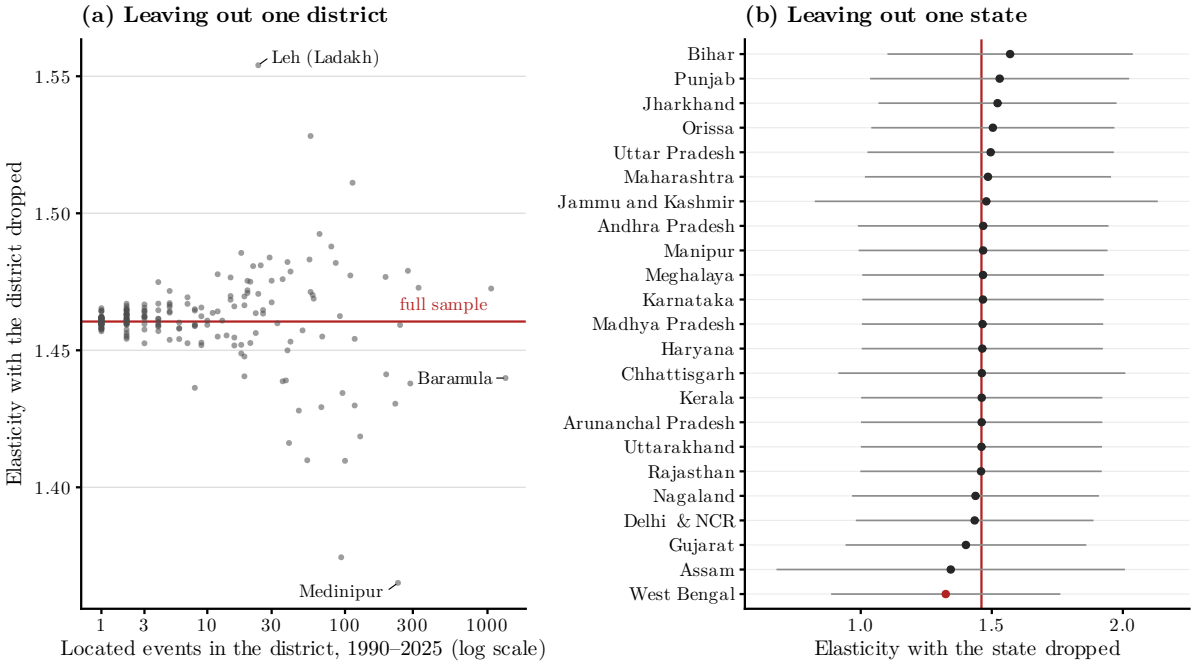


Figure E.11: Leave-one-out robustness, by district and by state

Notes: Each point re-estimates the specification of Table 2, column 4 on 1990–2025 with one district (panel a) or one state (panel b) removed; the red line is the full-sample estimate 1.46. Panel (a) plots the leave-it-out estimate against the dropped district's own count of located events, so the vertical spread shows influence and the horizontal position shows concentration; the three labelled districts are the two extremes and the most violent. Panel (b) shows 95 percent district-clustered intervals, states ordered by the resulting estimate, with West Bengal — the most influential — marked. Districts are dropped from the frame before estimation, so the spatial block is rebuilt on the reduced sample rather than carried over.

E.3. Corridor Weather and Road-Borne Trade

The exhibits below support Section B.6: the two placebo twins and the trade measure for one district, and the tests the text summarizes.

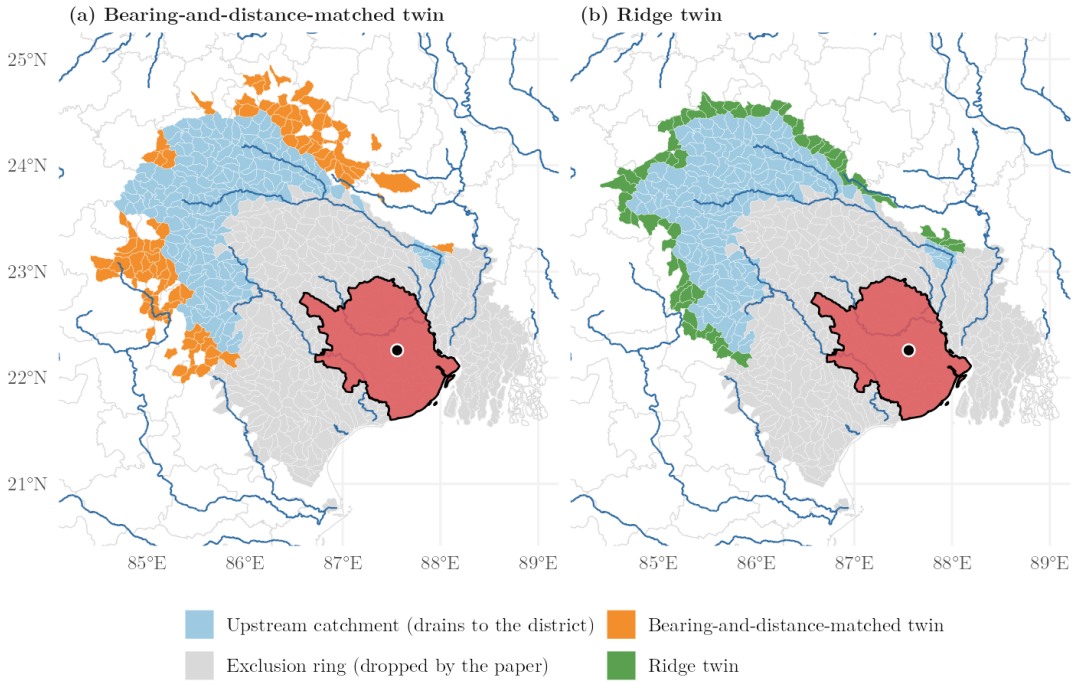


Figure E.12: The two sideways twins for one district

Notes: The example district is Medinipur (West Bengal), shown in red with its population centre (black point). Blue: the 183 level-10 basins of its upstream catchment after the neighbour exclusion (the treatment set); grey: the 439 basins the neighbour exclusion removes. Panel (a): the bearing-and-distance-matched twin, 92 basins chosen outside the catchment and the removed basins at the same distance (±15%) and bearing (±20°) as each catchment basin; 87% of the catchment's weight is matched, with a median distance gap of 0.5% and a median bearing gap of 15°. Panel (b): the ridge twin, the 56 basins sharing a boundary with the catchment but outside it and the removed basins, the nearest 119 km from the district. Blue lines: rivers draining more than 3,000 km^2 (HydroRIVERS).

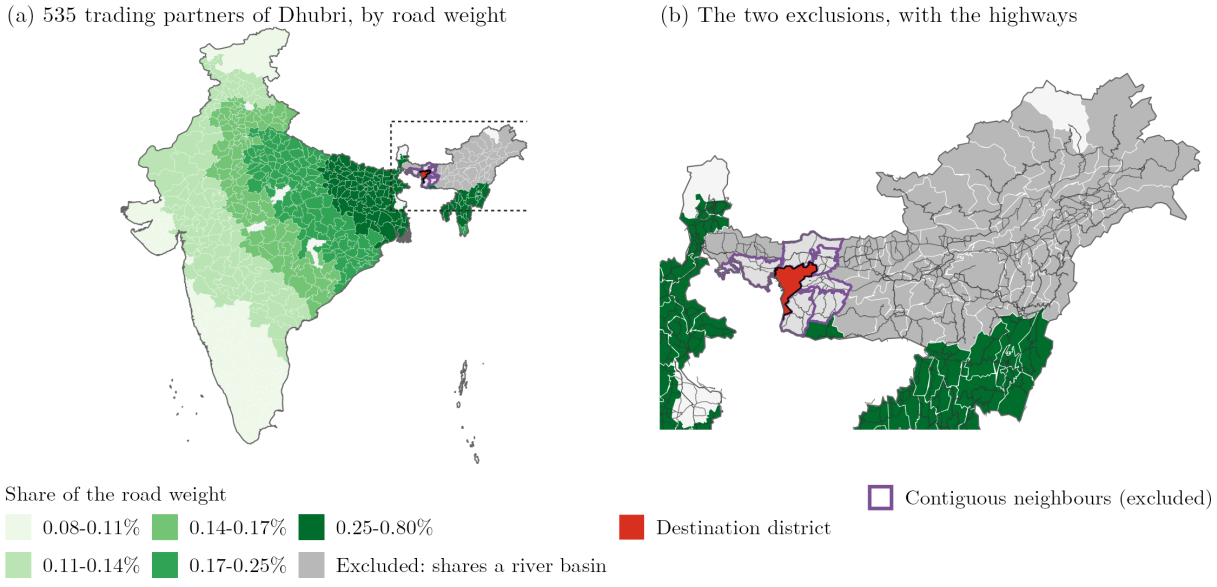


Figure E.13: The road-network trade measure, for one district

Notes: The example district is Dhubri (Assam), the district of Figure 2, shown in red. Panel (a): its 535 trading partners, shaded by their share of the weight $1/(1+h)$, where h is road travel time in hours, normalised over the partner set; 26% of the weight lies within 12 hours, and the median partner is 25 hours away. Panel (b): the 6 contiguous neighbours (purple outline) and the 41 districts with at least 5 percent of their area in the district's upstream, downstream or own sub-basins (grey) are excluded; dark lines are national and state highways (OpenStreetMap trunk and primary roads).

Table E.10: Sideways twins: rainfall on basins that do not drain to the district

	(1) Table 2 column 4	(2) + twin rainfall	(3) Twin replaces upstream rainfall
Panel A. Bearing-and-distance-matched twin			
log Upstream Precip. (t)	-0.088	0.945	
	(0.487)	(0.736)	
log Upstream Precip. ($t-1$)	1.467***	1.726**	
	(0.283)	(0.864)	
log Twin Precip. (t)		-1.301	-0.494
		(0.917)	(0.574)
log Twin Precip. ($t-1$)		-0.304	1.234***
		(1.022)	(0.356)
Upstream ($t-1$) − twin ($t-1$)		2.03 (1.86)	
Within-FE corr., twin and upstream ($t-1$)		0.94	
Observations	7,524	7,524	7,524
Number of Districts	209	209	209
Mean of dependent variable	1.00	1.00	1.00
Panel B. Ridge twin			
log Upstream Precip. (t)	-0.086	0.735	
	(0.487)	(0.639)	
log Upstream Precip. ($t-1$)	1.460***	1.862***	
	(0.283)	(0.715)	
log Twin Precip. (t)		-0.948	-0.329
		(0.718)	(0.515)
log Twin Precip. ($t-1$)		-0.453	1.195***
		(0.770)	(0.298)
Upstream ($t-1$) − twin ($t-1$)		2.31 (1.45)	
Within-FE corr., twin and upstream ($t-1$)		0.95	
Observations	7,560	7,560	7,560
Number of Districts	210	210	210
Mean of dependent variable	1.00	1.00	1.00

Notes: The outcome is the count of UCDP events located to a district or better, estimated by Poisson with district and year fixed effects on the Table 2 column 4 sample (1990–2025), with that column's full control set unless stated. Conley spatial-HAC standard errors (150 km) in parentheses; ***, **, and * indicate significance at 1%, 5% and 10%. The bearing-matched twin replaces each basin of the district's upstream catchment by a level-10 basin outside the catchment and the basins the neighbour exclusion removes at the same great-circle distance (±15%) and bearing (±20°) from the district's population centre, carrying the original basin's river-distance weight; the ridge twin is every basin sharing a boundary with the catchment but outside it and those removed basins, weighted by inverse straight-line distance (Figure E.12). Column 1 re-estimates column 4 of Table 2 on the district-years where the twin exists; column 2 adds the twin's rainfall at t and $t-1$; column 3 enters it in place of upstream rainfall, with the upstream temperature controls unchanged. The difference row is the contrast of the two lagged coefficients in column 2.

Table E.11: The district's own weather: rainfall intensity and the basins the neighbour exclusion removes

	(1)	(2)	(3)	(4)
Panel A. The district's own rainfall intensity				
	Table 2 column 4	+ intensity	Monthly totals replace JJAS	Monthly totals + intensity
log Upstream Precip. (t)	-0.086	-0.021	-0.005	0.046
	(0.487)	(0.483)	(0.464)	(0.482)
log Upstream Precip. ($t-1$)	1.460***	1.550***	1.427***	1.574***
	(0.283)	(0.268)	(0.262)	(0.271)
Local days >50 mm (t)		-0.017		-0.005
		(0.022)		(0.024)
Local days >50 mm ($t-1$)		-0.020		-0.007
		(0.035)		(0.034)
Local max 5-day total, mm (t)		0.0017		0.0013
		(0.0012)		(0.0011)
Local max 5-day total, mm ($t-1$)		0.0034*		0.0030
		(0.0020)		(0.0019)
log Local Precip., JJAS total ($t, t-1$)	✓	✓		
log Local Precip., June, July, August, September ($t, t-1$)			✓	✓
Observations	7,560	7,560	7,560	7,560
Number of Districts	210	210	210	210
Mean of dependent variable	1.00	1.00	1.00	1.00
Panel B. Rainfall on the basins the neighbour exclusion removes				
	Table 2 column 4	+ removed-basin rainfall	Removed-basin rainfall replaces upstream	
log Upstream Precip. (t)	-0.086	0.259		
	(0.487)	(0.581)		
log Upstream Precip. ($t-1$)	1.460***	1.351***		
	(0.283)	(0.410)		
log Removed-Basin Precip. (t)		-1.157	-0.891	
		(1.009)	(0.849)	
log Removed-Basin Precip. ($t-1$)		0.396	1.684**	
		(1.075)	(0.794)	
log Local Precip. (t)		0.384	0.180	
		(0.642)	(0.588)	
log Local Precip. ($t-1$)		0.327	-0.470	
		(0.896)	(0.860)	
Within-FE corr., ring and local ($t-1$)		0.93		
Within-FE corr., ring and upstream ($t-1$)		0.83		
Observations	7,560	7,560	7,560	
Number of Districts	210	210	210	
Mean of dependent variable	1.00	1.00	1.00	

Notes: The outcome is the count of UCDP events located to a district or better, estimated by Poisson with district and year fixed effects on the Table 2 column 4 sample (1990–2025), with that column's full control set unless stated. Conley spatial-HAC standard errors (150 km) in parentheses; ***, **, and * indicate significance at 1%, 5% and 10%. Panel A: days above 50 mm and the maximum five-day total are computed per level-10 basin and JJAS season from CHIRPS v3 daily rainfall and aggregated to the district as the overlap-area-weighted mean over its own basins. Columns 3 and 4 replace the JJAS local total by the four monthly totals (floored at 1 mm before the log), current and lagged. Panel B: removed-basin rainfall is JJAS rainfall over the basins of the district's upstream network that the neighbour exclusion removes (overlapping a bordering district or a foreign unit, excluding those overlapping the district itself), weighted by inverse straight-line distance.

Table E.12: Regime-explained and basin-specific upstream rainfall

	(1) Himalayan share	(2) Ganga, Brahmaputra and Indus shares
Regime part of log Upstream Precip. (t)	3.111*	0.565
	(1.864)	(0.634)
Regime part ($t-1$)	1.657	2.612***
	(1.081)	(0.673)
Residual part (t)	-0.136	-0.168
	(0.466)	(0.501)
Residual part ($t-1$)	1.505***	1.430***
	(0.307)	(0.282)
Regime − residual ($t-1$)	0.15	1.18**
	(1.03)	(0.58)
Within-FE variance share of the regime part	0.032	0.061
Observations	7,560	7,560
Number of Districts	210	210
Mean of dependent variable	1.00	1.00

Notes: The outcome is the count of UCDP events located to a district or better, estimated by Poisson with district and year fixed effects on the Table 2 column 4 sample (1990–2025), with that column's full control set unless stated. Conley spatial-HAC standard errors (150 km) in parentheses; ***, **, and * indicate significance at 1%, 5% and 10%. Log upstream rainfall is first regressed, over 1981–2025 with district effects, on the all-India JJAS rainfall anomaly, the JJAS Niño-3.4 anomaly and the July–August counts of active and break days, each interacted with the share of the catchment's weight in basins above 1,500 m (column 1) or in the Ganga, Brahmaputra and Indus macro-basins (column 2). The regime part is the index-explained fitted value (without the district mean); the residual is the rest. Both enter in place of upstream rainfall; year effects absorb the indices' main effects. Active and break days are reconstructed from CHIRPS v3 daily rainfall over the core monsoon zone by the rule of Rajeevan, Gadgil and Bhate (2010).

Table E.13: The downstream twin against its own sideways placebo

	(1) Table 2 column 5	(2) + downstream twin	(3) + upstream twin	(4) + both twins	(5) Upstream twin replaces upstream	(6) Downstream twin replaces downstream
log Upstream Precip. (t)	-0.275	-0.270	0.563	0.570		-0.276
	(0.473)	(0.479)	(0.762)	(0.778)		(0.471)
log Upstream Precip. ($t-1$)	1.273***	1.267***	1.775**	1.778**		1.268***
	(0.287)	(0.296)	(0.817)	(0.844)		(0.293)
log Upstream Twin Precip. (t)			-1.041	-1.041	-0.596	
			(0.884)	(0.904)	(0.532)	
log Upstream Twin Precip. ($t-1$)			-0.611	-0.618	0.989***	
			(1.012)	(1.021)	(0.377)	
log Downstream Precip. (t)	-1.347***	-1.441*	-1.292***	-1.458*	-1.372***	
	(0.464)	(0.778)	(0.442)	(0.760)	(0.470)	
log Downstream Precip. ($t-1$)	-0.935***	-0.799	-0.913***	-0.860	-1.021***	
	(0.344)	(0.994)	(0.348)	(1.023)	(0.349)	
log Downstream Twin Precip. (t)		0.113		0.196		-1.353***
		(0.818)		(0.827)		(0.480)
log Downstream Twin Precip. ($t-1$)		-0.159		-0.067		-0.917**
		(1.263)		(1.307)		(0.433)
Within-FE corr., upstream and its twin ($t-1$)				0.94		
Within-FE corr., downstream and its twin (t)				0.99		
Observations	6,300	6,300	6,300	6,300	6,300	6,300
Number of Districts	175	175	175	175	175	175
Mean of dependent variable	1.07	1.07	1.07	1.07	1.07	1.07

Notes: The outcome is the count of UCDP events located to a district or better, estimated by Poisson with district and year fixed effects on the Table 2 column 4 sample (1990–2025), with that column's full control set unless stated. Conley spatial-HAC standard errors (150 km) in parentheses; ***, **, and * indicate significance at 1%, 5% and 10%. The sample is Table 2 column 5's (districts with a downstream series) on the district-years where both twins exist. The upstream twin is the bearing-and-distance-matched twin of Table E.10; the downstream twin applies the same matching rule to the downstream set (pool outside the basins the district's water reaches and outside the basins the neighbour exclusion removes), weighted as the paper weights downstream basins. Downstream temperature controls enter every column; upstream temperature controls stay in column 5 and downstream ones in column 6.

Table E.14: Upstream rainfall and village road connectivity

	2001 paved-road share			1991 share
	(1)	(2)	(3)	(4)
log Upstream Precip. $(t-1)$	1.397***	1.412***	1.732***	-0.229
	(0.292)	(0.289)	(0.529)	(0.386)
× Paved-road share	-0.447*	-0.379	0.011	-0.724***
	(0.265)	(0.365)	(0.284)	(0.171)
× Electrified share			0.218	
			(0.226)	
× Irrigated share			-0.928***	
			(0.262)	
× log Rural density			-0.241	
			(0.297)	
× Agricultural worker share			0.572***	
			(0.203)	
× log Distance to town			-0.697***	
			(0.233)	
Local rainfall × road share		✓		
Observations	7,344	7,344	6,912	4,248
Number of Districts	204	204	192	118
Mean of dependent variable	1.02	1.02	1.01	0.67
Estimation	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓

Notes: The outcome and controls are those of column 4 of Table 2. Paved-road share: share of the district's rural population living in a village with a paved approach road, from the Census village directory (SHRUG), in 2001 or 1991. Each moderator is standardised across districts, so interactions are per standard deviation, and enters interacted with current and lagged upstream rainfall (lagged shown); its main effect is absorbed by the district fixed effect, and log Upstream Precip. is evaluated at the mean. Column 2 also interacts local rainfall with the road share; column 3 interacts upstream rainfall with five other 2001 district characteristics. Column 4 runs on the districts with 1991 village-directory data, which exclude Jammu and Kashmir (no 1991 census). Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table E.15: Upstream rainfall partitioned by dam regulation

	Aggregate	Regulated	Free-flowing
log Upstream Precip. (t)	-1.347		
	(1.206)		
log Regulated Upstream Precip. (t)		-1.082	
		(1.033)	
log Free-Flowing Upstream Precip. (t)			-1.192
			(1.130)
log Upstream Precip. $(t-1)$	1.284*		
	(0.668)		
log Regulated Upstream Precip. $(t-1)$		0.721*	
		(0.401)	
log Free-Flowing Upstream Precip. $(t-1)$			1.671*
			(0.944)
Observations	4932	4932	4932
Number of Districts	137	137	137
Mean of dependent variable	0.79	0.79	0.79
District & Year FE	✓	✓	✓

Notes: Poisson fixed-effects, UCDP event counts, 1990–2025, districts holding both a regulated and a free-flowing upstream component. Regulated water passes a dam completed by 1989, before the sample starts. Each column enters one component of the same upstream monsoon rainfall; local weather controls (current and lagged) included. The components share 76% of their within-district variation, so the contrast is estimated jointly in the text (total water plus composition) rather than by differencing these columns. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

E.4. The Dam Arm

Table E.16: The dam arm's first stage, and the routing counterfactual

	Free-flowing	Regulated	Gap (SE)
A. Flood-stage days per log point of upstream monsoon rain			
Routed days above the 2-year level	−1.51	9.27	−10.78 (2.42)
Districts		263	
Mean of dependent variable		2.52	
B. CaMa counterfactual: the same rain, dams on vs. off			
Conflict on water that arrived		0.0184	
Conflict on water held back		0.0411 (0.0197)	
Districts		162	
Mean of dependent variable		2.57	

Notes: Panel A regresses routed flood-stage days on log upstream monsoon rainfall separately for catchments with no dam (free-flowing) and with at least one (regulated), with district and year fixed effects; the gap is the difference. A buffering account requires the regulated catchment to deliver *less* flood per unit of rain; it delivers more, because dams sit on the largest rivers — the observational split has no usable first stage. Panel B holds the river fixed: CaMa-Flood is run twice on identical rainfall, with and without reservoir operation, and conflict is regressed on the water that arrived and the water held back, jointly. Attenuated delivery shows no differential conflict response. Panel A covers 1989–2025; Panel B covers 2001–2018, the years of the dam-operation run.

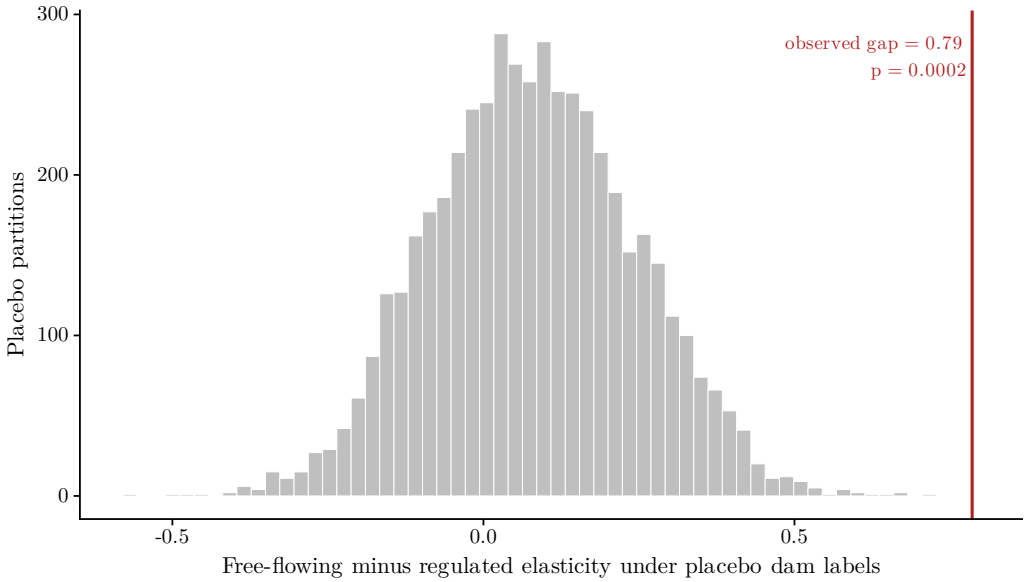


Figure E.14: Permutation distribution for the free-minus-regulated gap, full sample

Notes: Distribution of the gap across 5,000 within-district redistributions of dam labels: within each district, which upstream sub-basins are labelled dam-regulated and which free-flowing is reassigned at random, keeping the number of each and holding the sub-basins, their weights and rainfall, and the outcomes fixed. The red line is the observed gap, 0.79 ($p = 0.0002$). Excluding Assam the observed gap falls to 0.07 and 3,294 of the same 5,000 placebo draws exceed it.

E.5. The Flood Arm

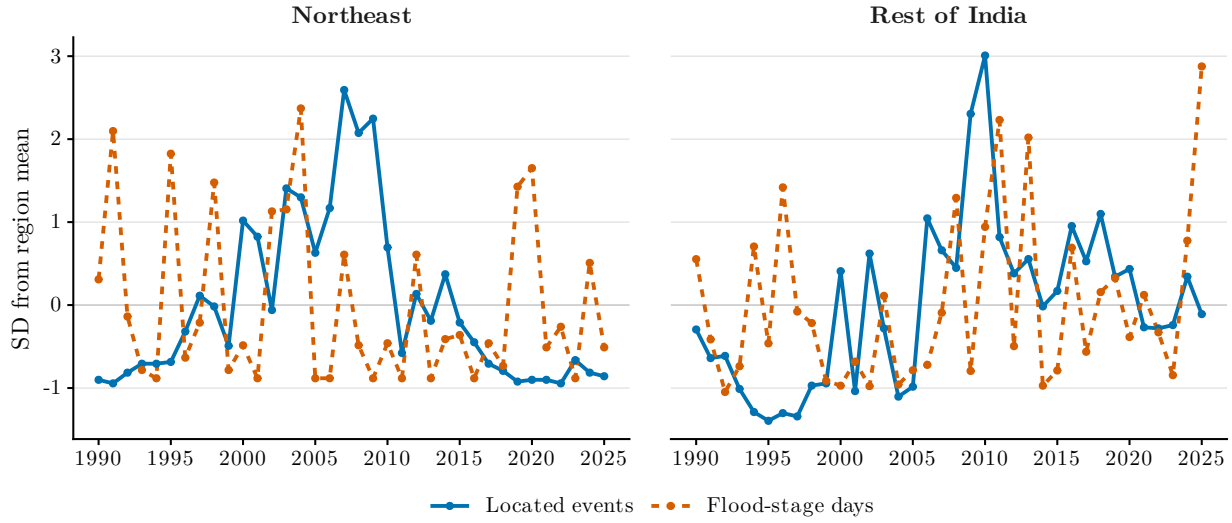


Figure E.15: Regional co-movement of conflict and routed flooding on the estimation sample, standardized within region (one axis; the two series have different units)

Notes: Located events per district (solid) and flood-stage days per district (dashed), 1990–2025. Jammu & Kashmir is omitted: its routed flood series is zero in every year but four, all outside 2000–2018.

References

Abatzoglou, J. T., Dobrowski, S. Z., Parks, S. A., and Hegewisch, K. C. (2018). TerraClimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958–2015. *Scientific Data*, 5:170191.

Adão, R., Kolesár, M., and Morales, E. (2019). Shift-share designs: Theory and inference. *Quarterly Journal of Economics*, 134(4):1949–2010.

Bates, P. D., Horritt, M. S., and Fewtrell, T. J. (2010). A simple inertial formulation of the shallow water equations for efficient two-dimensional flood inundation modelling. *Journal of Hydrology*, 387(1–2):33–45.

Borusyak, K. and Hull, P. (2023). Nonrandom exposure to exogenous shocks. *Econometrica*, 91(6):2155–2185.

Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1):181–213.

Emerick, K., de Janvry, A., Sadoulet, E., and Dar, M. H. (2016). Technological innovations, downside risk, and the modernization of agriculture. *American Economic Review*, 106(6):1537–1561.

Fetzer, T. (2020). Can Workfare Programs Moderate Conflict? Evidence from India. *Journal of the European Economic Association*, 18(6):3337–3375.

Funk, C., Peterson, P., Harrison, L., et al. (2026). The Climate Hazards Center infrared precipitation with stations, version 3. *Scientific Data*, 13:718.

Funk, C., Peterson, P., Landsfeld, M., et al. (2015). The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. *Scientific Data*, 2:150066.

Harari, M. and La Ferrara, E. (2018). Conflict, Climate, and Cells: A Disaggregated Analysis. *The Review of Economics and Statistics*, 100(4):594–608.

Harris, I., Osborn, T. J., Jones, P., and Lister, D. (2020). Version 4 of the CRU TS monthly high-resolution gridded multivariate climate dataset. *Scientific Data*, 7:109.

Lehner, B. and Grill, G. (2013). Global river hydrography and network routing: Baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15):2171–2186.

Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., et al. (2021). ERA5-Land: A state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13(9):4349–4383.

Patel, D. (2024). Floods. Working paper, Harvard University.

Schneider, U., Becker, A., Finger, P., Meyer-Christoffer, A., Ziese, M., and Rudolf, B. (2014). GPCC's new land surface precipitation climatology based on quality-controlled in situ data and its role in quantifying the global water cycle. *Theoretical and Applied Climatology*, 115(1–2):15–40.

Tellman, B., Sullivan, J. A., Kuhn, C., Kettner, A. J., Doyle, C. S., Brakenridge, G. R., Erickson, T. A., and Slayback, D. A. (2021). Satellite imaging reveals increased proportion of population exposed to floods. *Nature*, 596(7870):80–86.

Yamazaki, D., de Almeida, G. A. M., and Bates, P. D. (2013). Improving computational efficiency in global river models by implementing the local inertial flow equation and a vector-based river network map. *Water Resources Research*, 49(11):7221–7235.

Yamazaki, D., Ikeshima, D., Sosa, J., Bates, P. D., Allen, G. H., and Pavelsky, T. M. (2019). MERIT Hydro: A high-resolution global hydrography map based on latest topography dataset. *Water Resources Research*, 55(6):5053–5073.

Yamazaki, D., Kanae, S., Kim, H., and Oki, T. (2011). A physically based description of floodplain inundation dynamics in a global river routing model. *Water Resources Research*, 47(4):W04501.

Yamazaki, D., O'Loughlin, F., Trigg, M. A., Miller, Z. F., Pavelsky, T. M., and Bates, P. D. (2014a). Development of the global width database for large rivers. *Water Resources Research*, 50(4):3467–3480.

Yamazaki, D., Sato, T., Kanae, S., Hirabayashi, Y., and Bates, P. D. (2014b). Regional flood dynamics in a bifurcating mega delta simulated in a global river model. *Geophysical Research Letters*, 41(9):3127–3135.